The Folded Space

Fold, Return and Closure:
a Treatise on the FTM World Model

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Abstract

Space is described as a closed, folded three-dimensional manifold, pictured as a funnel. Mass is fold, $\Delta s = GM/(3c^2)$. Slope, clock rate and energy follow from it. Because space has no outside, every motion returns in an inner direction that closes at the closure length Λ. From this follow a threshold $a_0 = c^2/(2\Lambda)$ and a rule of motion, $a = a_N/(1 - e^{-\sqrt{(a_N/a_0)}})$, which is Newtonian above the threshold and gives flat rotation curves below it, without additional mass. The rotation curves of 141 galaxies (SPARC) read $a_0 = 1.11 \cdot 10^{-10}$ m/s^2, the scale of the early sound wave (DESI) 1.10, the shape of the distances of supernovae and BAO 1.09: the same threshold for part and whole. The breathing after Planck reads 1.065. Dark energy is the restoring of the three loops of the torus. From a_0, the measured densities of the baryons and the radiation and the measured weight of the mode follow $H_0 = 67.2$ km/s/Mpc and $\Omega_{DE} = 0.685$. All levels are ordered by three symmetries of the torus shape, mirror, threefold and full turn, and by two ladders. Nucleus, atom, substances, the docking of drugs, stars and galaxies show the same forms, tested on nuclear masses (AME2020), on the periodic table (PubChem) and on the galaxies.

Preface: Terms Are Tools

This treatise describes the world with a few images: funnel, fold, slope, clock rate, return, loop, closure. These words are tools, not new things. Behind each of them stands a mathematical object that geometry named long ago. The most important one is the **manifold**: a space that looks like ordinary flat space in the small, but may be curved, folded and closed in on itself in the large. The surface of the Earth is a two-dimensional manifold: every piece looks flat like a sheet of paper, while the whole is a sphere without an edge.

Fold and **funnel** are, in this model, the same as a curved three-dimensional manifold, only named more vividly. We keep our terms because they help to explain: a funnel can be drawn, a metric cannot. Wherever a term has its own precise meaning in the model, it is explained where it first appears. All symbols are collected in Appendix A.

One rule holds throughout: nothing is explained as an additional substance. Everything observed is a state or a rule of the one medium, space itself.

The course of the treatise. Part I starts with the funnel: what the fold is, and how slope, clock rate and energy follow from it. Part II closes space: because it has no outside, every motion returns, and the threshold and the rule follow from that return. Part III shows the symmetries that carry the whole, the mirror, the threefold and the full turn, and the two ladders on which all levels are ordered. Part IV goes through the levels from the particle to the galaxy, Part V through the whole with threshold, breathing and dark energy. Part VI explains how to read measurements without mistaking rules for things, what can be tested and what is set, as in the standard model. Every step builds only on the previous ones.

Part I: The Funnel

1 The Funnel as an Image of the Folded Manifold

We live in three spatial directions. We cannot look at a curvature of this three-dimensional space from outside, because there is no outside from which to look. One helps oneself with a cut: take a flat plane through the middle of a body, say the Sun, and draw how long the paths in this plane really are. Where space is folded, the paths inward are longer than flat geometry would lead us to expect. Drawing this extra length as depth pointing downward produces a **funnel**.

The funnel therefore shows the three-dimensional manifold in two dimensions. Its depth is not a place one could fall into, but a measure: a funnel that sinks deeper represents **more space within space**. Between two circles of equal circumference, the funnel holds a longer path than the flat plane. Space contains more there than its boundary reveals.

2 The Fold Δs

The **fold** Δs is the extra length in the centre of a body that folded space contains compared with flat space. It is the measure of mass:

$$\Delta s = GM/(3c^2)$$

Derivation. The fold grows with the mass, and it is distributed over the body: for a body of mass M, volume V, surface S and radius R, the extra radial length is $\Delta s = (GM/c^2)\cdot V/(S\cdot R)$, the length GM/c^2 weighted by the ratio of content to boundary. In the language of general relativity this is the excess radius of a homogeneous sphere. The ratio $V/(S\cdot R)$ is a pure shape number: for the sphere, $V = (4/3)\pi R^3$ and $S = 4\pi R^2$, so $V/(S\cdot R) = 1/3$. Hence the 3 in the denominator: it is the spatial dimension d = 3. A homogeneous sphere in d dimensions has $\Delta s = GM/(d\cdot c^2)$.

Examples. The Earth carries a fold of Δs = 1.478 mm, the Sun of Δs = 492.2 m, the Milky Way with $6\cdot 10^{10}$ solar masses of Δs = 197.4 AU. The fold depends only on the mass, not on the radius: a star that inflates keeps its fold.

3 The Slope g

The **slope** is the steepness of the funnel wall at a place, what we feel as weight:

$$g = GM/r^2 = 3c^2\cdot\Delta s/r^2$$

It follows from the fold by inserting $GM = 3c^2\Delta s$. The product $g\cdot r^2 = 3c^2\Delta s$ is fixed: the fold sits in the centre, and the slope falls off outward with the square of the distance. If a body expands at fixed mass, its edge flattens while the fold in the centre remains (the balloon effect): a red giant is deeply folded and only weakly sloped at its surface.

4 The Clock Rate and Time

The **clock rate** dτ/dt is the ratio in which space unrolls at a place, compared with flat outer space. It follows from the fold, since the clock rate at distance r is $\sqrt{(1 - 2GM/(rc^2))}$, and $2GM/c^2 = 6\Delta s$:

$$d\tau/dt = \sqrt{(1 - 6\Delta s/r)} \qquad \delta = 1 - d\tau/dt \approx 3\Delta s/r$$

Time is the difference in unrolling between folded and flat regions, not friction and not a flow of its own. It arises only where space is folded. Between two places it is $\delta_{AB} = d\tau/dt|_B - d\tau/dt|_A$. The small quantity δ is called the **spread**: the distance of the clock rate from its flat value of one.

At the **limit radius** $r_0 = 6\Delta s$ the clock stands still. For the Earth it lies at 8.87 mm, for the Sun at 2.95 km. A clock on the Earth's surface falls behind flat outer space by 0.022 s per year, on the Sun's surface by 67 s. In the Sun's funnel at the Earth's orbit the figure is 0.31 s. On a neutron star (1.4 M☉, 12 km) the clock rate is 0.81.

Light in the clock rate. c is the scale of the unfolded envelope: $c^2 = GM/(3\Delta s)$ links fold and slope. Measured at the place, light is always c, because clock and light follow the same rate. Counted from outside it is slower, $c_{eff} = c\cdot\sqrt{(1 - 6\Delta s/r)}$, a delay of the unrolling, not friction.

5 Energy Is Bound Fold

Mass and energy are the same, and both are fold:

$$E = Mc^2 = 3c^4\cdot\Delta s/G$$

Every energy, whether rest, motion or field, is bound fold. When a body binds to another, it gives up fold.

Derivation of binding. On a circular orbit in the funnel, $v^2 = GM/r = 3c^2\Delta s/r$, hence $\delta = 3\Delta s/r = v^2/c^2$: the spread of an orbit is the square of its speed relative to c. The binding energy of a circular orbit is $\frac{1}{2}mv^2$, that is $\frac{1}{2}\cdot\delta$ of the rest energy. **Binding gives up half the spread.**

Examples. In the hydrogen atom the electron circulates with v/c = α, the fine-structure constant. The binding is therefore $\alpha^2/2$ of the electron's rest energy, $2.661 \cdot 10^{-5}$ including the co-motion of the nucleus. The measured 13.598 eV (PubChem) is exactly this fraction. The iron nucleus ^{56}Fe carries 0.94 % less mass than its nucleons would carry separately (AME2020).

6 Orbits and Light in the Funnel

A planet follows the slope that the Sun's fold produces at its orbital radius:

$$v = c \cdot \sqrt{(3\Delta s_\odot / r)} \qquad T^2 = 4\pi^2 \cdot r^3 / (3c^2 \cdot \Delta s_\odot)$$

With $\Delta s_\odot$ the fold of the Sun this gives for the Earth the orbital speed v = 29.78 km/s and the period T of one year. Every motion in the funnel feels two things: the slower clock rate and the extra length. For light both parts are equal:

$$\text{deflection} = 6\Delta s/b \text{ (clock rate)} + 6\Delta s/b \text{ (fold)} = 12\Delta s/b \qquad \text{solar limb: } 0.8756'' + 0.8756'' = 1.7512''$$

Here b is the distance at which the ray passes the centre. Orbits feel both too: the perihelion of an orbit advances by $(12\pi + 12\pi - 6\pi) \cdot \Delta s/p = 18\pi\Delta s/p$ per revolution, with p the parameter of the orbit, its half width through the focus. For Mercury that is 42.98″ per century. Expressed with the spread $\delta = 3\Delta s/p$: the orbit misses the full turn by 3δ, and light is deflected by 4δ.

The funnel has fixed **steps**: at $r_0 = 6\Delta s$ the clock stands still, at 9Δs lies the photon sphere, at 18Δs the innermost stable orbit (Sun 4.43 and 8.86 km). The circles 6Δs, 9Δs, 18Δs are 2d, 3d, 6d with d = 3: the whole numbers of the funnel are multiples of the dimension. Every circle is the zero of a rate in $\delta = 3\Delta s/r$, and the multiples are the factors in front of δ. Around a body without twist the clock at rest runs, counted from outside, with $\sqrt{(1 - 2\delta)}$, and on the circular orbit $v^2 = GM/r = \delta \cdot c^2$ exactly. The revolution runs across the depth, without fold distance, and rate and speed add in their squares as for perpendicular quantities: the orbiting clock runs with $\sqrt{(1 - 2\delta - v^2/c^2)} = \sqrt{(1 - 3\delta)}$. A co-moving gyroscope turns back against the radius in its own time with $\Omega = \sqrt{(GM/r^3)}$ and runs ahead against the distant stars by $2\pi \cdot (1 - \sqrt{(1 - 3\delta)})$ per revolution, weakly 3πδ. A small deviation of the orbit, u = 1/r, follows in the orbital angle the equation $d^2\delta u/d\varphi^2 + (1 - 6\delta) \cdot \delta u = 0$, for bodies and light: the oscillation of the orbit against its revolution is $(\mathrm{Re}\ \tau_M)^2 = (\omega_{rad}/\omega_{azim})^2 = 1 - 6\delta$. At 6Δs the clock at rest stops. At 9Δs the orbiting one stops, there the orbit runs with c for the clock at rest at the place, $v_{loc}/c = \sqrt{(\delta/(1 - 2\delta))}$: the meeting point of the orbit. At 18Δs the radial oscillation stops, the orbit runs there with c/2 for the clock at rest, and inside a perturbed orbit does not stay on the circle. The energy of the orbit in rest energy is the square of the clock at rest divided by the orbiting one, $(1 - 2\delta)/\sqrt{(1 - 3\delta)}$, smallest at 18Δs and exactly one at 12Δs = 4d: inside, no circular orbit is bound. To first order $\mathrm{Re}\ \tau_M = 1 - 3\delta$: the orbit misses the full revolution by 3δ, the rotation of the perihelion. The fold lowers $(\mathrm{Re}\ \tau_M)^2$ below one, down to zero at the innermost stable orbit, and the pericentres run with the revolution. The rule raises it with 1 + χ above one, up to two in the deep regime, and the pericentres run against the revolution. At the photon sphere $(\mathrm{Re}\ \tau_M)^2 = -1$: the oscillation becomes growth, a deviation of light grows by the factor e per radian, and counted from outside revolution and decay have the same rate c/b_c with $b_c = 9\sqrt{3} \cdot \Delta s$, the edge of the shadow. For large l the ringdown is this circling light, $\omega = (c/b_c) \cdot ((l + \tfrac{1}{2}) - i \cdot (n + \tfrac{1}{2}))$ with the overtone n, its revolution $c/(2\pi \cdot b_c)$ at one solar mass 6.2 kHz.

The trace of the revolution. After one revolution, $\varphi \to \varphi + 2\pi$, the equation of the deviation, $d^2\delta u/d\varphi^2 + (\mathrm{Re}\ \tau_M)^2 \cdot \delta u = 0$, maps the state $(\delta u, \delta u')$ linearly, with determinant 1 and trace $2 \cdot \cos(2\pi \cdot \mathrm{Re}\ \tau_M)$. Its eigenvalues are a mirror with the fixed product one and the trace as their sum, $\lambda^2 - \mathrm{trace} \cdot \lambda + 1 = 0$. By its trace the map is of the same three kinds as every symmetry of the torus shape, as rotation, shear and stretching with no, one and two fixed directions of the deviation, like the symmetries with the fixed point inside, one and two fixed points on the boundary. Outside 18Δs the eigenvalues lie on the unit circle, $e^{\pm 2\pi i \cdot \mathrm{Re}\ \tau_M}$: the deviation turns by $2\pi \cdot \mathrm{Re}\ \tau_M$ per revolution, and no direction stays fixed, except at integer or half-integer $\mathrm{Re}\ \tau_M$, where every direction returns. At 18Δs they merge into one, and the map is the shear $(\delta u, \delta u') \to (\delta u + 2\pi \cdot \delta u', \delta u')$, the same form as the full revolution $\tau \to \tau + 1$: only the neighbouring circle of equal angular momentum stays fixed, every other deviation grows linearly. Inside they separate as real numbers, $e^{\pm 2\pi \cdot \sqrt{(6\delta - 1)}}$, with the growing and the shrinking deviation as fixed directions, at the photon sphere with the trace $2 \cdot \cosh 2\pi = 535.5$. At 24Δs = 8d, $\mathrm{Re}\ \tau_M = \tfrac{1}{2}$: the deviation returns after one revolution with reversed sign, the half turn, and the orbit near the circle closes after two revolutions. Newton and the held body, $\mathrm{Re}\ \tau_M = 1$ and 2, have the identity, for every orbit, not only near the circle. The rule alone lowers the trace from 2 at most to $2 \cdot \cos(2\pi\sqrt{2}) = -1.72$ and takes no circular orbit around a point mass out of the rotation. Far out, the restoring of the end state lowers $\mathrm{Re}\ \tau_M$ again to zero: at the last stable circular orbit around a bound region lies the second shear, with the neighbouring circle of equal angular momentum as fixed direction, and up to R_{max} the map stretches. Around a galaxy whose mode has been

handed over, the sequence runs back, with the identity at $R_{max}/\sqrt{3}$ and the half turn at $\sqrt{(7/15)}\cdot R_{max}$. Around a mass whose field stays GM/r^2, the half turn lies at $R_{max}/\sqrt[3]{5}$. The stable circular orbits thus lie between two shears, each preceded by the half turn, the inner one set by the folding, the outer one by the restoring.

The trace on one straight line. In the orbital angle an orbit obeys $d^2u/d\varphi^2 + u = \Sigma\, F_i(u)$, with L its angular momentum per mass in its own time. Every term $F_i \propto u^{d_i - 3}$ carries on the circle the share $w_i = F_i/u$, $\Sigma\, w_i = 1$, and a deviation oscillates with $(\mathrm{Re}\,\tau_M)^2 = 4 - d_{eff}$, $d_{eff} = \Sigma\, w_i \cdot d_i$. For a slope $F_i = g_i/(L^2u^2)$, and d_i is its flux dimension: Newton d = 3, the rule $3 - \chi$, the restoring d = 0 with a negative share. The folding adds the term $3GMu^2/c^2$, with d = 5. It is no slope at the place, it depends on the revolution: counted from outside the revolution stays Keplerian, in its own time it is faster, therefore Newton carries only $1 - 3\delta$, the square of the orbiting clock, and the folding the remainder 3δ. Thus $d_{eff} = 3 + 6\delta$, the point at the fraction 3δ on the segment from 3 to 5, and $(\mathrm{Re}\,\tau_M)^2 = 1 - 6\delta$. On the straight line $(\mathrm{Re}\,\tau_M)^2 = 4 - d_{eff}$ lie the three kinds of the trace, below 4 the rotation, at 4 the shear, above it the stretching. The identity lies at 3 and 0, Newton and the held body. The mirror of forces that carries one into the other keeps $\mathrm{Re}\,\tau_M \cdot \mathrm{Re}\,\tau_M' = 2$ fixed, its mirror point is the deep regime at 2. The half turn lies at 15/4. The rule lowers d_{eff}, folding and restoring raise it, both past 15/4 up to 4. Inside, the shear lies in the middle of the segment, at $18\Delta s$ Newton and folding each carry half. At its end, the photon sphere $9\Delta s$, $d_{eff} = 5$ and $(\mathrm{Re}\,\tau_M)^2 = -1$: the orbiting clock stands still, Newton carries nothing, and the folding alone holds the orbit. Light has in the orbital angle only its term and therefore circles only there.

The generators on the line. Every symmetry of the torus form has an integer trace. A rotation therefore has the form of such a symmetry only at trace 0 or ±1, at any other trace between −2 and 2 it is no symmetry of the form. On the line, between integer and half-integer $\mathrm{Re}\,\tau_M$, where every direction of the deviation returns, the trace is zero in the middle, at $\mathrm{Re}\,\tau_M = \frac{1}{4}$ and $\frac{3}{4}$: the map has the form of the mirror $\tau \to -1/\tau$, every direction returns after two revolutions, and the orbit near the circle closes after four. At one third and two thirds, at $\frac{1}{6}$ and $\frac{1}{3}$, $\frac{2}{3}$ and $\frac{5}{6}$, the trace is ±1: the map has the form of the threefold rotation, every direction returns after three revolutions, and the orbit closes after three or six. Between two identities thus lie threefold rotation, mirror and threefold rotation, and with the shear all three generators stand on the line. The fold runs through two such sequences, from Newton to the half turn at $24\Delta s$ and from there to the shear at the innermost stable orbit. The rule around a point mass, $\mathrm{Re}\,\tau_M$ from 1 to $\sqrt{2}$, runs through one sequence and does not reach the identity at 3/2: the threefold rotation at $a_N = 3.17 \cdot a_0$, the mirror at $1.12 \cdot a_0$ and the threefold rotation at $0.23 \cdot a_0$.

The winding on the line. With the energy of the carrier at any speed, $W = \sqrt{(p^2c^2 + m^2c^4)} - K_{el}/r$, an orbit around a winding follows in the orbital angle $d^2u/d\varphi^2 + u = K_{el} \cdot W/(p_\varphi \cdot c)^2 + \alpha_\varphi{}^2 \cdot u$, with $K_{el} = |Q_1Q_2|/(4\pi\varepsilon_0)$ the coupling of opposite windings, p_φ the angular momentum and $\alpha_\varphi = K_{el}/(p_\varphi \cdot c)$. On the circle α_φ is exactly its speed relative to c, at any speed, like $\alpha = v/c$ in the atom. Like Newton, Coulomb carries the square of the orbiting clock, $1 - \alpha_\varphi{}^2$, and the second term the remainder $\alpha_\varphi{}^2$ with d = 4. It is no slope, it comes from the energy of the carrier, $W + K_{el} \cdot u$, which grows inwards. Its share depends only on the angular momentum, not on the depth, so the term is linear in u, and every orbit, not only one near the circle, turns as a rosette by the same angle, $(\mathrm{Re}\,\tau_M)^2 = 1 - \alpha_\varphi{}^2$. Around the winding the orbiting clock carries no clock rate, only the speed: at equal speed the share is v^2/c^2, a third of the 3δ of the folding, which also carries the clock rate 2δ, and the segment, from 3 to 4, is half as long as the one from 3 to 5. Per revolution the orbit therefore advances weakly by $\pi\alpha_\varphi{}^2$, a sixth of the folding. The shear lies at $K_{el} = p_\varphi \cdot c$, with angular momentum $\hbar$ at $|Q_1Q_2| = q_P{}^2$, the meeting point of the winding, $Z\alpha = 1$. Beyond it the map stretches, and every orbit falls into the centre. Thus the orbit around a winding reaches c at its shear, the one around a folding only at the end of its segment.

Part II: The Closed Space

7 No Outside: the Torus

Space has no edge and no outside. It loops back into itself. The simplest form of such a space in three directions is the **torus T^3**: take a cube and declare opposite faces to be the same. Whoever walks out on the right comes back in on the left. Every direction therefore has a finite **loop**, a closed path that leads around space. The torus T^3 has three independent loops.

Because there is no outside, fold cannot drain away anywhere. When funnels fold extra length into their centres, the closed medium answers with a stretching of the loops:

$$\sigma = \Sigma \Delta s_{\text{centres}} - \Delta L_{\text{space}} = 0$$

The medium thus runs a cycle: flat matrix ($\Delta s = 0$), vortex (slope g), fold (extra length Δs in the centre), return through the closed whole. Charges, by contrast, cancel on the closed space, $\Sigma Q_{el} = 0$. And the form of the slope is constrained: the torus has Euler characteristic zero, so every slope obeys funnels − filament saddles + wall saddles − voids = 0 (Section 28).

8 The Inner Direction and the Return

Because space is closed, every motion returns. This **return** does not run in space, but in an **inner direction** ϖ, orthogonal to space: a half-line from the place, closed at the **closure length** Λ. It is not a spatial direction. One cannot travel in it. It is the path on which the state of a place finds its way back to itself.

The return obeys a single equation, a random walk with loss, for the share u of the state that is at depth ϖ at time t:

$$\partial_t u + (\Gamma + \theta)\cdot u = D\cdot \partial_\varpi{}^2 u \qquad \Gamma = a_N/c \qquad D = GM/c$$

Here $a_N = GM/r^2$ is the Newtonian slope at the place, Γ the loss from the clock-rate gradient, θ the loss from the expansion of the region, and D the spreading in the inner direction.

Derivation of the return fraction. Stationary and without expansion, for the profile $u = G(\varpi)$ the equation $D\cdot G'' = \Gamma\cdot G$ remains, with $G(0) = 1$ (G here this profile, not the gravitational constant). The solution falls off exponentially, and at the return depth ϖ_R

$$q = G(\varpi_R) = e^{-\varpi_R\cdot\sqrt{(\Gamma/D)}} = e^{-s} \qquad s^2 = \Gamma\cdot\tau_D \qquad \tau_D = \varpi_R{}^2/D$$

q is the **return fraction**: the share of the state that comes back to the place.

The state space. A point of the state space is a complete state of a system, not a place and not a particle. Every independent direction of circulation is a degree of freedom, and n circulations span a torus T^n. Every force acts in it in one of three roles: **tangential** (it conserves energy and entropy and produces circulations and phases, like the electromagnetic one), **transverse** (it produces entropy and pulls onto the torus, like heat and radiation) or **parametric** (it slowly shifts the torus itself). In one equation: $\dot z = \mathscr{L}(z)\nabla E + \mathscr{M}(z)\nabla S + \varepsilon\cdot F_{par}$, with $\mathscr{L} = -\mathscr{L}^T$ and $\mathscr{L}\nabla S = 0$ (tangential), $\mathscr{M} = \mathscr{M}^T \geq 0$ and $\mathscr{M}\nabla E = 0$ (transverse). Gravity is parametric: it is not a direction of circulation, but fixes which orbits exist and which degrees of freedom are independent. It acts through two channels, the pre-tension (it changes the stiffness and with it which circulations exist) and the clock rate (it shifts all frequencies alike). A slow change keeps the action E/ω fixed, the torus moves to the neighbouring fibre of equal action. Every inner coupling term appears twice in the energy balance with opposite signs, it only shifts energy. Only gravity and externally driven parameters can supply it. A tangential coupling alone can never lock in. Locking needs loss.

Derivation of the threshold. An exactly open direction would have an infinite mean return time. The inner direction is closed at Λ. By Kac's theorem the mean return time to the place is then the path there and back, $\tau_D = 2\Lambda/c$, the same for every mass. With $\Gamma = a_N/c$ we get $s^2 = a_N\cdot\tau_D/c$, and with the **threshold**

$$a_0 = c/\tau_D = c^2/(2\Lambda)$$

it follows that $s^2 = a_N/a_0$. With $D = GM/c$ the depth of the return is $\varpi_R = \sqrt{(D\cdot\tau_D)} = \sqrt{(GM/a_0)} = r^*$: the inner direction reaches as deep as the threshold reaches far in space, and $s = r^*/r$. Kac's theorem holds for every closed state space: for every partition into regions, $\Sigma_i\ 1/\langle T_{A_i}\rangle = 1$, the mean return time into a region being the total measure divided by its measure. Closure thus distributes a fixed budget of recurrence. An open space sets it to zero. For the mean return times of a closed, stationary system into its states, counted from every moment in them, $\Sigma\ 1/\langle T\rangle = 1$. If a part never returns, the sum stays below one: a direct test of closedness. If the region expands, the loss θ from the expansion is added. Whether the rule acts on a perturbation is then decided by the sum $s^2 = a_N/a_0 + \tilde\theta\cdot\tau_D$, not by a_N/a_0 alone, with $\tilde\theta$ the expansion rate of the region, the loss θ of the return equation. The model works with $a_0 = 1.06\cdot10^{-10}$ m/s^2. This belongs to $\Lambda = 13.7$ Gpc and $\tau_D = 89.6$ billion years. τ_D is longer than the age of the universe: the return is not a process along the orbit, but a stationary loop at a fixed place. Its depth follows from $\tau_D = \varpi_R{}^2/D$ with $D = GM/c$: $\varpi_R{}^2 = 2GM\Lambda/c^2 = r_0\cdot\Lambda$, and that is r^{*2} (Section 10). The return depth is the threshold radius.

9 The Rule

Counted at the place, all returns add up: the state comes back with the fraction q, of that again q, and so on. The renewal $u(0) = a_N + u(\varpi_R)$, each return an event in one sense of circulation, gives the **rule**:

$$a = \nu \cdot a_N \qquad \nu = \Sigma\, q^n = 1/(1-q) = 1 + n_B(s) \qquad n_B = 1/(e^s - 1) \qquad q = e^{-\sqrt{(a_N/a_0)}}$$

ν is called the **enhancement**. It has the form of one plus a Bose occupation. What is counted are returns, not particles. In general a feedback loop of gain q sums all returns to $\Sigma q^n = 1/(1-q)$: $q < 0$ damps, $0 < q < 1$ enhances, $q \geq 1$ runs away. The sign belongs to the whole closure, not to the single link.

The two limits. If the slope is large compared with the threshold ($a_N \gg a_0$), s is large, $q \to 0$, $\nu \to 1$: Newton holds exactly. If it is small ($a_N \ll a_0$), $\nu \approx 1/s$, so

$$a \to \sqrt{(a_N \cdot a_0)} = \sqrt{(GM \cdot a_0)}/r \qquad v^2 = a \cdot r = \sqrt{(GM \cdot a_0)} \qquad v^4 = GM \cdot a_0$$

The orbital speed becomes independent of the radius: this is the **flat rotation curve** of galaxies, and $v^4 \propto M$ is the Tully-Fisher relation. What the observer reads as dark matter is this changed rule of motion, not additional mass. The rule has exactly one branch: $d \ln a / d \ln a_N = 1 - \chi/2$ lies strictly between ½ and 1, without jump and without hysteresis, with the switch $\chi = s/(e^s - 1)$ (Section 15), and $d \ln a / d \ln a_0 = \chi/2$. In the deep regime the acceleration of the orbit (tangential) is the mean of slope (parametric) and threshold (transversal), $a = \sqrt{(a_N \cdot a_0)}$, and v_{flat} is at every place the mean of the Newtonian speed, $\sqrt{(GM/r)}$, and the speed of the threshold, $\sqrt{(a_0 \cdot r)}$, both equal at r*. The deep regime is scale-invariant ($r \to \lambda r$, $t \to \lambda t$, $a \to a/\lambda$). From this follows $v^4 \propto M$.

The transition lies at the **threshold radius**, where $a_N = a_0$:

$$r^* = \sqrt{(GM/a_0)}$$

There $\nu = 1.582$. **Example:** the Milky Way as a point mass of $6 \cdot 10^{10}$ solar masses has r* = 8.88 kpc. The rule gives v = 249 km/s at 5 kpc, 214 at r*, 196 at 15 kpc, 178 at 50 kpc, against the deep value of 170.4 km/s. The Sun has its threshold far outside, at r* = 7480 AU. The planetary system lies deep in the Newtonian regime.

The balance. The rule is a balance like that between emission and absorption of light: emission $\propto 1 + n_B$, absorption $\propto n_B$, and their ratio is q. Newton is the spontaneous part, the return the stimulated one. Decomposed, $\nu = \frac{1}{2} + \frac{1}{2} \cdot \coth(s/2)$: a fixed half and the fluctuating part of an oscillator.

Light and orbits. The rule acts on the generator of motion (Section 29), over the return steps along the orbit: on a circular orbit $a = \nu \cdot g_N$ with the slope in the plane of the orbit. The metric carries its curl-free part, $\nu \cdot g_N = -\nabla\Phi + \nabla \times A$: the slope that light follows obeys $\nabla^2\Phi = \nabla \cdot [\nu \cdot \nabla\Phi_N]$, and both potentials of the metric are equal, $\Psi = \Phi$. The vortex comes from $\nabla \times (\nu \cdot g_N) = \nabla\nu \times g_N$: it vanishes where ν is constant on the surfaces of equal Newtonian depth, for sphere, filament and wall, and there orbits and light read the same. In a thin disc the field form reads ν at the full magnitude of the slope at the surface of the layer, with its vertical field $2\pi G\Sigma$. The orbits read only the slope at their place: in SPARC the field form as orbit rule lies $\Delta\chi^2 = 368$ above $\nu \cdot g_N$. Inside discs the lensing mass is therefore smaller than the dynamical mass. In the triad $-\nabla\Phi$ is the centre of the rule and $\nabla \times A$ its vortex: the orbit reads both, light only the centre. Far outside r* the deflection of light tends to the fixed value $\alpha_\infty = 2\pi \cdot (v_{flat}/c)^2$, for the Milky Way 0.42″. A round lens images the source into a ring where its mean surface density inside reaches $c^2/(4\pi G \cdot D_E)$, with $D_E = D_l \cdot D_{ls}/D_s$ from the angular distances of lens and source. For the mass enclosed in the image, therefore, $G \cdot M/R_E^2 = c^2/(4D_E)$ at the ring, for every mass and every round distribution: the ring is the place where the length of this slope reaches the distance $4D_E$, the threshold the place where it reaches ℓ_0. For a point in the Newton regime $R_E^2 = 2r_0 \cdot D_E$, like $r^{*2} = r_0 \cdot \Lambda$. Read as surface density, ring and threshold are the same form, $g = \pi G \cdot \bar{\Sigma}$: the threshold encloses on average $a_0/(\pi G)$, the ring $c^2/(4\pi G \cdot D_E)$. Even in front of the background radiation $4D_E$ stays below 5.1 Gpc (lens at z = 1, in the form part with $a_0 = 1.06$): every ring lies inside the threshold of the mass it encloses, at $G \cdot M/R_E^2 \geq 5.3 \cdot a_0$. Strong lenses lie in the Newton regime of this mass. The beam reads the rule there only on its way in front of and behind the lens, where it crosses r*: it counts every return, $\nu - 1 = \Sigma q^n$, the n-th with $1/n^2$, and at a point mass the deflection grows thereby by $(\pi^2/6) \cdot (b/r^*)^2$ for $b \ll r^*$. At the ring $R_E = \sqrt{(2r_0 \cdot D_E)}$ this is $(\pi^2/6) \cdot 2D_E/\Lambda$, the same for every mass: at $4D_E = 5.1$ Gpc the deflection there grows by 32 %, and the ring lies at $1.20 \cdot R_E$. The weak lens beyond r* reads the rule entirely. Conversely, the flat rotation reads the fold directly, $\Delta s = v_{flat}^4/(3c^2 a_0)$. A generator that does not depend on time and direction conserves energy and angular momentum. As a force at the place, without a generator, no cold disc would hold: with $\nu \cdot g_N$ as a force on each body a disc at the threshold (dispersion

$0.15 \cdot v_c$) expands, as soon as it forms arms and bars, to 2 to 3 times its half-mass radius in 3 Gyr and gives off 4 to 29 % of its angular momentum (seven runs with 20 000 to 80 000 bodies). With a generator, here the field form, it stays at the threshold, its half-mass radius goes from $1.17 \cdot r^*$ to $0.96 \cdot r^*$, and its angular momentum is conserved to 0.2 % as under Newton. In the solar system the return vanishes to every order. At the Earth's orbit the addition is e^{-7480}: the known tests of gravity remain untouched ($\gamma = \beta = 1$), and gravitational waves travel at c, with two polarisations transverse to the direction, from the changing quadrupole of the fold. Their sources always lie far above the threshold frequency $\Omega(r^*)/\pi$, so they are single funnels. In the strong field the return is without effect, waveform and ringdown are those of general relativity. The rise of the frequency, $\dot{f} = (96/5) \cdot \pi^{8/3} \cdot (3\Delta s_c/c)^{5/3} \cdot f^{11/3}$ with Δs_c the fold of the chirp mass of the pair, reads the fold of the source, just as the flat rotation reads that of a galaxy. Two equal masses at separation a radiate $P = (64/5) \cdot (c^5/G) \cdot \delta^5$ with $\delta = 3\Delta s/a$: the power is a power of the spread, and its meeting point c^5/G, the rest energy per light time of the depth, depends on no mass. Therefore every merger of equal folds reaches the same peak power. The fold itself radiates as a black body $(c^5/G)/(15360\pi \cdot \alpha_G)$. The breathing is global and does not radiate.

Travel time. The travel time carries the two halves of the deflection as well. Passing the funnel it lengthens by $\Delta t = (r_0/c) \cdot \ln(4r_1r_2/b^2)$, half from the clock rate, half from the fold, with r_1 and r_2 the distances of source and observer, as long as $b \ll r_1, r_2 \ll r^*$. The deflection is its gradient across the beam, $c \cdot |\partial \Delta t/\partial b| = 2r_0/b = 12\Delta s/b$. Via an image at distance $b = x \cdot R_E$ from a point lens, light including the detour needs, up to a time common to all images, $(r_0/c) \cdot ((x - y)^2 - 2 \cdot \ln|x|)$, with y the distance of the source from the axis in the plane of the lens, counted in R_E. An offset by R_E costs the detour $R_E^2/(2c \cdot D_E) = r_0/c$: detour and funnel count in the same unit, the light time of the limiting radius, 9.85 µs for the Sun, counted at the lens. The images are the stationary points (Fermat), $x^2 - y \cdot x - 1 = 0$. Their sum is the position of the source, their product -1: the two images lie on either side of the lens, $b_+ \cdot |b_-| = R_E^2$, mirror images at the ring, and at $y = 0$ they meet in the ring. Their difference in travel time, $(r_0/c) \cdot (y \cdot \sqrt{(y^2 + 4)} + 2 \cdot \ln((\sqrt{(y^2 + 4)} + y)/(\sqrt{(y^2 + 4)} - y)))$, depends besides r_0/c only on the ratio of their distances from the lens, $y = \sqrt{(b_+/|b_-|)} - \sqrt{(|b_-|/b_+)}$: it reads the fold of the lens without any distance, as the rise of the frequency reads the fold of the source, as long as the rule along the beam stays small, $2D_E/\Lambda \ll 1$.

The rule along the beam as a sheet. Its increase of the deflection at a point mass, $(\pi^2/6) \cdot (b/r^*)^2 \cdot 12\Delta s/b = (\pi^2/3) \cdot b/\Lambda$, is linear in b and depends on no mass: it is the deflection of a uniform sheet. Read as a surface density it is $(1/\pi G) \cdot \int (a - a_N) \cdot d \ln r = a_0/(\pi G) \cdot \int \chi \, ds = (\pi^2/6) \cdot a_0/(\pi G) = 398 \, M\odot/pc^2$, the surface density of the threshold times the integral of the switch, for every mass and every $b \ll r^*$. Read as matter, a uniform layer of this surface density would lie in front of every compact galaxy whose mode has been handed over, within $b \ll r^*$. Against the surface density of the ring, $c^2/(4\pi G \cdot D_E)$, it is $\kappa = (\pi^2/6) \cdot 2D_E/\Lambda$, the increase at the ring. The images read it only as a heavier lens (mass sheet degeneracy), with the fold $r_0/(1 - \kappa)$ and the ring at $R_E/\sqrt{(1 - \kappa)}$. The time delay is shorter by the factor $1 - \kappa$ than that of this heavier lens and therefore, up to terms of higher order in b/r^*, still reads the fold of the lens itself, with y from the distances of the images. For a lens at $z = 0.5$ in front of a source at $z = 2$, in the form part, $\kappa = 0.20$, and the images read 25 % more fold than the lens carries. Reading the time delay with the fold of the images gives H_0 too high by $1/(1 - \kappa)$. Around a cluster whose gas never cooled there is no such sheet, its field is $(1 + Q) \cdot$Newton.

10 The Triad and the Three Radii

Every field on the closed space splits into three parts, the **triad**: the **background** (harmonic, without sources, the loops), the **vortex** (rotation and remaining slope, the return at the place) and the **centre** (the flux out of the mass). The field of the parent level is background for the child level. In this way the triad repeats on every level.

Every body has, in addition, three radii:

$$r_0 = 6\Delta s \text{ (centre)} \qquad R \text{ (envelope, vortex)} \qquad r^* = \sqrt{(GM/a_0)} = c \cdot \sqrt{(3\Delta s/a_0)} \text{ (transition to the background)}$$

The envelope R is the surface of the body. Between r_0 and R runs the vortex, outside r^* the background carries. At the threshold radius the spread is the square of the flat rotation speed, $\delta(r^*) = 3\Delta s/r^* = (v_{flat}/c)^2$. Compact bodies have their threshold far outside their envelope and remain Newtonian. Galaxies have it within their extent. Atom, star, galaxy and cell are the same funnel. They differ only in two ratios: fold to envelope ($\Delta s/R$: clock rate and edge) and envelope to threshold (R/r^*: Newton or return). Both are linked through the whole: $\delta \cdot (R/r^*)^2 = R/\ell_0$ with $\delta = 3\Delta s/R$. The return therefore carries where a body is larger against the closure than its spread, $R/\ell_0 > \delta$. Dark matter is the second ratio, not a substance. The whole lies at its own threshold with r^*/Λ just below one.

Part III: Symmetries and Ladders

11 The Three Generators

The torus has not only loops, it has a **shape**: a flat torus can be square, rectangular or skewed. This shape is described by a single complex number, the **modulus** τ. Different values of τ can describe the same torus. The conversions between them are the symmetries of the torus shape. They form a group, PSL(2,$\mathbb{Z}$), and this group has exactly two **generators** from which every other symmetry is composed:

mirror	$\tau \to -1/\tau$	applied twice: the identity
threefold rotation	$\tau \to -1/(\tau + 1)$	applied three times: the identity
	$\mathrm{PSL}(2,\mathbb{Z}) = \mathbb{Z}_2 * \mathbb{Z}_3$	

The sign $*$ is the free product: every symmetry is a chain of the two generators. To these belongs the **full turn** $\tau \to \tau + 1$: once all the way around, the closure per turn.

Their fixed points are the two distinguished shapes: the square torus (90°, fixed point of the mirror) and the hexagonal one (60°, 120°, fixed point of the threefold rotation). Only these rotations carry the shape of the torus into itself. This is why the images of the same points that arrive along several paths through the cell, the cube of the torus (Section 7), can only be rotated against each other by 60°, 90°, 120° and 180°. The cell carries only the modes $k = 2\pi \cdot n/L$. At the best point of all precision data ($Q = 2\pi - 1$, maximum of the dark energy at saturation) L is 0.998 times the diameter of the last scattering surface, and the cube lowers the quadrupole of the background radiation by 13 % and raises the octupole by 2 %. On Planck's data below $\ell = 30$ this changes χ^2 by only 0.08: the cell is consistent with the data. If L is shorter than the diameter D_S of the last scattering surface, the observer sees the same points of the scattering surface twice: three pairs of opposite circles of radius α, $\cos\alpha = L/D_S$, on three perpendicular axes, the threefold of the loops on the sky. At the best point $L/D_S = 0.9984 \pm 0.0007$, so $\alpha = 3.2°$ (2.4° to 3.9°). The whole path that light covers comoving from the start into the end state is $c \cdot \eta_\infty = \int c \cdot dt/a$, particle and event horizon together, in the form part today 0.51·L and 0.18·L. Against the cell it is a pure number: with $L^3\rho_m = 3c^4/(2Ga_0)$ and the end value $\Theta_\infty = \sqrt{15}/\pi$, a_0 and ρ_m cancel, $c \cdot \eta_\infty/L = (\Theta_\infty/8\pi^2)^{1/3} \cdot \int_0^\infty du/\sqrt{(u + u^4)} = 0.70$, u the stretching a in units of its value at equality of form part and fold. With radiation it is 0.69. More dark energy shortens the path, and the state part is never negative. So no light ray closes a turn of the cell up to the end state. The radius of the last scattering surface grows to 0.68·L: for every a_0, L becomes shorter than its diameter, the three pairs of circles appear and grow to at most $\alpha = 43°$. The diameter of the scattering surface, 1.36·L, stays below the distance $\sqrt{2}$·L of the images across the diagonals of the faces: the threefold of the loops stays on the sky.

Every symmetry of the shape is, by its **trace**, of one of three kinds:

Trace	Fixed point	Kind	In the model
below 2	inside	mirror, threefold rotation	**meeting points**
equal to 2	one, on the boundary, rational	full turn	**closure**, resonance
above 2	two, on the boundary, irrational	passing through, simplest: fixed point at the golden ratio	**open winding**

Meeting point, closure and open winding are the three kinds of one and the same group. Orbits run on tori with irrational winding, and no loop ever closes exactly. The golden winding is the most stable under perturbation, because it lies furthest from every rational resonance. Mirror, threefold and full turn, which recur on every level of the model, are therefore not separate findings, but the whole symmetry of the torus shape.

The trace of a motion. Every small deviation that returns linearly, $x'' + K \cdot x = 0$ in its own course λ, has the same three kinds. After the course λ its map has determinant 1 and trace $2 \cdot \cos(\sqrt{K} \cdot \lambda)$, for $K < 0$ it is $2 \cdot \cosh(\sqrt{-K} \cdot \lambda)$. For $K > 0$ it turns (trace below 2), for $K = 0$ it shears (trace 2), for $K < 0$ it stretches (trace above 2). For the orbit $K = (\mathrm{Re}\ \tau_M)^2$, counted in the orbital angle. A wave runs in time with $K = \omega^2 = \omega_{free}^2 \cdot (1 - q)$: the share q that acts back on it through the wave takes from its free oscillation. Its response $\Sigma q^n = 1/(1 - q)$ is ω_{free}^2/ω^2. In an isothermal gas $\omega_{free}^2 = (v/\ell)^2$ with $v^2 = kT/m$, and q is $(\ell/\ell_J)^2$ for the folding and $-(\ell/\ell_D)^2$ for the winding, which therefore always turns. In the disk $\omega_{free}^2 = \kappa^2 + \sigma^2 k^2$, and q is largest at $|k| = \kappa/\sigma$, equal to $1/Q_T$ (Toomre) there. Every limit of stability is the shear, $K = 0$: the innermost stable orbit, the Jeans length, $Q_T = 1$ and

the tidal density, for in the tide $\mathscr{T}$ a uniform sphere with $K = (4\pi G/3)\cdot\rho - \mathscr{T}$ is torn apart below $\rho = 3\mathscr{T}/(4\pi G)$. For the waves the sum of returns becomes infinite there. The shell has the same three kinds. With the energy E per mass and without restoring, $\frac{1}{2}\dot{r}^2 - GM/r = E$ with M the mass of the field inside the shell, it becomes in the mirror $r = z^2$ with the course $d\lambda = dt/r$ a held body, $(dz/d\lambda)^2 - (E/2)\cdot z^2 = GM/2$: energy and coupling exchange their roles, $K = -E/2$. The bound region turns, $z \propto \sin(\sqrt{K}\cdot\lambda)$, the rolling circle with $\theta = 2\sqrt{K}\cdot\lambda$. After its full revolution, the crossing in the centre, z stands with reversed sign: z and −z are the same shell. The mean shell shears, $E = 0$, $z \propto \lambda$ and $r^3 = (9/2)\cdot GM\cdot t^2$. The void stretches, $z \propto \sinh(\sqrt{-K}\cdot\lambda)$. The mirror $\Phi \to -\Phi$ reverses E and K, exchanges turning and stretching, and its fixed point is the mean shell.

Breathing of a held body. A held body whose distances all breathe alike, $r \to (1 + \varepsilon)\cdot r$, has the same three kinds. The energy of its carriers, $\propto p^n$, falls as in the breathing of the whole, with $(1 + \varepsilon)^{-n}$, and with it $3\int P\, dV = n\cdot E_{kin}$. Every slope g_i that goes as $(1 + \varepsilon)^{1 - d_i}$, d_i its flux dimension, carries $\langle r\cdot g_i\rangle \propto (1 + \varepsilon)^{2 - d_i}$, averaged over the mass. In equilibrium $n\cdot E_{kin} = M\cdot\Sigma\langle r\cdot g_i\rangle$, and with $\langle r^2\rangle$ follows $\ddot{\varepsilon} + K\cdot\varepsilon = 0$,

$$K = \Sigma\,(n + 2 - d_i)\cdot\langle r\cdot g_i\rangle/\langle r^2\rangle$$

Plotted over $\ln(1 + \varepsilon)$, holder and slope are two straight lines, $\ln(n\cdot E_{kin})$ with the gradient −n and $\ln(M\cdot\langle r\cdot g\rangle)$ with $2 - d$. The equilibrium is their intersection, and K is the difference of their gradients times $\langle r\cdot g\rangle/\langle r^2\rangle$: the holder pushes like a slope of dimension $n + 2$. If both run parallel, $d = n + 2$, the body shears: it rests at one mass only, but then at every radius. A circular orbit is such a body, its holder is the revolution, L^2/r^3 with $n = 2$, and K are its epicycles: the 4 in $\kappa^2 = (4 - d_{eff})\cdot\Omega^2$ is $n + 2$, which is why the restoring, $d = 0$, counts fourfold. After Newton, $d = 3$, $K = (n - 1)\cdot\langle r\cdot g\rangle/\langle r^2\rangle$, and the binding per mass is $-((n - 1)/n)\cdot\langle r\cdot g\rangle$. Gas, $n = 2$, turns and binds with half, and when uniform it breathes like the epicycles at its hull, $K = GM/R^3$. The holder at c, $n = 1$, runs parallel to Newton: its mass does not depend on the radius, and its binding is zero. If light holds together with the heat of the gas, $n - 1 = \beta\cdot(4 - 3\beta)/(8 - 7\beta)$ with the share β of the gas in the pressure, 5/18 at $\beta = \frac{1}{2}$ and β/2 at small β: the heavier the star, the closer to the shear it breathes. The folding along of the pressure makes the slope steeper, d above 3, and the shear comes before the holder reaches c. There lies the limiting mass: along a sequence of held bodies the mass is largest where two equilibria coincide like two roots, and there $K = 0$. With the rule, $d = 3 - \bar{\chi}$, $\bar{\chi}$ the switch averaged over r·g. In the deep regime, $d = 2$, r·g does not depend on the radius, and $K = n\cdot\langle r\cdot g\rangle/\langle r^2\rangle$: every holder turns, the deep regime has no limiting mass. Under the restoring, a uniform body of gas whose field is GM/r^2, a cluster with $M = (1 + Q)\cdot M_b$ among them, shears at $R_{max}/\sqrt[3]{4}$, like the last stable circular orbit around a cluster, where its mean density is four times the tidal density of the whole.

Every phenomenon is thus the return of a closed path, counted at a place, weighted with a nome $q = e^{2\pi i\tau}$. Its modulus |q| is the loss, with one sense of circulation: gravity, rule of return, weight of the closure rule, vacuum part. Its phase is the winding, lossless, in both senses of circulation: charge, spin, orbit, entanglement.

The generating form. Every length of a funnel is a point in the triangle of its three directions (Section 14), and the winding adds α^n. The ratio of two points is a coupling. Read at a place, against the distance r, it becomes the nome: $s = r^*/r$, $|q| = e^{-s}$ with $\mathrm{Im}\,\tau = s/2\pi$, and the phase of the winding is $\mathrm{Re}\,\tau$. Reflected about the place, the depth becomes the length of the slope, $g = c^2/(r^2/3\Delta s)$, the reach of the own horizon of the observer at rest at the place (Rindler), and the threshold is the place where this mirror image reaches the closure ℓ_0. Where it reaches $2r_0$ lies the horizon of the fold, where it reaches $4D_E$ the Einstein ring, and its turn is the heat of the slope. What returns is counted by Σq^n, the rule, all multiples together by the η function. The generators act on this: the mirror exchanges the corners and reflects about the midpoints, two mirrors stretch by a coupling, the full revolution counts per radian. The content the form takes from measurement, as the standard model takes its numbers: the amount of baryons, which with Q sets the closure ℓ_0 and thereby ε_L, α and the masses of the carriers, and what holds a body (N, λ_H, J).

12 The Mirror on Every Level

A **mirror** is a fixed product of two quantities: if one grows, the other shrinks in the same ratio, and both lie symmetrically about their geometric mean. The basic case is the fold itself. Every mass has two lengths: the fold $3\Delta s = GM/c^2$, which grows with the mass, and the **quantum length** $\bar{\lambda} = \hbar/(Mc)$, which falls with the mass. Their product does not depend on the mass:

$$\bar{\lambda}\cdot 3\Delta s = (\hbar/Mc)\cdot(GM/c^2) = \hbar G/c^3 = \ell_P{}^2$$

The **Planck length** ℓ_P is the centre of the mirror. Both lengths meet at the Planck mass m_{Pl}. The coupling of gravity, $\alpha_G = (M/m_{Pl})^2$, is the distance from the meeting point.

The same form recurs on every level. Fanned out:

Level	Mirror pair	Fixed product (centre of the mirror)	Meeting point
every mass	fold ↔ quantum length	$\bar{\lambda}\cdot 3\Delta s = \ell_P^2$	Planck mass, $3\Delta s = \bar{\lambda} = \ell_P$
star	Planck length ↔ stellar fold	$\bar{\lambda}_p^2 = \ell_P\cdot 3\Delta s_*$	the holder reaches c (Section 21)
atom	electron radius ↔ Bohr radius	$\bar{\lambda}_e^2 = r_e\cdot a_B$	innermost electron at $Z\alpha = 1$
charge	electric ↔ magnetic charge	$e\cdot g_m$ fixed (Dirac)	Planck charge $q_P = e/\sqrt{\alpha}$
heat	thermal length ↔ quantum length	$\lambda_{th}\cdot\bar{\lambda} = 8\pi\ell_P^2$	Planck temperature
rotation	inner ↔ outer horizon	$a_J^2 = r_+\cdot r_-$	spin number $j = 1$
funnel	limit radius ↔ closure length	$r^{*2} = r_0\cdot\Lambda$	the whole, $r^*/\Lambda = 0.971$
part and whole	fold of the funnel ↔ fold of the whole	r^* as geometric mean	the threshold of every funnel
return	mass ↔ mirror mass	$M\cdot\tilde{m} = m_{Pl}^2/2$	space ↔ inner direction
loops (all multiples)	Newton ↔ deep	$s \leftrightarrow 4\pi^2/s$	$s = 2\pi$, one full turn
whole and place	step integral ↔ local coefficient	$\zeta(k) \leftrightarrow \zeta(1-k)$	$(2\pi)^{-k}$, per radian
forces	Newton (r^{-2}) ↔ held body (r^1)	$(a+3)(A+3) = 4$	r^{-1}, the deep regime
vacuum	Planck density ↔ quantum density of the closure length	$\rho_s \propto \sqrt{(\text{Planck density}\cdot \hbar c/\ell_0^4)}$	ε_L^2 between 1 and ε_L^4

How each row arises.

- *Star:* with $3\Delta s_* = GM_*/c^2$, $M_* = m_{Pl}^3/m_p^2$ and $\ell_P\cdot m_{Pl} = \hbar/c$, one gets $\ell_P\cdot 3\Delta s_* = \hbar^2/(m_p^2c^2) = \bar{\lambda}_p^2$.
- *Atom:* the electron radius is $r_e = \alpha\cdot\bar{\lambda}_e$, the Bohr radius $a_B = \bar{\lambda}_e/\alpha$. Their product is $\bar{\lambda}_e^2$, and α drops out.
- *Charge:* by Dirac, the product of the electric charge e and the magnetic charge g_m is a whole multiple of the quantum of action. The meeting point of both is the Planck charge $q_P = e/\sqrt{\alpha}$.
- *Heat:* a fold has the temperature $kT = \hbar c/(24\pi\Delta s)$ (k the Boltzmann constant), its thermal length is $\lambda_{th} = \hbar c/kT = 24\pi\Delta s$. Multiplied by $\bar{\lambda} = \hbar/(Mc)$ this gives $8\pi\cdot\hbar G/c^3 = 8\pi\ell_P^2$.
- *Rotation:* a rotating funnel has two horizons $r\pm = GM/c^2 \pm \sqrt{((GM/c^2)^2 - a_J^2)}$ with the twist $a_J = J/(Mc)$, J the angular momentum. Their product is $r_+\cdot r_- = a_J^2$, their sum $6\Delta s$. At spin number $j = a_J/(3\Delta s) = 1$ both coincide. Above it they are mirrored complex, $r\pm = 3\Delta s \pm i\cdot\sqrt{(a_J^2 - 9\Delta s^2)}$, and lie on the circle $|r\pm| = a_J$ around the node, at the angle $\cos\varphi = 1/j$ to the depth. For a particle they stand perpendicular, $r\pm \approx 3\Delta s \pm i\cdot a_J$, the imaginary displacement of the node where spin and moment sit. Since $a_J/\bar{\lambda} = J/\hbar$ is the number of quanta of rotation, $j = (J/\hbar)/\alpha_G$: the rotation reaches c when the number of its quanta reaches α_G, $J = GM^2/c$. A particle with spin ½ has $j = m_{Pl}^2/(2M^2)$, the electron $2.9\cdot 10^{44}$, and the meeting point $j = 1$ only at $m_{Pl}/\sqrt{2} = 1.54\cdot 10^{-8}$ kg: a particle is far overspun, therefore it has no horizon, and next to it the medium stays flat. It is winding, hardly fold. The Earth ($J = 5.86\cdot 10^{33}$ kg·m²/s, input) has $a_J = 3.27$ m against $3\Delta s = 4.4$ mm, so $j = 738$, and its horizons stand at $\varphi = 89.92°$. If a held body carries itself entirely by rotation, $v = v_K$ at the hull, its spin number is largest, $j = k/\sqrt{(3\Delta s/R)}$ with $k = I/(MR^2)$: the meeting point of the twist, $j = 1$, lies at $3\Delta s/R = k^2$. A star of form $n = 3$ ($k = 0.075$) reaches at most $j = 52$ (Sun), the white dwarf at its limiting mass (1.417 M☉, $R = 1.03\cdot 10^3$ km, $k = 0.095$) 2.1, the neutron star at the limit of its holder (0.710 M☉, $R = 9.16$ km, $k = 0.174$) only 0.51. Between the barrier of the electrons and that of the nucleons the twist falls below the meeting point: a white dwarf can be overspun, a neutron star never, every rotation it carries lies in the Thales circle of its fold. A galaxy with a flat curve carries at its threshold $a_J = (v_{flat}/c)\cdot r^*$, the mean of threshold and depth, $a_J^2 = r^*\cdot 3\Delta s$. Its spin number is $j = \sqrt{(r^*/3\Delta s)} = c/v_{flat}$, for the Milky Way ($6\cdot 10^{10}$ M☉) 1759, its horizons stand at 89.97° to the depth, a disc at the threshold with $J/M = 2R_d\cdot v_{flat}$ has $\sqrt{2}\cdot c/v_{flat}$. The galaxy too is overspun, because its orbit is slower than c: its spin number falls with mass as $M^{-1/4}$ and reaches one only at the meeting point M_U. A cluster brings its twist along from the infall, linearly as the commutator of the shear T with its form I, $J_i = a^2\dot{D}\cdot\varepsilon_{ijk}\cdot(T\cdot I)_{jk}$. The whole does not twist, its restoring is pure trace and commutes with every form. In the equatorial plane the clock rate of resting clocks remains exactly $\sqrt{(1 - 6\Delta s/r)}$. Co- and counter-rotating orbits differ by $4\pi\cdot a_J/c$, independent of the radius, and the twist separates the vertical cycle of an orbit from the revolution: the orbital plane precesses, the orbital torus goes from T^2 to T^3.

- *Funnel:* $r^{*2} = GM/a_0 = (2GM/c^2)\cdot(c^2/(2a_0)) = r_0\cdot\Lambda$.
- *Return:* Section 16. *Whole and place:* Section 15. *Vacuum:* Section 25. *Part and whole:* Section 23.
- *Loops:* if one counts not only the returns of a loop but all its multiples, the sum is $Z = \prod 1/(1 - q^n)$, up to $q^{1/24}$ the reciprocal of the η-function of the torus. Its mirror $\eta(-1/\tau) = \sqrt{(-i\tau)}\cdot\eta(\tau)$ maps the Newtonian regime onto the deep one, $s \leftrightarrow 4\pi^2/s$. In the deep regime $\ln Z = \pi^2/(6s) + \tfrac{1}{2}\cdot\ln(s/2\pi) - s/24$, and the leading part is the first step of the ladder.
- *Forces:* a force r^a and a force r^A are mirror partners when $(a + 3)(A + 3) = 4$: the map $z \to z^{(a+3)/2}$ of the plane (z the place as a complex number) carries the orbits of one into the orbits of the other. With $z \to z^2$, the ellipse about the centre, the orbit inside a uniformly held body (r^1), becomes the Kepler ellipse about the focus (r^{-2}). The mirror point of bound orbits is the exponent −1: exactly the force of the deep regime, $\sqrt{(GMa_0)}/r$. The second, the exponent −5, leads to the fall into the centre. The rule runs from Newton to the meeting point of the force laws.

Together they show: the levels of the world are mirror images of one another, reflected about fixed centres, and the whole stands to the part in the same form as the proton to the Planck point.

The turn. Besides the mirror a second form recurs, the count per radian: a length meets its loop after one full turn. So the wave of an orbit closes after one turn, $M\cdot v\cdot r = n\hbar$, in the atom at $a_B = \bar{\lambda}_e/\alpha$, so heat is a loop and entropy a number of turns (section 27), a bound region runs as a rolling circle, $r = (r_U/2)\cdot(1 - \cos\theta)$, with the turnaround after half a turn and the crossing at the centre after a full one, without restoring linearly at $\delta_c = (3/20)\cdot(12\pi)^{2/3} = 1.686$ and compressed to $18\pi^2$ times the mean, the whole compares its loops in $\Theta = 2\pi a_0/(cH)$, and its mean return is $1 + Q = 2\pi$.

Recurring forms. Mirror, threefold and turn recur in a few forms, each at many places:

Form	Places	What it fixes
Fixed product, transition at equality (geometric mean, mirror)	$g{\cdot}r^2 = 3c^2\Delta s \cdot \ell_P{}^2 = 3\Delta s{\cdot}\bar{\lambda} \cdot \bar{\lambda}_p{}^2 = \ell_P{\cdot}3\Delta s_* \cdot$ $r^{*2} = 3\Delta s{\cdot}\ell_0 \cdot r^*{\cdot}\bar{r} = \ell_P{\cdot}\ell_0 \cdot \bar{\lambda}_e{}^2 = r_e{\cdot}a_B \cdot \lambda_{th}{\cdot}\bar{\lambda} = 8\pi\ell_P{}^2 \cdot$ $a_J{}^2 = r_+r_- \cdot r_0(M_{irr})^2 = r_0{\cdot}r_+$ (leg theorem) $\cdot$ $b_+{\cdot}b_- = -R_E{}^2$ (images of a point lens) $\cdot$ $e{\cdot}g_m$ (Dirac) $\cdot$ $s^2 = \Gamma{\cdot}\tau_D$	transitions and meeting points where the motion reaches c: Planck mass ($3\Delta s = \bar{\lambda}$), Planck charge, j = 1, holder of the star at c, threshold radius ($a_N = a_0$), couplings and spreads as distance from the meeting point
Roots as a triangle	horizons of the rotating fold in the Thales circle over r_0, height a_J, legs $r_0(M_{irr})$ and $r_0(M_{rot})$ $\cdot$ horizons of the end state as projections of an equilateral triangle, $\cos 3\theta = -M/M_N$ $\cdot$ two roots with the depth as sum and the mirror as product: horizons $r_+ + r_- = 6\Delta s$, $r_+{\cdot}r_- = a_J{}^2$, turning points of an orbit with the energy $-\frac{1}{2}c^2$, images of a lens $x^2 - y{\cdot}x - 1 = 0$, eigenvalues of the trace $\lambda^2 - \text{trace}{\cdot}\lambda + 1 = 0$	limits where two roots coincide, and there the form shears: the rotation at j = 1, the images in the ring, the innermost stable orbit, the limiting mass of two equilibria, the mass at M_N
Sum of returns	$\Sigma q^n = 1/(1 - q)$ $\cdot$ Kac $\langle T\rangle = \mu(X)/\mu(A)$ $\cdot$ $\sigma = \Sigma\Delta s - \Delta L = 0 \cdot a = a_N + n_B{\cdot}a_N$,B at the nome of the place (galaxies) or of the whole (clusters, $n_B = Q$) $\cdot$ along the ray the n-th return with $1/n^2$, together $\pi^2/6$ $\cdot$ fluctuation $\langle\Delta n^2\rangle = n_B{\cdot}\nu = -d\nu/ds$ $\cdot$ in the gas fold $q = (\ell/\ell_J)^2$, winding $q = -(\ell/\ell_D)^2$	binding and enhancement without a source
Turn (count per radian)	the wave closes after one turn, $M{\cdot}v{\cdot}r = n\hbar$: $a_B = \bar{\lambda}_e/\alpha$, $\bar{\lambda}/\alpha_G$, the phase of a locked pair with $\Phi_0 = h/(2e)$ $\cdot$ thermal length as a loop: $2\pi{\cdot}c^2/g$ of the slope (Unruh), $4\pi{\cdot}r_0$ of the fold, $2\pi c/H$ of the horizon $\cdot$ $S \le 2\pi k{\cdot}R/\bar{\lambda}$ (Bekenstein) $\cdot$ $s^2 = 2\pi{\cdot}T_U/T_I$ $\cdot$ $\Theta = 2\pi a_0/(cH)$, $H{\cdot}\tau_D = 2\pi/\Theta \cdot 1 + Q = 2\pi$ $\cdot$ the bound region as a rolling circle, $r = (r_U/2){\cdot}(1 - \cos\theta)$: turnaround after half a turn, crossing in the middle after the whole, $\omega{\cdot}t = \pi/(2\sqrt{2})$ after the turnaround, without restoring linear $\delta_c = (3/20){\cdot}(12\pi)^{2/3} = 1.686$ and compressed to $18\pi^2$ times the mean $\cdot$ $s \leftrightarrow 4\pi^2/s$ with the mirror point $s = 2\pi$	heat and entropy as the number of turns of a length, the whole counted per radian: a length meets its loop after one full turn ($T = T_I$ at $r_0 = \Lambda/2\pi$)
Dimension as a factor	$\Delta s = (GM/c^2){\cdot}V/(S{\cdot}R) \cdot d_{eff} = 3 - \chi$ $\cdot$ spherical mean $\langle\cos^2\theta\rangle = 1/3$ per direction: fold per direction 1/3, emission of the winding across the oscillation 2/3, coupling of winding and twist at the nucleus $4\pi{\cdot}2/3 = 8\pi/3$, diamagnetism $\frac{1}{4}{\cdot}2/3 = 1/6$, pressure of light p = u/3, heat $\frac{1}{2}$kT per direction, (3/2)kT in space	the integers 3, 6, 9, 12, 18 of the model, the same third in all three roles
Square for the perpendicular	$\kappa^2 + v_z{}^2 = (2 + \chi)\Omega^2$ $\cdot$ clock rate and speed of the orbiting clock, $(d\tau/dt)^2 = 1 - 2\delta - v^2/c^2$ $\cdot$ fold and twist, $M^2 = M_{irr}{}^2 + M_{rot}{}^2$ $\cdot$ twist and winding, $a_J{}^2 + r_Q{\cdot}3\Delta s$ $\cdot$ slope in and above the disc, $\sqrt{(g_R{}^2 + (2\pi G\Sigma)^2)}$ $\cdot$ energy of the carrier, $W^2 = p^2c^2 + m^2c^4$	the perpendicular adds in the square, the parallel linearly: fold and field of the winding, $M_{irr} + M_{el}$

Form	Places	What it fixes
One function, two countings (mirror about the middle)	return of the fold $\nu = 1 + n_B(s) = \frac{1}{2} + \frac{1}{2}\cdot\coth(s/2)$, $\nu(s) + \nu(-s) = 1$ · field of the envelope in log r $\nu\cdot f(1-f)$, $f = 1/(e^x + 1) = \frac{1}{2} - \frac{1}{2}\cdot\tanh(x/2)$, $f(x) + f(-x) = 1$ · shell and void, $K \to -K$, cos ↔ cosh · web, $\Phi \to -\Phi$: funnel ↔ void, filament ↔ wall, $\lambda_k \to -\lambda_{4-k}$ · emission, $k - 1 \to 1 - k$	the threshold of the orbits (magnitude, one sense of circulation) and the ladder of the periods (winding, both senses). The middle stays fixed: the middle shell, which shears, and the middle axis, growth at the limit, k = 1
Exponent of a ratio	$q = e^{-s}\cdot\Lambda_{QCD}/M = e^{-2\pi/(\beta_0\alpha_s)}$	large differences from moderate numbers (α_G)
Closing form, opening coupling	$1/r^2$ slope (great circles on S^3, degree n) · mirror field of the envelope (spherical harmonics on S^2, degree n + l, order 2l + 1) · held body · opened by fold, rule, screening, twist	perihelion, rosettes, closure numbers of envelope and nucleus

The fold stands between two counter-lengths: small the quantum length $\bar{\lambda}$ (at the place), large the length of action $\ell_0 = 2\Lambda$ (inner direction). The same form carries every level.

13 Meeting Points and Spread

The fixed points of the mirror are the **meeting points**: the places where motion reaches c. There, centre, mirror and envelope coincide. At the Planck point this happens completely ($3\Delta s = \bar{\lambda} = \ell_P$), and the funnel is degenerate. At the other end of the ladder the whole lies almost the same way, $r_0/\Lambda = 6/(1 + Q)$, $r^*/\Lambda = \sqrt{(6/(1 + Q))}$, with Q the weight of the mode of the whole (Section 24).

The **spread** of every level is the speed of its motion relative to c, the distance from the meeting point:

- Around the quantum length it is linear in the speed, a phase, an angle: for the atom $\alpha = v/c$, and likewise for the fold in $3\Delta s : \bar{\lambda} : \bar{\lambda}/\alpha_G$ with $\alpha_G = v/c$, because at $\bar{\lambda}/\alpha_G$ the quantum length closes after one revolution around its own fold.
- Around the threshold it is quadratic, a magnitude, a clock rate: for the funnel $\delta(r^*) = 3\Delta s/r^* = (v_{flat}/c)^2$, the ratio of the radii $r_0/r^* = 2\delta(r^*)$, like $\delta = v^2/c^2$. Linear or quadratic depends on the centre of the ladder, not on winding or fold.

Coupling constants are thus not arbitrary numbers, but distances from the meeting point of their level.

Which direction carries. Windings cancel ($\Sigma Q_{el} = 0$), folds add. The winding therefore carries until the fold of N nucleons reaches the winding, $N^{2/3}\cdot\alpha_G = \alpha$ (binding per nucleon, up to factors of order one, independent of the holder):

$$M = m_p\cdot(\alpha/\alpha_G)^{3/2} = 2.30\cdot10^{27}\ \text{kg} \qquad \text{depth there } \delta = \alpha^2\cdot m_e/m_p = 2.9\cdot10^{-8}$$

The same limit read in the depth: below it the depth of a body is smaller than the binding of its envelopes in rest energy of the nucleons, the atoms hold it (Earth $\delta = 7.0\cdot10^{-10}$). Above it the fold holds it with its heat, and the envelopes dissolve (Sun $\delta = 2.1\cdot10^{-6}$). The fold reaches the meeting point of its holder at $N^{2/3}\cdot\alpha_G = 1$, on the mass scale of the stars $M_* = m_{Pl}^3/m_p^2$. The two limits stand in the ratio $\alpha^{3/2} = 6.2\cdot10^{-4}$. The return carries where the envelope reaches the threshold, $R \geq r^*$, below the surface density $a_0/(\pi G) = 242\ M\odot/pc^2$: in the galaxies. In the whole all three directions meet, r_0, r^* and Λ lie almost together.

So on every level a different direction of the same funnel carries: particle and atom the winding, planet to star the fold, the compact funnels the fold at the meeting point of their holder, held by the blocking of the half turn (the white dwarf by the windings of the electrons, the neutron star by those of the nucleons), the galaxy the return, the whole all three.

The levels in the forms. Every level has the same forms, none stays empty:

Level	carried by	centre	spread	meeting point	holder	returns
particle	winding	$\bar{\lambda}\cdot 3\Delta s = \ell_P{}^2$	$\alpha_G = (M/m_{Pl})^2$, $j = (J/\hbar)/\alpha_G$	$m_{Pl}/\sqrt{2}$: $j = 1$, $r_0 = \bar{\lambda}$, $\tilde{m} = M$	its own winding, spin ½ on the double path	$\mu/\mu_B = 1 + \alpha/2\pi$
atom	winding	$\bar{\lambda}_e{}^2 = r_e\cdot a_B$	$\alpha = v/c$	Planck charge, $\alpha = 1$	closure per revolution, $2\pi\cdot a_B = h/(m_e\cdot v)$	Fermi form of the envelope
planet, star	fold	$\bar{\lambda}_p{}^2 = \ell_P\cdot 3\Delta s_*$	$\delta = \bar{\lambda}_p/\lambda_H$, for the star $kT/(m_p\cdot c^2)$	M_*: $N^{2/3}\cdot\alpha_G = 1$	atoms, heat	$q = e^{-r^*/R} \approx 0$, Sun $r^*/R = 1.6\cdot 10^6$
white dwarf, neutron star, fold	fold at the meeting point	$r_0 = \bar{\lambda}(\tilde{m})$	$\delta = m_e/m_p$ up to one	the holder reaches c	blocking of the half turn	$q \approx 0$
galaxy	return	$r^{*2} = 3\Delta s\cdot\ell_0$	$(v_{flat}/c)^2 = \sqrt{(\alpha_G\cdot\alpha_I)}$	M_U: $r^* = \ell_0$	closure of the whole, $\delta = r^*/\ell_0$	$\nu = \Sigma q^n$, $q = e^{-r^*/r}$
cluster	return at the nome of the whole	seam: $q = q_{cosm}$	$\delta(R_{200}) = 2.4\cdot 10^{-5}$ (X-COP)	R_{200} at the seam, at $M_{200} = 8\cdot(\ln q)^6\cdot\Theta^3\cdot M_H/200^2$	pressure of the never-cooled gas	$1 + Q$
whole	all three	$r^{*2} = r_0\cdot\Lambda$	$r_0/\Lambda = 6/(1 + Q)$	r_0, r^* and Λ almost in one point	itself, $\Lambda = (1 + Q)\cdot\Sigma\Delta s_b$	$1 + Q = 2\pi$

14 The Ladder of Lengths

Five lengths place every mass: the quantum length $\bar{\lambda}$ (at the place), the Planck length ℓ_P, the fold $3\Delta s$, the threshold radius r^* (in space) and the **length of action** $\ell_0 = 2\Lambda$ (in the inner direction). They form a **ladder**. Where two lengths become equal lies a distinguished mass. Their positions hang on a single number, the ratio of the Planck length to the length of action:

$$\varepsilon_L = \ell_P/\ell_0 = 1.9\cdot 10^{-62}$$

$M_{min} = \varepsilon_L\cdot m_{Pl} = \hbar/(c^2\tau_D)$	$\bar{\lambda} = \ell_0$
$M_c = \varepsilon_L{}^{1/3}\cdot m_{Pl} = 5.8\cdot 10^{-29}$ kg	$r^* = \bar{\lambda}$
m_{Pl}	$3\Delta s = \bar{\lambda} = \ell_P$
$M_U = m_{Pl}/\varepsilon_L = c^3\tau_D/G = 1.1\cdot 10^{54}$ kg	$r^* = 3\Delta s = \ell_0$

These are not bodies, but mass scales at which two lengths become equal. M_c divides the ladder at one third: there $\bar{\lambda} = r^* = \ell_P{}^{2/3}\cdot\ell_0{}^{1/3}$. Both ends are set by the return time τ_D, the lower one through $\hbar$, the upper one through G. Without closure the ladder would have no ends. The mirror $M \leftrightarrow m_{Pl}{}^2/M$ exchanges $\bar{\lambda}$ and $3\Delta s$ and maps M_{min} onto M_U. The map $r \to r^{*2}/r$ is a mirror of the ladder as well: it maps the limit radius r_0 onto the closure length Λ, space onto the inner direction. In units of the upper rung M_U the Tully-Fisher relation is simply $M = M_U\cdot(v_{flat}/c)^4$, and the fold of this mass is $\Delta s(M_U) = ⅔\cdot\Lambda$. As a length the vacuum part lies in the middle of the ladder, at the threshold of the Planck mass: $r^*(m_{Pl}) = \sqrt{(\ell_P\cdot\ell_0)} = 0.117$ mm, and $\rho_s\cdot c^2 = (\pi^3/30)\cdot\hbar c/r^*(m_{Pl})^4$. The one number of the ladder counts the baryons of the cell: with $\Lambda = (1 + Q)\cdot\Sigma\Delta s_b$, $\varepsilon_L = 3m_{Pl}/(2(1 + Q)\cdot N_b\cdot m_p)$, so with $1 + Q = 2\pi$, $\varepsilon_L\cdot N_b = (3/4\pi)\cdot m_{Pl}/m_p$. The cell holds $N_b = 1.6\cdot 10^{80}$ baryons, counted from $\rho_b\cdot L^3$ alike.

By directions. The three basic lengths of every funnel belong to the three directions of the torus: $\bar{\lambda}$ at the place tangential, $3\Delta s$ the depth parametric, ℓ_0 the inner direction transversal. Their means taken in pairs join two directions: $\ell_P = \sqrt{(\bar{\lambda}\cdot 3\Delta s)}$, $r^* = \sqrt{(3\Delta s\cdot\ell_0)}$ and $\tilde{r} = \sqrt{(\bar{\lambda}\cdot\ell_0)}$. Here $\tilde{r}^2 = \hbar\cdot\tau_D/M$: this is how far the quantum motion of the body, $D = \hbar/(2M)$, reaches into the inner direction in one return time, $\tilde{r} = \sqrt{(2D\cdot\tau_D)}$, as its fold with $D = GM/c$ reaches $r^* = \sqrt{(D\cdot\tau_D)}$. The mirror $M \leftrightarrow m_{Pl}{}^2/M$ exchanges $\bar{\lambda}$ and $3\Delta s$ and with them $\tilde{r}$ and r^*, $r^*/\tilde{r} = M/m_{Pl}$, and both cross at m_{Pl} at the length of the vacuum part. To M_c belongs the mirrored $m_{Pl}{}^2/M_c = 8.1\cdot 10^{12}$ kg, where $3\Delta s = \tilde{r}$ at the length $\ell_P{}^{2/3}\cdot\ell_0{}^{1/3}$. Below M_c the threshold lies within the quan-

tum length: for the electron $\bar{\lambda}_e = 510 \cdot r^*$, its field outside the quantum length lies entirely below a_0. Above it the threshold lies outside, for the proton $r^* = 154 \cdot \bar{\lambda}_p$. Below m_{Pl}^2/M_c the body's own quantum motion reaches further than its depth, $\tilde{r} > 3\Delta s$. So the ladder has five meeting masses, $\varepsilon_L^{\,k} \cdot m_{Pl}$ with k = 1, ⅓, 0, −⅓, −1, pairwise mirror images about m_{Pl}. At the ends five of the six lines meet: at M_{min} $\bar{\lambda} = \tilde{r} = \ell_0$ and $r^* = \ell_P$, at M_U $3\Delta s = r^* = \ell_0$ and $\tilde{r} = \ell_P$. At the bottom the quantum length fills the closure and the threshold is one Planck length. At the top the fold fills the closure and the body's own quantum motion reaches only one Planck length in a return time: there lies the whole, its baryons carry $(3/4\pi) \cdot M_U$, with $\tilde{r} = 2.05 \cdot \ell_P$. In the plane of mass and hull, particles and atoms lie below the line $\tilde{r}$ and Earth and galaxy above it. A body held by atoms, $R = a_B \cdot N^{1/3}$, crosses $\tilde{r}$ at $2.0 \cdot 10^{-8}$ kg and r^* at $3.1 \cdot 10^{-8}$ kg, almost at the crossing point of both: lighter grains lie like the atoms below $\tilde{r}$ and above r^*, heavier ones like Earth and star above $\tilde{r}$ and below r^*.

The ratio of two basic lengths is a **coupling** on the edge between their directions: $\alpha_G = 3\Delta s/\bar{\lambda} = (M/m_{Pl})^2$ between parametric and tangential, $3\Delta s/\ell_0 = (v_{flat}/c)^4$ between parametric and transversal and $\alpha_I = \bar{\lambda}/\ell_0 = kT_I/(Mc^2)$ between tangential and transversal, the period of the inner direction in rest energy. The triangle closes, $\alpha_G \cdot \alpha_I = (v_{flat}/c)^4$: Tully-Fisher is the product of the two couplings that carry ħ, and ħ drops out. Each coupling becomes one at a rung, α_G at m_{Pl}, α_I at M_{min}, Tully-Fisher at M_U, and at M_c α_G and α_I are equal. The mirror inverts α_G and exchanges the other two. The winding has the same edges to the place, $\alpha = r_e/\bar{\lambda}_e$ and α_I, its third edge stays empty, the winding has no threshold. On every edge lies a phenomenon: between parametric and tangential the fold at the Planck point, between parametric and transversal the rule of galaxies, between tangential and transversal entanglement, the phase of the winding at a shared node of the inner direction.

One formula for the ladder. Every length of the ladder is $\bar{\lambda}^p \cdot (3\Delta s)^q \cdot \ell_0^{\,r}$ with p + q + r = 1, a point in the triangle of the three directions. A body has no further lengths: from M, ħ, G, c and a_0 only two pure numbers can be formed, M/m_{Pl} and ε_L. Each of its lengths is therefore a point times a number of the form, the horizon $2 \cdot 3\Delta s$, the thermal length of the fold $8\pi \cdot 3\Delta s$ and c times its lifetime $5120\pi \cdot \alpha_G \cdot 3\Delta s$. Every further length needs content, a second mass, a charge, a rotation or a holder. Equivalently $\ell = \ell_P \cdot (M/m_{Pl})^k \cdot \varepsilon_L^{\,-r}$ with k = q − p, parametric against tangential, and r, the transversal share. The corners are the basic lengths, $\bar{\lambda}$ (−1, 0), $3\Delta s$ (1, 0) and ℓ_0 (0, 1), the midpoints of the edges ℓ_P (0, 0), r^* (½, ½) and $\tilde{r}$ (−½, ½), the centroid is $\ell_P^{2/3} \cdot \ell_0^{1/3}$ (0, ⅓). Measuring ln M from m_{Pl} and ln ℓ from ℓ_P in units of $\ln(1/\varepsilon_L)$, x and y, every length is the line $y = r + k \cdot x$, and two lengths meet at $x = (r_2 - r_1)/(k_1 - k_2)$. In the plane x, y the ladder is a rectangle from M_{min} to M_U and from ℓ_P to ℓ_0: r^* and $\tilde{r}$ are its diagonals and cross in the middle, at the length of the vacuum part, $\bar{\lambda}$ and $3\Delta s$ run as a V from the Planck point in the middle of the lower edge into the upper corners. The rule lies on the diagonal r^*: from the mirror point of the loops, $r^*/2\pi$, to the seam, $5.8 \cdot r^*$, it is 0.025 of the height wide. The electron lies at x = −0.36, left of M_c, the proton at −0.31, Earth and Sun at 0.53 and 0.62, the Milky Way at 0.79, a cluster of 10^{15} M☉ at 0.86 and the whole at 0.99. The meeting masses are x = ±1, where $\bar{\lambda}$ and $3\Delta s$ reach the corner ℓ_0, ±⅓, where they reach the centroid, which is why M_c divides the ladder at one third, and 0 at m_{Pl}, where they reach each other. The mirror $M \leftrightarrow m_{Pl}^2/M$ is $k \to -k$: every pair (k, r) and (−k, r) has a fixed product, $\bar{\lambda} \cdot 3\Delta s = \ell_P^2$ and likewise $r^* \cdot \tilde{r} = \ell_P \cdot \ell_0$. Threshold and quantum return of every body are mirror partners about the length of the vacuum part (0, ½). The mirrors of lengths are point reflections: about r^* it maps $3\Delta s$ onto ℓ_0, about $\bar{\lambda}$ the depth onto $\bar{\lambda}/\alpha_G$ (−3, 0). Every coupling is an arrow between two points, $(M/m_{Pl})^{\Delta k} \cdot \varepsilon_L^{\,-\Delta r}$: α_G (2, 0), Tully-Fisher (1, −1) and α_I (−1, −1), and the arrows of the triangle close. Two mirrors in succession stretch every length by the square of the distance between their centres. The couplings $\alpha_G = r^{*2}/\tilde{r}^2$, $\alpha_I = \ell_P^2/r^{*2}$ and Tully-Fisher $= \ell_P^2/\tilde{r}^2$ are these squares. The stretch has its fixed points at the boundary, at zero and infinity: two meeting points make an open winding, the same group. The winding adds an axis, α^n at the point $\bar{\lambda}$: r_e, $\bar{\lambda}_e$ and a_B are n = 1, 0 and −1. The rotation adds the second, j at the point $3\Delta s$, $a_J = j \cdot 3\Delta s$: at the meeting point j = 1, $a_J = 3\Delta s$, for a particle with spin ½, $a_J = \bar{\lambda}/2$ and $j = 1/(2\alpha_G)$. Both axes are tangential, one at the quantum length, the other at the depth, and both meet the fold at m_{Pl}: winding at $Q_{el}/q_P = M/m_{Pl}$, rotation of spin ½ at $m_{Pl}/\sqrt{2}$.

Points and lines of the ladder. The arrow of two lengths, $(M/m_{Pl})^{\Delta k} \cdot \varepsilon_L^{\,-\Delta r}$, becomes one at $x = -\Delta r/\Delta k$: where two lengths meet depends only on the slope of their connection in the triangle. Points and lines thus exchange their roles. If points of the triangle lie on the line $r = y_0 - x_0 \cdot k$, their lines in the plane x, y pass through the point (x_0, y_0). Numbers of the form shift a meeting mass only by a factor of the form, and a decade of mass is 0.016 in x. The lines $\bar{\lambda}$, $3\Delta s$ and ℓ_0 themselves form a triangle there, with the corners (M_{min}, ℓ_0), (m_{Pl}, ℓ_P) and (M_U, ℓ_0). Each edge of the triangle of directions becomes one of these corners: the edge from $3\Delta s$ to ℓ_0 carries the lengths without ħ, the one from $\bar{\lambda}$ to ℓ_0 those without G, the one from $\bar{\lambda}$ to $3\Delta s$ those without a_0, for ħ appears only in $\bar{\lambda}$, G only in $3\Delta s$ and a_0 only in ℓ_0. Each corner thus lacks one constant, $M_U = c^4/(G \cdot a_0)$ without ħ, $M_{min} = \hbar a_0/c^3$ without G, $m_{Pl} = \sqrt{(\hbar c/G)}$ without a_0, and without ħ the quantities M, G, c and a_0 form only the one number $GMa_0/c^4 = M/M_U$. Corners and midpoints of the triangle are connected only along the three edges and the three medians, hence

the ladder has exactly five meeting masses: parallel to the edges, in turn, M_U, M_{min} and m_{Pl}, parallel to the median from $\bar{\lambda}$ to $r^* M_c$, to the one from $3\Delta s$ to $\bar{r}\, m_{Pl}^2/M_c$. The third, from ℓ_0 to ℓ_P, holds k fixed: lengths connected in this way run parallel and keep the same ratio for every mass, a power of ε_L.

What a line holds fixed. On the line $y = r + k \cdot x$ the length grows as M^k, and its steepness says which quantity stays fixed along it: $k = 1/n$ holds GM/R^n. Steepness 1 holds the spread $GM/(Rc^2)$, ½ the slope GM/R^2, ⅓ the tide GM/R^3 and with it the mean density, and in the deep regime ¼ holds the orbit, $\Omega^4 = GMa_0/R^4$. On 1 lie $3\Delta s$ and the circles $6\Delta s$ to $18\Delta s$, on ½ r^* and the Einstein ring at fixed D_E, on ⅓ the held body $R = \lambda_H \cdot N^{1/3}$, the smoothing length and R_{max} after Newton, on ¼ R_{max} under the rule. Steepness 0 holds a time, $\ell_0 = c \cdot \tau_D$. The mirror $k \to -k$ exchanges GM/R^n for $M \cdot R^n$: on -1 $\bar{\lambda}$ holds the action $M \cdot \bar{\lambda} \cdot c = \hbar$, on $-½$ $\bar{r}$ holds the time of the body's own quantum motion, $M \cdot \bar{r}^2/\hbar = \tau_D$. Parallel lines read the same quantity at a different value. A fixed ratio Z of two lines lies at the mass of their meeting point times $Z^{1/(k_1 - k_2)}$, with the exponent 2 between spread and slope, 3/2 between spread and tide, 6 between slope and tide and 4 between slope and the orbit of the deep regime. Therefore 3/2 stands wherever a held body reaches a fixed spread: in $m_p \cdot (\alpha/\alpha_G)^{3/2}$ at $\delta = \alpha^2 \cdot m_e/m_p$, in M_V and in $M_* = m_p \cdot \alpha_G^{-3/2}$ at $\delta = \bar{\lambda}_p/\lambda_H$. This line and the held body shift equally with λ_H, so their meeting point lies at M_* for every holder. From M_V, by contrast, the mass of the star drops out, because its holder holds the density fixed on steepness ⅓: with $R = \lambda_H \cdot N^{1/3}$, $M_V = \sqrt{5} \cdot m_p \cdot (\lambda_H/(12\Delta s_p))^{3/2}$, on the circular orbit $2 \cdot m_p \cdot (\lambda_H/(18\Delta s_p))^{3/2}$, and the fold reads only its holder.

15 The Ladder of Steps

The function of the rule itself forms a ladder. Each step is the previous one times s:

Step d	$s^d \cdot n_B(s)$	Meaning	Integrated over all s
0	n_B	sum of returns	(grows without bound)
1	$\chi = s/(e^s - 1)$	switch from Newton (0) to the deep regime (1)	$\pi^2/6$
2	$s^2 \cdot n_B$	additional acceleration, peak at $a_N = 2.54 \cdot a_0$	$2\zeta(3)$
3	$s^3 \cdot n_B$	structure action, vacuum part	$\pi^4/15$

At r^* ($s = 1$) all steps are equal. Their peaks are Wien's displacement constants in d dimensions, $d(1 - e^{-s}) = s$. The deep regime is the Rayleigh-Jeans limit, the Newtonian regime the Wien tail: the rule has the same shape as black-body radiation, because both count returns within a period.

Whole and place. The same function has two readings. Integrated over all s, for the whole, each step gives $\Gamma(k)\zeta(k)$ with $k = d + 1$, $\Gamma(k) = (k - 1)!$ the gamma function (here not the loss Γ) and ζ Riemann's zeta function, $\zeta(k) = \Sigma\, 1/n^k$. At the place, expanded near the deep regime, it has the coefficients $\zeta(0) = -½$, $\zeta(-1) = -1/12$, $\zeta(-3) = 1/120$:

$$n_B(s) = 1/s + \Sigma\, \zeta(-n) \cdot (-s)^n/n! \qquad \zeta(1 - k) = 2 \cdot (2\pi)^{-k} \cdot \cos(\pi k/2) \cdot \Gamma(k)\zeta(k)$$

Riemann's mirror connects the two. Thus the constant of the vacuum part, $\Gamma(4)\zeta(4) = \pi^4/15$, is the mirror image of $\zeta(-3) = 1/120$, which the observer at the place reads as the correction $-(r^*/r)^3/720$, and $\pi^2/6$ is the mirror image of $-1/12$. What carries the whole and what appears at the place as a correction are the same quantity seen from two sides.

The half turn. With the half turn of the winding, $q \to -q$, every step of the ladder $s^d \cdot n(s)$ is weighted with $1 - 2^{-d}$: ½, ¾ and ⅞ for d = 1, 2, 3. The step d = 0, the sum of all returns, stays finite, $\int n_F\, ds = \ln 2 = 0.693$: one bit, the two senses of a winding. For the fold it grows without limit, that is the deep regime. The returns of a winding add up to at most ½: even with return the winding would have no deep regime. Counting all multiples, windings that block each other count $\prod(1 + q^n)$, deep $\ln Z = \pi^2/(12s) - ½ \cdot \ln 2 + s/24$ against $\pi^2/(6s) + ½ \cdot \ln(s/2\pi) - s/24$ for the fold, both up to terms in $e^{-2\pi^2/s}$ and $e^{-4\pi^2/s}$. The half turn halves the leading part and reverses the zero point, $+s/24$ against $-s/24$: as charges cancel and masses add, the zero points of winding and fold cancel. The mirror maps the blocking onto the odd modes, $\prod(1 + q^n) = (q \cdot \tilde{q})^{-1/24} \cdot \prod(1 - \tilde{q}^{2n-1})/\sqrt{2}$ with $\tilde{q} = e^{-2\pi^2/s}$. The mirror point of the winding lies at $s = \pi\sqrt{2}$, that of the fold at 2π. Windings that block each other occupy each mode at most once, $\prod(1 + q^n)$. Returns that come only in odd modes count $\prod 1/(1 - q^{2n-1})$. Both are equal (Euler, at $q = ½$ both 2.384): the blocking of the half turn is a full turn on the doubled path that changes sign after one turn.

16 The Return as a Mirror Image

If one reads the time of the return as imaginary time, the return equation becomes the Schrödinger equation in the inner direction ϖ, with $D = \hbar/(2\tilde{m})$ and the slope as the potential, $V = \hbar\Gamma = \hbar a_N/c$.

Derivation. From $D = GM/c$ and $D = \hbar/(2\tilde{m})$ follows the **mirror mass**

$$\tilde{m} = \hbar c/(2GM) = m_{Pl}^2/(2M) \qquad \bar{\lambda}(\tilde{m}) = \hbar/(\tilde{m}c) = 2GM/c^2 = r_0$$

The funnel M returns like the quantum motion of its mirror mass, and the quantum length of that mass is its limit radius. The return fraction $q = e^{-s}$ is the tunnelling factor of this mirror image through the slope, the same form as $e^{-\sqrt{(E_G/E)}}$ when two nuclei of energy E pass through their Coulomb barrier (E_G the Gamow energy). This is why the period of the return is a temperature, $kT_I = \hbar/\tau_D = \hbar a_0/c$, and why the rule has the Bose form: heat is periodicity in imaginary time. The geometric distribution of the number of returns has, at fixed mean, the largest entropy, $H = \ln \nu + \chi$. The threshold itself compares two such quantities: the Unruh temperature T_U of the observer held in place against the slope, with the period of the inner direction, $s^2 = 2\pi \cdot T_U/T_I$. This is a period, not heat at the place. Likewise the loss from expansion is a temperature against the same period: an expansion at the rate H gives $H \cdot \tau_D = 2\pi \cdot T_H/T_I = 2\pi/\Theta$, with the temperature of the horizon $kT_H = \hbar H/(2\pi)$. The fluctuation of the number of returns is $\langle \Delta n^2 \rangle = n_B \cdot \nu = n_B + n_B^2 = 1/(4 \cdot \sinh^2(s/2))$, Einstein's form for the fluctuation of radiation: n_B counts the returns singly, like independent events, n_B^2 their bunching. The enhancement is the ratio of fluctuation to mean, $\nu = \langle \Delta n^2 \rangle / n_B$, and Newton ($\nu = 1$) is the limit of independent returns. The fluctuation is the response of the rule to the depth, $\langle \Delta n^2 \rangle = -d\nu/ds = -d(\frac{1}{2} \cdot \coth(s/2))/ds$: only the oscillator part responds, the fixed half does not. Hence $d \ln \nu / d \ln s = -s \cdot \langle \Delta n^2 \rangle / \nu = -\chi$: the switch that sets the slope of the relation is the fluctuation of the returns. It belongs to the distribution in the inner direction, at the place only its mean ν acts. Both parts are equal where half returns, $q = \frac{1}{2}$, at $a_N = (\ln 2)^2 \cdot a_0 = 0.48 \cdot a_0$. There the stimulated part carries as much as the spontaneous one, the return as much as Newton, $\nu = 2$. At larger a_N the single returns dominate, at smaller a_N the bunching. With the blocking of the half turn, each mode occupied at most once, it is $n_F \cdot (1 - n_F) = -dn_F/ds \leq \frac{1}{4}$: the winding fluctuates within a bound, the fold without one. Gravity and the quantum are two views of the same mirror, M in the spatial direction, $\tilde{m} = m_{Pl}^2/(2M)$ in the inner one.

The meeting point of the ladder of steps and the mirror point of the loops also count the same loop: with $s = -\ln|q| = 2\pi \cdot \mathrm{Im}\ \tau$, the meeting point $s = 1$ lies at one radian, the mirror point $s = 2\pi$ at the full turn. The threshold counts per radian, the mirror per turn.

Two nomes. The binding runs only through the magnitude of the nome. If the phase entered, the enhancement would fall below 1 outside. The funnel has a second nome, the core nome $|q_{\Delta s}| = \Delta s/r$, and the cusp form $\Delta = q_{\Delta s} \cdot \prod(1 - q_{\Delta s}^n)^{24}$ vanishes for $q_{\Delta s} \to 0$. The author leaves open which nome carries the imaginary part, $|q| = e^{-r^*/r}$ or the core nome $|q_{\Delta s}| = \Delta s/r$. With the first, $\mathrm{Im}\ \tau = r^*/(2\pi r)$ is the depth of the return against the circumference at the place, the modulus of the torus of turn and inner direction. With the second, $|q_{\Delta s}|$ is a third of the depth $\delta = 3\Delta s/r$ of the funnel, $\mathrm{Im}\ \tau = \ln(r/\Delta s)/2\pi$, and the addition is $\chi \cdot \Delta(q_{\Delta s}) \cdot 3c^2/r^*$ with the cusp form $\Delta(q_{\Delta s}) = q_{\Delta s} \cdot \prod(1 - q_{\Delta s}^n)^{24}$. At the place both are linked one to one, $(r^*/r)^2 = 3 \cdot (\Delta s/r) \cdot \ell_0/r$, and the switch of the second carries the nome of the first, $\chi/s = \Sigma_{n \geq 1}\ e^{-ns}$. Both thus give the same rule up to the factor $\prod(1 - q_{\Delta s}^n)^{24} \approx 1 - 8\delta$ in front of the addition. On the rule it acts as $1 - 8(v/c)^2 \cdot q(1 - q)$, with the speed v of the circular orbit and the nome q of the place or, in clusters, of the whole, so by at most $2(v/c)^2$. In SPARC (2688 points), in the lensing around isolated galaxies (KiDS-1000), in HIFLUGCS and X-COP this is at most $2 \cdot 10^{-5}$ dex, $\Delta\chi^2 = 0.01$ in SPARC and 0.06 in HIFLUGCS, with the whole mass of the field as fold $1 \cdot 10^{-4}$ dex: on the data the two cannot be told apart. They are separated in the form. With $e^{-r^*/r}$ the threshold lies at $\mathrm{Im}\ \tau = 1/2\pi$ for every mass, the whole has the nome $Q/(1 + Q)$, and the half turn $q \to -q$ gives the Fermi form of the winding without a deep regime. With $\Delta s/r$, $|q_{\Delta s}|$ stays below 1/6 outside $r_0 = 6\Delta s$ and never reaches the nome of the whole, the threshold lies, depending on the mass, at $\mathrm{Im}\ \tau = \ln(3c^2/v_{flat}^2)/2\pi$ (Sun 4.5, Milky Way 2.6), and the half turn only reverses the sign of the addition, $\nu \approx 1 - n_B$: the winding would have a deep regime with reversed sign, through zero at $s = \ln 2$.

Part IV: The Levels

The same formulas hold on every level. Only mass and radius change. Every level receives its position from its neighbours: the end of the nucleus from the coupling α of the shell, the window of matter from the kind of bond of the atom, the spread of the star from what holds it, the burning stages from the iron of the nucleus, the smallest gas-bearing galaxy from the heat of its gas, the position of the galaxy relative to the threshold from the whole.

17 Particle, Coupling, Charge

An elementary particle is an oscillating state of the medium. Next to it, space remains practically flat. Gravity and electromagnetism have the same form, a **length of action divided by the quantum length**:

$$\alpha_G = 3\Delta s_p/\bar{\lambda}_p = Gm_p^2/(\hbar c) = 5.906\cdot10^{-39} \qquad \alpha = r_e/\bar{\lambda}_e = 1/137.036$$

Two ways of imprinting space at a node must be distinguished. The **fold** is a magnitude: it changes the clock rate and has one sense of circulation, that of loss. The **winding** is a phase: it does not change the clock rate, is lossless and counts in both senses of circulation. Mass is fold, **charge is winding**: the phase of a node has a whole-number winding n around its circulation, the charge is $Q_{el} = n\cdot e$ with e the elementary charge, and the sign is the handedness. On the torus, n is a topological number (the first Chern number). It changes only when the amplitude passes through zero. This is why Coulomb's law remains $1/r^2$ in every field, and why there is no electromagnetic threshold effect: the argument of the rule of return is the clock-rate gradient, never an electric force. In the state space the Lorentz force acts tangentially: on a moving charge its magnetic part does no work, it conserves energy and entropy and produces circulations and phases. Joule heat J·E is the channel into the transverse role. By Onsager and Casimir the transport splits into a symmetric part (conductivity, heat conduction, Seebeck, Peltier: transverse) and an antisymmetric one (Hall, Nernst: tangential). The magnetic field decides which share circulates and which is lost.

Twist and three rates. Fold and winding sit at the same node. The twist $a_J = J/(Mc)$ is an imaginary displacement of this place. Both multipole series are $M\cdot(i\cdot a_J)^l$ and $Q_{el}\cdot(i\cdot a_J)^l$ (i the imaginary unit, l the order), hence $J = Mc\cdot a_J$ and $\mu = Q_{el}\cdot c\cdot a_J$: from this follows, for every mass, charge and rotation, the same ratio of magnetic moment to angular momentum, g = 2. For the electron this is the Bohr magneton $\mu_B = e\hbar/(2m_e) = 9.274\cdot10^{-24}$ J/T. For spin ½, $a_J = \bar{\lambda}/2$: the moment is charge times c times half the quantum length, $\mu = e\cdot c\cdot\bar{\lambda}/2 = e\hbar/(2M)$. For the proton it is $\mu_N = \mu_B\cdot m_e/m_p$, the ratio $m_p/m_e = 1836.2$ is content, not form. Every mass has three rates: the fold rate $c/(3\Delta s)$, the Compton rate $c/\bar{\lambda}$ and the folding rate $\alpha_G\cdot c/\bar{\lambda}$. The Compton rate is their geometric mean, once more a mirror. Below the Planck mass the fold rate is the fastest, above it the reverse, at the Planck mass all three are equal. The rate of the inner direction, $1/\tau_D = a_0/c$, is the same for every mass and meets the Compton rate at M_{min}, the fold rate at M_U and the folding rate at M_c. The length of the folding rate is the depth mirrored about $\bar{\lambda}$, $\bar{\lambda}^2/(3\Delta s)$, like the hull of the winding $a_B = \bar{\lambda}_e^2/r_e$.

Return of the winding. In the field B the twist turns at the rate eB/M, for g = 2 exactly as fast as the orbit goes round, the orbit as a whole only half as fast. The difference the measurement reads is made by the return of the winding onto itself: counted per radian, with the weight α, once back, $\mu/\mu_B = 1 + \alpha/(2\pi) = 1.001161$, measured 1.001160 (CODATA 2018). Independently, the hydrogen line, winding times twist at the place of the nucleus, reads the first return as $0.88\cdot\alpha/2\pi$: without return 1418.95 MHz, with the first 1420.60 MHz, measured 1420.41 MHz (Hellwig et al. 1970), with $g_p = 5.586$ as content (CODATA 2018). When the rate eB/M reaches the Compton rate, the field lies at the meeting point of the winding, $B = M^2c^2/(e\hbar) = 4.41\cdot10^9$ T, $E = B\cdot c = 1.32\cdot10^{18}$ V/m (Schwinger field): there pairs arise from the medium.

Matter in the field. The three roles give three responses. Tangentially every orbit turns against the field without loss, the diamagnetism of every substance (for the hydrogen atom with $\langle r^2\rangle = 3a_B^2$, per mole $-\mu_0\cdot N_A\cdot e^2\cdot a_B^2/(2m_e) = -2.99\cdot10^{-11}$ m³/mol, that is $-2.38\cdot10^{-6}$ cm³/mol). Open windings align against the loss kT, paramagnetism, with two senses of circulation in the Fermi form $m = \tanh(\mu B/kT)$, small the Curie law. If the windings of neighbouring atoms lock in the same sense, the field of the others is a feedback with $q = T_C/T$: the susceptibility is $(C/T)\cdot\Sigma q^n = C/(T - T_C)$, the tipping point q = 1 is the Curie temperature, ferromagnetism. Locking in opposite senses is q < 0, the antiferromagnet. Fully locked is the superconductor: two locked windings obey $\dot{\psi} = \Delta\omega - K\cdot\sin\psi$, the equation of the locking of two cycles (Josephson), the phase of the pair closes per circuit with $\Phi_0 = h/(2e) = 2.068\cdot10^{-15}$ Wb, and the transverse channel is closed: no loss, no resistance, the field penetrates only to the closing length λ_L (Meissner).

Twist of the fold. Because the fold carries the same multipole series as the winding, a rotating funnel turns along around itself: the plane of an orbit turns with $\Omega = 2GJ/(c^2r^3) = 6\Delta s \cdot c \cdot a_J/r^3$, the axis of a gyroscope on a polar orbit on average with $GJ/(2c^2r^3) = 3\Delta s \cdot c \cdot a_J/(2r^3)$, a quarter of that. That is the magnetism of the fold: twist without charge. Conversely the twist of the fold carries no magnetic moment, $\mu = Q_{el} \cdot c \cdot a_J$ is zero without charge. The field of a planet or star is therefore winding, not fold: locked currents (the dynamo as the locking of the flows in the coupling window), no form measure of the rotation. A relation $\mu \propto J$ through $\sqrt{G}$ (Blackett) would mean $Q_{el}/M = \sqrt{G}$, the meeting point of charge and mass. The Earth lies far below. In the triad the rotation of the galaxy is its vortex. Its magnetic field is winding of the gas, not twist of the fold, and belongs to the state, not to the form. The magnetism contributes nothing to the restoring of the whole, because $\Sigma Q_{el} = 0$.

18 The Nucleus: a Held Funnel

A **held funnel** is a body whose building blocks are held at the distance of a **holding length** λ_H. Its envelope is then $R = \lambda_H \cdot N^{1/3}$ for N building blocks. The atomic nucleus is such a funnel, held by its nucleons themselves.

Derivation of the holding length from nuclear masses. Through the mutual repulsion of its protons, the binding of a nucleus with Z protons and mass number A loses the Coulomb energy $a_C \cdot Z(Z-1)/A^{1/3}$ with $a_C = (3/5) \cdot \alpha\hbar c/\lambda_H$. Fitting the liquid-drop formula to all nuclei of AME2020 gives $\lambda_H = 1.25$ fm. Independently of that, **mirror nuclei**, in which Z and N are exchanged, differ only in winding. Their difference in binding is pure Coulomb energy and reads the holder as $\lambda_H = 1.33$ fm (36 pairs, AME2020).

The nucleons move inside the nucleus with $v/c \approx \bar{\lambda}_p/\lambda_H = 0.17$. Half the spread of the binding of ^{62}Ni independently gives 0.14. Size and binding read the same speed up to a factor of order one. Nuclear transitions run by themselves towards the smallest fold per nucleon, at ^{56}Fe. The largest binding per nucleon belongs to ^{62}Ni (AME2020).

The tipping point. The fissility $x = E_C/(2E_S)$ compares the Coulomb energy E_C of the protons (winding) with the surface energy E_S of the holding. It is the gain of a feedback that tips over at $x = 1$: the restoring force of a deformation is $\propto 1 - x$. The binding per nucleon is largest at $x = ¼$ (fitted to the measured nuclei of AME2020: $x = 0.25$). The tipping point is the meeting point of the nucleus. The heaviest known nuclei lie there ($Z = 118$, $A = 294$: $x = 0.98$), while the innermost electron of the same atoms is only at $Z\alpha = 0.86$: matter ends at the meeting point of the nucleus before the shell reaches its own. Because x grows with α, the coupling of the shell sets the end of the nucleus.

The limit of the shell. The heaviest known nuclei are poor in neutrons. The shell holds up to $Z = 170$: $Z\alpha = 1$ ($Z = 137$) holds for a point-like nucleus. Around the nucleus with $R = \lambda_H \cdot A^{1/3}$ the binding of the innermost electron reaches twice the fold energy of the electron, $6c^4 \cdot \Delta s_e/G$, only there (Dirac equation).

The shells. The model's closure numbers give the liquid-drop formula a shell term, and it follows from the funnel itself. The levels of the held funnel with twist ($\kappa = 0.12$, $\mu = 0.03$) close at 2, 8, 20, 28, 50, 82 and bind against their averaged filling with the shell spacing $\hbar\omega = (5/6) \cdot (3/2)^{4/3} \cdot \hbar^2/(m_N \cdot \lambda_H^2) \cdot A^{-1/3} = 38.0\ \text{MeV} \cdot A^{-1/3}$. AME2020 reads independently 39.3, hence $\lambda_H = 1.23$ fm. Only the binding part counts, for where the shells do not bind, the nucleus deforms. With no free number in the shell term the binding scatters by 1.83 MeV, the deformed nuclei by 1.26 instead of 1.93 MeV.

19 The Atom, Closing and Opening

An atom is a small, fast-circulating vortex: tiny fold, large steepness. The shell of hydrogen circulates with $v/c = \alpha$ on the Bohr radius $a_B = 0.529 \cdot 10^{-10}$ m, at $6.58 \cdot 10^{15}$ Hz. The atom too has its three radii: the threshold radius of the proton lies at $r^* = 32$ fm, between nucleus and shell ($a_B = 1631 \cdot r^*$). As a compact funnel, its interior remains Newtonian. The binding of the atom comes from the geometry of its shell, not from the depth of its fold.

Closing forms. The pure $1/r^2$ slope closes every orbit, and orbits of equal energy bind equally, whatever their shape. This has a reason in form: the momenta of a Kepler orbit lie on a circle, and mapped onto the sphere S^3 every orbit is a great circle. In the space of motions, not in position space, the funnel is a closed sphere. The n^2 states of a shell (n its number) are the vibrations of this sphere, $2n^2$ with both senses of circulation. Only two slopes close every orbit (Bertrand's theorem): the $1/r^2$ slope outside a funnel and the linear one inside a uniformly held body. Their shells close after

funnel ($1/r^2$):	2, 10, 28, 60	$2n^2$ per shell
held body (r^1):	2, 8, 20, 40, 70, 112	$O(k) = (k+1)(k+2)(k+3)/3$

Opening couplings. Every deviation from the pure form opens the closure, on every level: in the solar system the fold (3δ per revolution), in the galaxy the rule and the extended mass (rosettes, in the deep regime 254.6° lie between the pericentres), in the atom the inner shells that spread out the centre, in the nucleus the twist. Shell and nucleus open in mirrored fashion:

- In the shell, the orbits with the **smallest** orbital winding l bind more strongly, because they dive deep. Winding here means the number of nodes of the oscillation around the centre, not the winding of the charge (Section 17). The shells fill by n + l and close after 2, 10, 18, 36, 54, 86 instead of after 28 and 60.
- In the nucleus, the twist pulls down, in every shell, the state with the **largest** orbital winding, j = l + ½ (j here the angular momentum of the nucleon, not the spin number), from the next shell. The nucleus then closes after O(k) + 2(k + 2), with k the number of the shell, that is after 28, 50, 82, 126 (k = 2 to 5). The next closure would be 184.

The orbit in a galaxy has a torus too. Its modulus τ_M carries in its real part the ratio of radial to azimuthal frequency, in its imaginary part the binding. For the point mass $(\mathrm{Re}\,\tau_M)^2 = 1 + \chi$: $\chi = 0$ gives the closed Kepler orbits, $\chi = 1$ rosettes. The rule lifts the Kepler degeneracy $\kappa = \Omega$ (Ω the frequency of the orbit, κ that of the radial oscillation about the circular orbit) and replaces it by local resonances at rational κ/Ω.

In the data. The six strongest drops of the ionisation energy lie after Z = 2, 10, 18, 36, 54, 86 (factor 4.6 at He down to 2.8 at Rn, PubChem), the six strongest jumps of the two-nucleon binding at N = 16, 20, 28, 50, 82, 126 and Z = 14, 20, 28, 38, 50, 82 (AME2020). The series meets 20, 28, 50, 82 and 126. The jumps at N = 16 and Z = 14, 38 lie at subshells within the series, and 8 does not show in the two-nucleon binding. The closure numbers of both levels thus follow from one closing form each and one coupling that opens it.

The ladder of periods. With the noble gas Og the seventh period closes at Z = 118 (PubChem). The periods are 2, 8, 8, 18, 18, 32, 32 long. The length of period p is the number of states with n + l = p + 1, that is $2k^2$ with k the rounded-up half of p + 1: every step $2k^2$ of the $1/r^2$ form comes twice, because n + l = 2k − 1 and 2k have the same largest winding l = k − 1. The ladder is itself a closing form: a field that is mirrored in log r about a radius R ($r \leftrightarrow R^2/r$) binds at the threshold E = 0 exactly at the couplings N(N + 1) with N = n + l, each step with $2k^2$ states, k the rounded-up half of N. Its states at the threshold are spherical harmonics, $u = \sqrt{r}\cdot P_N^{2l+1}((r - R)/(r + R))$ with P the associated Legendre function: the step n + l is the degree on a sphere whose poles are nucleus and infinity and whose equator is R. The winding l gives the order 2l + 1, and the mirror $r \leftrightarrow R^2/r$ is the reflection at the equator. In log r this field is $v\cdot f(1 - f)$ with $f = 1/(e^x + 1)$, $x = \ln(r/R)$: the same function as the rule, only in the other counting, the winding in both senses of circulation, $f(x) + f(-x) = 1$, instead of the magnitude in one, $\nu(s) + \nu(-s) = 1$. The shell is a funnel held by the counting rule of the electrons, like the white dwarf, with $R \propto Z^{-1/3}$ (Thomas-Fermi): its field follows the mirror form inside to 0.6 % and opens outside. Its height $v = 3.44\cdot Z^{2/3}$ lies from Ne to Og in the window $N(N + 1) < v < (N + 1)(N + 2)$ with N = p + 1, the field binds exactly the occupied steps.

20 Matter: the Coupling Window

Two coupled cycles with phase difference ψ obey $\dot\psi = \Delta\omega - 2k\cdot\cos\varphi\cdot\sin\psi$, with Δω the difference of their eigenfrequencies, the coupling k (here not the Boltzmann constant) and its phase φ. They lock in when $|\Delta\omega| \le 2k|\cos\varphi|$, and their torus then contracts onto a circle. Only the lossy part $k\cdot\cos\varphi$ of the coupling locks, the rest merely shifts the frequencies. Too weak a coupling lets the cycles drift apart. In general, for cycles θ and slow structure Z: $\dot\theta = \omega(Z) + K(\theta, Z)$, $\dot Z = G(\theta, Z) - D(Z)\cdot Z$, with eigenfrequencies ω, cross-coupling K, feed G and loss D. A common medium with pair coupling k carries at most $N \approx 1 + 2W/(\pi k)$ free cycles in a frequency band W. Above that they lock in. If a feedback grows to q = 1, Σq^n diverges at this tipping point. Beyond it the hysteresis window opens with the distance from the tipping point to the power 3/2, and before each edge of the window the return rate 1 − q falls with the square root of the distance to the edge (critical slowing down). From T^2 to T^1 lead three ways: locking (both cycles go on oscillating, bound), kinking, and the loss of a cycle. The number of independent cycles is the number of neutral Lyapunov exponents, transverse to the torus they are negative. Stability is motion without an obligation to repeat: the feedback must take hold before a perturbation builds up. Bifurcations are changes of role: the birth of a cycle makes a transverse direction tangential, locking makes a tangential one transverse. Hysteresis needs no store, the state is only the sequence of the extremes not yet erased. Noise and loss come from the same damping. In between lies the **coupling window** $k_{min} < k < k_{lock}$. A substance is the coupling window of its atoms: solid is locked (the lattice as identification), liquid is inside the window, gaseous is frayed. Melting gives up the identification of the lattice, not the cohesion.

The width of the window depends on the level below: across 93 elements, T_{boil}/T_{melt} ranges from 0.81 (As, which sublimes, so the window is closed) to 8.2 (Ga), and the more firmly the atom binds, the narrower the window (rank correlation with the ionisation energy −0.48, PubChem). The trend runs between kinds of bond (noble gases and halogens narrow, metals wide), not within a group.

Docking. The same window carries the docking of a drug to its receptor. At the molecule the winding carries, not the fold, and drug and receptor are not things that meet but nodes of the winding that close a common loop. For the drug the body is a closed space whose only outside is excretion. The circulation brings it back with the fraction $q = 1 - E$ per pass, E the fraction that liver and kidney take per pass, and the passes add up to the Bose form of the rule, $\nu = 1/E$. The threshold lies, as at r*, where loss time and circulation time are equal, at $E = 1 - 1/e$, and the clearance of the whole sets the threshold of every binding site, $\ell_* = \sqrt{(D/\Gamma)}$. On every level stands the triad: at the site contact, mouth of the pocket and threshold of return, in the field of receptors bound, near and far, in the body binding site, circulation and excretion.

A site carries at most one partner. With $q = [L]/K_d$ it reads the return with two senses of rotation, $\theta = q/(1 + q)$, the Fermi form. A drug that attaches to itself reads the same return with one sense, $q/(1 - q)$, and tips into stacks at $q = 1$. Rebinding follows the one return equation, and the dimension of the return sets its form: in a channel the residence time grows as $1/\sqrt{\Gamma}$, on a surface only logarithmically, in free solution it stays finite. Uptake and release are mirror images in time, so the field of receptors stretches both rates equally. It sets the clock of binding, not its depth K_d. Whoever reads the release in tissue and the uptake in the test tube reads a time of the inner direction as a depth at the place.

Bound, the form is closed and gives nothing off, it gives off only at a change. The cell reads the changes $S = k_{off}\cdot\theta$ with its own clock k_Z in the Fermi form, and from this follows the effect $W = \tau\cdot[L]/(K_d + (1 + \tau)\cdot[L])$ with $\tau = k_{off}/k_Z$: efficacy is the ratio of two clocks. Because the free receptor changes by itself with the basal clock S_0, all modes of action stand on one ladder, mirrored about S_0. Faster than S_0 a drug acts, equal to S_0 it only blocks, slower it lowers the effect below the basal level. The receptor answers the drug as a substance answers a field, in the three roles. The reactive part of the coupling only shifts frequencies, this is allostery: it strengthens only detuned drugs, and mirrored detunings answer with opposite sign. Occupancy sets itself against kT. Coupled sites lock in, with the steepness $n_H = T/(T - T_K)$ after Curie and Weiss and with hysteresis below T_K. A tissue of many such groups keeps only the extremes of the doses that no later dose has exceeded. The control loop of part and whole is one feedback in two readings: under the dose the effect $1/(1 - q)$ remains, after withdrawal $q/(1 - q)$ runs on. With negative feedback, habituation and rebound add up to the full effect like occupancy and vacancy, with positive feedback the effect grows towards the tipping point and decays ever more slowly. Repeated dosing is the return in time with the fixed half of the rule, and dosing and the cycles of the body lock in rational ratios, with windows that shrink with their order m as K^m.

21 Stars

Jeans and Debye. In the Newtonian range, every disturbance of a gas over the length ℓ (wavelength divided by 2π) acts back on itself, for each kind of carrier with mass m and charge e, density n and temperature T. A compression of the mass draws in through the fold, $q = (\ell/\ell_J)^2$, a separation of the charges pushes back through the winding, $q = -(\ell/\ell_D)^2$, with $\ell_J^2 = kT/(4\pi G\cdot n\cdot m^2)$ and $\ell_D^2 = \varepsilon_0\cdot kT/(n\cdot e^2)$. The response is $\Sigma q^n = 1/(1 - q)$. The winding damps and screens beyond the Debye length. The fold builds up at $q = 1$, beyond the Jeans length heat no longer holds the gas. At both lengths the binding of a carrier to its surroundings reaches the heat, $4\pi G\cdot m^2\cdot n\cdot\ell_J^2 = kT = (e^2/\varepsilon_0)\cdot n\cdot\ell_D^2$. The ratio of the two lengths is therefore set by the couplings alone, $\ell_J/\ell_D = \sqrt{(\alpha/\alpha_G)}$, at every density and temperature and without ħ, for the mass and charge of the proton $1.1\cdot10^{18}$. The Jeans sphere holds $N = (\ell_J/\ell_D)^3$ Debye spheres, $\ell_J = \ell_D\cdot N^{1/3}$ like the envelope $R = \lambda_H\cdot N^{1/3}$ of a held funnel: the fold of all N binds as strongly as the winding of one, $N^{2/3}\cdot\alpha_G = \alpha$, the same condition at which the fold of N nucleons reaches the winding. Because the windings cancel beyond the Debye sphere, each counts as one carrier, and the Jeans mass is $\rho\cdot\ell_J^3 = M\cdot n\cdot\ell_D^3$ with $M = m\cdot(\alpha/\alpha_G)^{3/2}$.

Mass scale. The fold of a star is fixed by the fold and the quantum length of the proton:

$$M_* = m_{Pl}^3/m_p^2 = 1.853\ M\odot \qquad \bar{\lambda}_p^2 = \ell_P\cdot 3\Delta s_*$$

The star is the mirror image of the Planck point at the quantum length of the proton: its fold stands to the quantum length of the proton as the latter stands to the Planck length.

Held stars. What holds the star sets its spread. If the $N = M/m_p$ nucleons are held at the distance of a holding length λ_H, then $R = \lambda_H \cdot N^{1/3}$, and on the mass scale M_* the spread is

$$\delta = \bar{\lambda}_p/\lambda_H$$

the ratio of the quantum length of the carrier of the fold to that of the holder (up to factors of order one). Three cases follow from this one form:

- If heat holds it ($\lambda_H = \hbar c/kT$), this is the central temperature of the star, $kT_c = f \cdot \delta \cdot \bar{m}c^2$ with $\bar{m}$ the mean mass of the particles and a factor f of order one.
- If the **counting rule** of the electrons holds it, the rule that every state holds only one winding per sense of circulation (the closure of the shells, Section 19), at the distance $\bar{\lambda}_e$, then $\delta = m_e/m_p$: the white dwarf. The same balance, the pressure of the counting rule against the slope, gives the limiting mass $M = \omega_3 \cdot (\sqrt{(3\pi)}/2) \cdot m_{Pl}^3/(2m_u)^2 = 1.456\ M\odot$ ($\omega_3 = 2.018$ the coefficient of the form of the funnel, two nucleons per electron, m_u the mass unit).
- If the counting rule of the nucleons themselves holds it, δ is of order one: centre and envelope almost coincide, the meeting point at which the holder reaches c.

The same form carries the next stages. The galaxy is held by the closure of the whole: at its threshold $\delta = 3\Delta s/r^* = r^*/\ell_0 = (v_{flat}/c)^2$, which is Tully-Fisher as a held funnel. At the meeting point of this holder, $r^* = \ell_0$, stands the whole itself, as the neutron star stands at the meeting point of the holder of stars. It holds itself, since its closure is the fold of its parts, $\Lambda = (1 + Q) \cdot \Sigma \Delta s_b$: therefore it lies, counted per radian with $1 + Q = 2\pi$, at the same place for any amount of baryons, $r^*/\Lambda = \sqrt{(6/(1 + Q))} = 0.977$, and the amount drops out.

Limit of the holder. How close centre and envelope come is set by the holder itself, for its pressure is energy and folds as well: $dP/dr = -(3c^2\Delta s_r/r^2) \cdot (\varepsilon + P)/c^2 \cdot (1 + 4\pi r^3 P/(M_r \cdot c^2))/(1 - 6\Delta s_r/r)$, with the fold Δs_r of the mass M_r within r and the energy ε per volume. Whether the holder reaches c first or the depth counts first sets the limiting mass. If the counting rule of the electrons holds and the nucleons carry the fold, $2m_u$ per electron, the holder comes first: the folding along lowers the limiting mass only to 1.417 M⊙, when the holder already runs at 0.999·c in the centre. If the counting rule of the nucleons bears pressure and fold at once, both come together and the depth cuts first: the limiting mass is $0.384 \cdot m_{Pl}^3/m_n^2 = 0.710\ M\odot$ with R = 9.16 km and $3\Delta s/R = 0.114$, the holder runs at 0.64·c in the centre. By Newton, without the folding along, it would grow to $\omega_3 \cdot (\sqrt{(3\pi)}/2) \cdot m_{Pl}^3/m_n^2 = 5.73\ M\odot$, the folding along cuts it by a factor of 8. If one carrier bears pressure and fold alone, 0.384 is a pure form number, the same for every mass of the carrier: the form sets the number, the mass of the carrier is content. What the nucleons alone hold, holds above 0.710 M⊙ only with more than their blocking. Read from outside, every energy carries the clock rate of its place, like the light that arrives from there, and in equilibrium it is, read this way, the same throughout the body, at uniform density $(\rho c^2 + P) \cdot d\tau/dt$. The pressure at the centre in rest energy is therefore the ratio of the clock rates, $P/(\rho c^2) = (d\tau/dt|_R)/(d\tau/dt|_0) - 1$, weakly their difference $\frac{1}{2} \cdot \delta$, half the depth: an inner difference of clock rates in rest energy like the core temperature. The clock at the centre then runs at $(3/2) \cdot \sqrt{(1 - r_0/R)} - \frac{1}{2}$. At the same R it runs faster in no body at rest whose holder carries equally in every direction and whose density does not rise outward, in the neutron star of 1.4 M⊙ and 12 km at most at 0.71. If it stands still, no finite pressure carries any more. Hence the envelope of every such body runs faster than $\frac{1}{3}$, its light arrives far away reddened at most to z = 2, and it stays outside $(9/8) \cdot r_0 = 27\Delta s/4$, so $3\Delta s/R$ stays below 4/9 against $\frac{1}{2}$ at the fold, at 1.4 M⊙ above 4.65 km (Buchdahl). Between there and r_0 no held funnel is at rest. The whole lies at its threshold with $3\Delta s/r^* = \frac{1}{2} \cdot \sqrt{(6/(1 + Q))} = 0.49$ between 4/9 and $\frac{1}{2}$: it is not at rest and holds itself by its closure, not by pressure.

The cap. If a uniformly held body rests, its interior is the cap of a sphere S^3 of radius A, $A^2 = 3c^2/(8\pi G\rho) = R^3/r_0$. The cap reads only the energy per volume ρc^2, the pressure folds along only in the clock rate. A measures the curvature of the interior, not a distance in space: for the Earth 1.14 AU. The shell lies at the angle ψ_R from the pole, the centre, $R = A \cdot \sin \psi_R$, and twice the spread is $\sin^2\psi_R = 2\delta$. Along the path l outwards the circumference $2\pi r$ grows as $dr/dl = \cos \psi$, at the shell this is the clock rate, and outside, in the field GM/r^2 of every untwisted funnel, $d\tau/dt = dr/dl$ holds. Inside, the clock runs at $(3/2) \cdot \cos \psi_R - \frac{1}{2} \cdot \cos \psi$, from the growth of the circumference at the shell and at the place. At the centre it stops at $\cos \psi_R = \frac{1}{3}$: the limit of the holder lies 70.5° from the pole. A neutron star of 1.4 M⊙ and 12 km, read as uniform, reaches 36° from the pole. The arc from the pole to the shell, $A \cdot \psi_R$, is longer than R, weakly by $R^3/(6A^2) = \Delta s$: the fold depth is the excess of the arc over the sine. At 90° the shell would lie on the equator, $R = A = r_0$, at the horizon of the fold: there the circumference no longer grows, and the clock stops. With the restoring, $c^2/A^2 = (8\pi G/3) \cdot \rho + H_\infty^2$, and at the shell $\cos \psi_R$ remains the clock rate of the end state. Without

mass the part of the end state at rest around every place is thus the half sphere of radius c/H_∞, its equator the horizon of the whole. A bound region whose field stays GM/r^2 has the same cap at the moment of its turnaround, for then it rests: the holder holds a body in the form that a region has only at its turnaround.

Holder at c. If the holder runs at c, its pressure is a fixed multiple of the third $p = u/3$, $P = b\cdot\hbar c\cdot(\rho/m)^{4/3}$ with the mass m per particle and a coefficient b of the way of counting. The funnel then has the form $n = 3$, and its mass does not depend on the radius, $M = (4\omega_3/\sqrt{\pi})\cdot b^{3/2}\cdot m_{Pl}{}^3/m^2$. The counting rule of the electrons counts according to Fermi, $b = (3\pi^2)^{1/3}/4$, and with $m = \mu_e\cdot m_u$ gives the limiting mass. Light counts according to Bose, $u = aT^4$ with $a = (\pi^2/15)\cdot k^4/(\hbar c)^3$. If it holds together with the heat of the gas, $P = \rho kT/(\mu m_H) + aT^4/3$, then $(1 - \beta)/\beta = (\pi^2/45)\cdot(\ell/\lambda_H)^3$ with the share β of the gas in the pressure, the spacing $\ell = (\mu m_H/\rho)^{1/3}$ of the particles and $\lambda_H = \hbar c/kT$. At fixed β the thermal length is everywhere a fixed multiple of the spacing, the funnel held by heat, with $b = \ell/(\beta\cdot\lambda_H) = (45\cdot(1 - \beta)/(\pi^2\beta^4))^{1/3}$ and $M = 9.726\cdot\sqrt{(1 - \beta)}/(\mu^2\beta^2)\cdot m_{Pl}{}^3/m_H{}^2$. The 45 is the 3 of the third times the 15 from $\pi^4/15 = \Gamma(4)\zeta(4)$, the stage $d = 3$ of the Bose form. The share of light thus depends only on $\mu^2 M$ against the mass scale of the stars. Light and gas hold equally, $\beta = ½$, at 146 M⊙ for gas of the primordial mixture ($Y_p = 0.2449$, $\mu = 1/(2X + 3Y_p/4) = 0.590$ with $X = 1 - Y_p$). The same form $n = 3$ carries both ways of counting.

The burning stages lie as a ladder in the depth as long as the binding per nucleon grows, up to $x = ¼$: in the valley by the liquid-drop formula $Z = 26.7$, between ^{56}Fe and ^{62}Ni (AME2020). If the counting rule of the electrons holds the funnel, it reaches at most $kT = 0.096\cdot(\mu/\mu_e)\cdot m_{ec}{}^2$ (μ and μ_e the mean mass per particle and per electron in units of m_u), for carbon and oxygen $5.0\cdot10^8$ K. Beyond the iron group only the capture of neutrons leads on. That the star on the mass scale lies at the border between radiative and convective envelopes is the condition $\alpha_G \approx \alpha^{12}\cdot(m_e/m_p)^4$ (Carter 1974: pressure balance, ignition, radiative transport with the Eddington limit $L_{Edd} = (9/2)\cdot m_p\cdot c^3\cdot\Delta s/r_e{}^2$ and emission together), with the numbers $2.0\cdot10^{-39}$ against $5.9\cdot10^{-39}$, equal up to a factor of order one. Gravity acts here only collectively, through $\alpha_G{}^{-3/2}$ nucleon folds.

The fold. When the shell falls onto the centre, $R = r_0 = 6\Delta s$, the funnel becomes a **fold**, the black hole. Its depth remains the finite fold Δs. Only someone who takes the depth of the funnel for a place reads a singularity. Of the three radii two remain, r_0 and r*, with $r^{*2} = r_0\cdot\Lambda$: mirrored at the threshold radius, the horizon is the closure length. At the horizon $s^2 = M_U/(4M)$. The fold lies deep in the Newtonian regime there, and inside a galaxy it remains a single funnel (10^6 M⊙: $s^2 = 1.4\cdot10^{17}$ at the horizon, r* = 36 pc). Only at $M_U/4 = 1.4\cdot10^{23}$ M⊙ do horizon, threshold and closure length coincide, the meeting point that the whole almost reaches.

The fold in the end state. The rate $\sqrt{(1 - 6\Delta s/r)}$ counts against flat outer space. In the end state there is none, the restoring $H_\infty{}^2\cdot r$ acts in the rate of every clock at rest, $H_\infty = c\cdot\sqrt{\Lambda_s}$. The count is against t, without mass the clock at the centre. M is the mass of the field GM/r^2, for a cluster $(1 + Q)\cdot M_b$:

$$d\tau/dt = \sqrt{(1 - 6\Delta s/r - (H_\infty\cdot r/c)^2)} \qquad \text{horizons: } u^3 - u + u_0 = 0 \qquad u = r\cdot H_\infty/c \qquad u_0 = 6\Delta s\cdot H_\infty/c$$

The clock at rest runs fastest where slope and restoring cancel, at $r^3 = GM/H_\infty{}^2$, the largest turnaround R_{max}, where a body rests freely. There its rate is that of the orbital clock at the same place, $\sqrt{(1 - 9\Delta s/R_{max})} = \sqrt{(1 - (M/M_N)^{2/3})}$, around a cluster with $(1 + Q)\cdot M_b = 10^{15}$ M⊙ $6.4\cdot10^{-6}$ below one. Around a galaxy, with the potential of the rule, $v_{flat}{}^2\cdot\ln r$, the fastest clock at rest likewise lies at its largest turnaround. The orbital clock itself does not feel the restoring: it takes from v^2/c^2 of the circular orbit, $v^2 = GM/r - H_\infty{}^2\cdot r^2$, as much as from the square of the rate. The orbital clock stays $\sqrt{(1 - 9\Delta s/r)}$, the photon sphere at $9\Delta s$. The horizons are the zeros of the rate. Their sum is zero and their sum of squares 2, hence they are $u_k = (2/\sqrt{3})\cdot\cos(\theta - 2\pi k/3)$ with $\cos 3\theta = -M/M_N$, the projections of an equilateral triangle with circumradius $2/\sqrt{3}$, which the mass turns from $\theta = 30°$ without mass to 60° at M_N. Without mass they are $u = 0$ and ±1, with mass the horizon of the fold r_F, the horizon of the whole r_H and the negative root $-(r_F + r_H)$, linked by $r_F{}^2 + r_F\cdot r_H + r_H{}^2 = (c/H_\infty)^2$. At

$$M_N = c^3/(3\sqrt{3}\cdot G\cdot H_\infty) = (\sqrt{5}/6\pi^2)\cdot M_U = 2.2\cdot10^{22}\ M\odot \qquad r_N = c/(\sqrt{3}\cdot H_\infty) = 3.11\ \text{Gpc}$$

the horizon of the fold, the photon sphere, R_{max} and the horizon of the whole coincide in r_N, and the fastest clock stands still. M_N/M_U is a pure number because $H_\infty\cdot\tau_D = 2\pi/\Theta_\infty = 2\pi^2/\sqrt{15}$. Only there are both horizons equally warm, the slope of $(d\tau/dt)^2$ is the same at both. Against t both become cold there, as always when two horizons coincide. For the freely resting clock at R_{max}, whose rate goes to zero with them, the thermal length stays $2\pi\cdot r_N$, one turn of the common horizon. The radius r_N lies at $0.58\cdot r^*$, where the rule of bound baryons would strengthen the field by $\nu = 1.22$: the limiting mass holds for a source whose field stays GM/r^2 out to there. On the ladder the lines of the fold (1, 0), of R_{max} (⅓, ⅔) and of the horizon of the whole (0, 1) run together in the corner at M_U. M_N lies before it, at $x = 0.98$, where the photon sphere $9\Delta s$ reaches the largest turnaround. Above M_N a fold in the end state has no region at rest. The meeting point of the lengths at $M_U/4$ lies 6.6 times above: a meeting point

of the ladder, not a fold of the end state. Without the circles of the fold four lines of the ladder meet in one point. The restoring $H_\infty{}^2 \cdot r$ reaches a_0 at $r_\times = a_0/H_\infty{}^2 = (\Theta_\infty/2\pi)^2 \cdot \ell_0 = 1.06$ Gpc, the mirror image of the length of action at the horizon of the end state without mass, $r_\times \cdot \ell_0 = (c/H_\infty)^2$. The largest turnaround lies on the way from the threshold r^* (½, ½) to $r_\times$ (0, 1) at the fraction $1/d_{eff}$, $R_{max} = r^{*1 - 1/d_{eff}} \cdot r_\times{}^{1/d_{eff}}$: after the rule, $d_{eff} = 2$, their mean $\sqrt{(r^* \cdot r_\times)}$ (¼, ¾), after Newton, $d_{eff} = 3$, $(r^{*2} \cdot r_\times)^{1/3}$ (⅓, ⅔). The number of the form thus lies entirely in $r_\times$. At $M_\times = a_0{}^3/(G \cdot H_\infty{}^4) = (\Theta_\infty/2\pi)^4 \cdot M_U = 8.5 \cdot 10^{20}$ M⊙, x = 0.95, $r^* = r_\times = R_{max}$ in both forms: slope, threshold and restoring are equal there, $GM/r^2 = a_0 = H_\infty{}^2 \cdot r$, at $v_{flat} = a_0/H_\infty = 0.196 \cdot c$. Counted from the threshold, $R_{max}/r^* = (M_\times/M)^{1/(2d_{eff})}$, around the Milky Way ($6 \cdot 10^{10}$ M⊙) after the rule 345. Only a region with $v_{flat} \ll a_0/H_\infty$ therefore has a deep regime, $r \gg r^*$.

The fold is its own mirror: its horizon is the quantum length of its mirror mass, $r_0 = \bar{\lambda}(\tilde{m})$, and the horizon of the mirror mass is its own quantum length, $r_0(\tilde{m}) = \bar{\lambda}(M)$. The fold is its own mirror image, $\tilde{m} = M$, at $m_{Pl}/\sqrt{2}$, where $r_0 = \bar{\lambda}$: at the same mass at which a particle with spin ½ reaches the spin number j = 1. Its temperature is the rest energy of the mirror mass divided by 4π, $kT = \tilde{m}c^2/(4\pi) = \hbar c/(24\pi\Delta s)$. If the fold radiates as a black body at its temperature, it lives $t = 5120\pi \cdot \alpha_G \cdot 3\Delta s/c$, one solar mass $2.1 \cdot 10^{67}$ years. The length $c \cdot t$ is the point (3, 0) of the ladder, mirrored about ℓ_P to $\bar{\lambda}/\alpha_G$ (−3, 0): $\bar{\lambda}/\alpha_G$, $\bar{\lambda}$, $3\Delta s$ and $\alpha_G \cdot 3\Delta s$ follow one another in the measure α_G. What falls into it releases, down to the innermost stable orbit ($18\Delta s$, $\delta = 1/6$), $1 - (1 - 2\delta)/\sqrt{(1 - 3\delta)} = 1 - \frac{2}{3}\sqrt{2} = 5.7$ % of its rest energy, six times as much as nuclear binding up to iron. At the meeting point of rotation (j = 1) it is $1 - 1/\sqrt{3} = 42$ %, and the fold is cold there because both horizons coincide.

Tearing and swallowing. A star of mass m with envelope R stays whole at a fold of mass M as long as R lies below its smoothing length in the fold's tide, $R^3 < Gm/\mathscr{T}$, that is, as long as its mean density exceeds the fold's tidal density. The tide counts in the star's own frame. Moving across the radius, the star feels $\mathscr{T} = (3\gamma^2 - 1) \cdot GM/r^3$, with γ the rate of the resting clock against the star's, far out $2GM/r^3$. What falls in from far away has energy one. At its turning point $\gamma^2 = 1/(1 - 2\delta)$, and it can turn around only outside $12\Delta s$, where $\gamma^2 = 2$ and $\mathscr{T} = 5GM/r^3$. The star is torn apart if it turns around inside r_T, where its smoothing length drops to R, far out $r_T = R \cdot (2M/m)^{1/3}$. If r_T lies inside $12\Delta s$, the star falls in whole. Far out r_T rises with $M^{1/3}$, the circle with M, and the circle reaches r_T at $M_V = \sqrt{5} \cdot (¼/\delta_m)^{3/2} \cdot m$: the limit is a power of the ratio between the fold's spread at the circle, ¼, and the star's own, $\delta_m = 3\Delta s_m/R$. For the Sun $M_V = 9.0 \cdot 10^7$ M⊙, and a fold of 10^6 M⊙ tears it apart at 0.6 AU. On a circular orbit the revolution adds to this, since a co-moving gyroscope turns against the radius at $\sqrt{(GM/r^3)}$. With $\gamma^2 = (1 - 2\delta)/(1 - 3\delta)$ the tide with the revolution is $3\gamma^2 \cdot GM/r^3$, by Newton $3GM/r^3$ as in the Jacobi radius, and at the innermost stable orbit $18\Delta s$ equal to $4GM/r^3$. If the star spirals in this way, $M_V = 2 \cdot (⅙/\delta_m)^{3/2} \cdot m$, for the Sun $4.4 \cdot 10^7$ M⊙. This is a prediction.

Tide at a planet. A planet is a single funnel up to its smoothing length in the tide of the Sun including the orbit, $\ell^3 = Gm/(3\Omega^2)$, the Earth out to $1.50 \cdot 10^6$ km: only within it can its moons orbit. A moon stays whole as long as it is on average denser than the tidal density of its planet at the place of its orbit. On a circular orbit the tide including the orbit is $d_{eff} \cdot \Omega^2$, and the limit reads only the moon's own density ρ: it is torn apart when its period becomes shorter than $\sqrt{(3\pi \cdot d_{eff}/(G\rho))}$, around every planet $3 \cdot \sqrt{(\pi/(G\rho))}$, however heavy the planet is, at the mean density of the Earth 2.4 hours. The same limit holds on every stage where the tide follows Newton: a star of the Sun's density is torn apart far out at a fold on a circular orbit that runs shorter than 4.8 hours. Within the limit no moon gathers.

Growth at the limit. If what falls into a fold gives off the fraction ε of its rest energy down to the innermost stable orbit, and this release reaches the Eddington limit L_{Edd}, the fold grows as $dM/dt = ((1 - \varepsilon)/\varepsilon) \cdot L_{Edd}/c^2$, by a factor e in $t_e = (\varepsilon/(1 - \varepsilon)) \cdot t_{Edd}$. The time of the limit is the depth of the proton, mirrored at r_e: $c \cdot t_{Edd} = \sigma_T/(4\pi \cdot 3\Delta s_p) = \frac{2}{3} \cdot r_e{}^2/(3\Delta s_p)$ with $\sigma_T = 4\pi \cdot \frac{2}{3} \cdot r_e{}^2$, the release of the winding transverse to the oscillation. The light acts on the winding of the electron, the slope on the folding of the proton. Mirrored at the threshold of the proton, the same depth gives ℓ_0, $r^*(m_p)^2 = 3\Delta s_p \cdot \ell_0$, hence $t_{Edd} = \frac{2}{3} \cdot (r_e/r^*(m_p))^2 \cdot \tau_D = 0.45$ billion years, for every mass, and a_0 drops out. With the energy E of the innermost orbit per rest energy, $\varepsilon/(1 - \varepsilon) = 1/E - 1$, without rotation $3/(2\sqrt{2}) - 1 = 0.061$ with $t_e = 27$ million years, at the meeting point of rotation (j = 1) $\sqrt{3} - 1 = 0.73$ with $t_e = 330$ million years. Since the clock Θ also counts the time of the co-breathing observer in τ_D, a fold at rest in the breathing that starts from a seed M_K reaches by Θ at most $\ln(M/M_K) = ((1 - \varepsilon)/\varepsilon) \cdot (r^*(m_p)/r_e)^2 \cdot \Theta_\infty \cdot \operatorname{artanh}(\Theta/\Theta_\infty)/2\pi$, for small Θ $((1 - \varepsilon)/\varepsilon) \cdot (r^*(m_p)/r_e)^2 \cdot \Theta/2\pi$. Up to $\Theta = 0.1$, after 0.95 billion years, a non-rotating fold thus grows by at most a factor e^{35}, one at j = 1 by a factor of 18: the seed counts as a factor, the rotation in the exponent.

For $j \le 1$ the lengths of the rotating fold form a right triangle in the Thales circle over $r_0 = 6\Delta s$, the horizon of the non-rotating fold of the same mass. Its height is the twist a_J, its foot divides the diameter into the two horizons r_+ and r_- (altitude theorem), and its legs are the horizons of two partial masses, $r_0(M_{irr})^2 = r_0 \cdot r_+$ and $r_0(M_{rot})^2 = r_0 \cdot r_-$ (leg theorem). Hence $M^2 = M_{irr}{}^2 + M_{rot}{}^2$, and the product of the two partial masses depends only on the rotation, $M_{irr} \cdot M_{rot} = Jc/(2G)$. At fixed rotation the mass is smallest where both are equal, $M_{irr} = M_{rot} = M/\sqrt{2}$: this is j = 1, the greatest height in the circle, $a_J = 3\Delta s$. Beyond it there is no triangle.

The area of the outer horizon is that of the non-rotating fold of mass M_{irr}, $4\pi\cdot(r_+^2 + a_J^2) = 4\pi\cdot r_0(M_{irr})^2$, and the entropy reads only it, $S = 4\pi k\cdot(M_{irr}/m_{Pl})^2$. What falls into the fold never decreases this area. Without its radiation, therefore, at most $(M - M_{irr})\cdot c^2$ can be extracted (Penrose, Christodoulou), at the meeting point $j = 1$ that is $1 - 1/\sqrt{2} = 29$ % of the rest energy of the fold, and the entropy there is half that of the non-rotating fold of the same mass. The temperature $\hbar c\cdot(r_+ - r_-)/(4\pi k\cdot r_0\cdot r_+)$ reads the distance of the foot from the centre.

The horizons as an orbit. The equation of the horizons, $r^2 - 6\Delta s\cdot r + a_J^2 = 0$, is that of the turning points of a Kepler orbit around M with the energy $-\frac{1}{2}c^2$ and the angular momentum J/M per mass, $c^2r^2 - 2GM\cdot r + (J/M)^2 = 0$. The orbit reaches only to the outer horizon: a form of the equation, not the path of a body. Its major axis is r_0, its auxiliary circle the Thales circle, pericentre and apocentre are r_- and r_+, its focus is the node and thus the foot of the altitude, its semi-minor axis is a_J and its eccentricity $e = \sqrt{(1 - j^2)} = (r_+ - r_-)/r_0$. Without rotation it turns at r_0, where the escape speed reaches c. At $j = 1$ it is the circle at $3\Delta s$, traversed at c, $J/M = 3\Delta s\cdot c$: so the rotation reaches c at the meeting point. For $j > 1$, $-\frac{1}{2}c^2$ lies below the floor of its potential, $-\frac{1}{2}c^2/j^2$, and there is neither orbit nor triangle. The ellipse about the centre with the legs $r_0(M_{irr})$ and $r_0(M_{rot})$ as semi-axes, the orbit inside a held body, goes over into this Kepler orbit under $z \to z^2/r_0$, the mirror of the two slopes that close every orbit. The leg theorem is this map of the vertices onto the turning points, $r_0(M_{irr})^2 = r_0\cdot r_+$, and the altitude is the semi-minor axis of the image, $a_J\cdot r_0 = r_0(M_{irr})\cdot r_0(M_{rot})$. $M^2 = M_{irr}^2 + M_{rot}^2$ counts like the energy of the inner orbit, $M_{irr}\cdot M_{rot}$ like its angular momentum, and in both orbits j is the angular momentum against that of the circle of equal energy. Temperature and entropy read the eccentricity, $T = T_0\cdot 2e/(1 + e)$ and $S = S_0\cdot(1 + e)/2$ with T_0 and S_0 of the unrotated fold of equal mass: at $j = 0.6$, $e = 0.8$, $T = 0.89\cdot T_0$ and $S = 0.9\cdot S_0$. Likewise the horizons of the end state are the turning points of a radial orbit with the energy $-\frac{1}{2}c^2$ per mass in the potential of slope and restoring, $-GM/r - \frac{1}{2}H_\infty^2 r^2$.

The winding at the fold. The folded field of the winding lies outside the node: within r the folding $M - Q_{el}^2/(8\pi\varepsilon_0 c^2 r)$ remains. So the resting clock goes with

$$(d\tau/dt)^2 = 1 - 6\Delta s/r + r_Q\cdot 3\Delta s/r^2 \qquad r_Q = Q_{el}^2/(4\pi\varepsilon_0 M c^2) \qquad r_Q\cdot 3\Delta s = (\ell_P\cdot Q_{el}/q_P)^2$$

Not the phase enters the clock but the folding of its field, and that depends on no mass: like ℓ_P, $\sqrt{(r_Q\cdot 3\Delta s)}$ lies at the middle of the edge between tangential and parametric, times the charge in q_P. The horizons are the roots of $r^2 - 6\Delta s\cdot r + a_J^2 + r_Q\cdot 3\Delta s = 0$. Their sum remains the depth, their product is carried by twist and winding: the height in the Thales circle becomes the hypotenuse of a_J and $\sqrt{(r_Q\cdot 3\Delta s)}$, and both horizons meet at $j^2 + r_Q/(3\Delta s) = 1$. Twist and winding count against the same coupling, $j = (J/\hbar)/\alpha_G$ and $r_Q/(3\Delta s) = (Q_{el}/q_P)^2/\alpha_G$. Without rotation the meeting point lies at $Q_{el}/q_P = M/m_{Pl}$, the meeting point of charge and mass. With $r_e/(3\Delta s_e) = \alpha/\alpha_G(m_e) = 4.2\cdot 10^{42}$ the electron lies far above it in the winding as well.

Without rotation M_{irr} is the folding within the horizon, $M = M_{irr} + Q_{el}^2/(8\pi\varepsilon_0 c^2 r_+)$, and only the field outside can be extracted. With rotation $M^2 = (M_{irr} + M_{el})^2 + M_{rot}^2$ with $M_{el} = Q_{el}^2/(16\pi\varepsilon_0 G\cdot M_{irr})$: the field of the winding adds to M_{irr} linearly as folding to folding, the twist stands across both and counts in the square. As $M_{irr}\cdot M_{rot}$ depends only on the rotation, $M_{irr}\cdot M_{el} = (m_{Pl}\cdot Q_{el}/(2q_P))^2$ depends only on the charge, in lengths $r_0(M_{irr})\cdot r_0(M_{el}) = r_Q\cdot 3\Delta s$. Without rotation and at fixed charge the mass is smallest where both are equal, $M_{irr} = M_{el} = M/2$, at the meeting point $r_Q = 3\Delta s$. There half of the rest energy can be extracted, and the entropy is a quarter of that of the uncharged fold of equal mass. In general the temperature reads the distance of the horizons against the area, $\hbar c\cdot(r_+ - r_-)/(4\pi k\cdot r_0(M_{irr})^2)$, without winding with $r_0(M_{irr})^2 = r_0\cdot r_+$.

Loss per turn. Every orbit runs tangentially. Left open, it gives off a part into the transversal direction per turn, and how much depends on the number of senses of circulation. The winding has two senses, the centre of its charge can oscillate, it emits as a dipole. The fold has one sense, its centre stands still, it emits only as a quadrupole. Per turn, against the binding of the orbit:

winding: $\Delta E/E = (8\pi/3)\cdot(r_e/r)\cdot(v/c)$	hydrogen: $(8\pi/3)\cdot\alpha^3 = 3.3\cdot 10^{-6}$	
fold:	$\Delta E/E = (128\pi/5)\cdot(m/M)\cdot(v/c)^5$	Earth around the Sun: $2.3\cdot 10^{-24}$

Left open, the winding of hydrogen would fall onto the nucleus in $1.6\cdot 10^{-11}$ s. It does not fall, because its wave closes after one turn: the closure keeps the transversal direction shut, emission happens only at the change between closed forms. The same number of senses of circulation that lets windings cancel and folds add lets the winding emit as a dipole and the fold only as a quadrupole.

Fall and braking. The lowest order of emission, $l = 1$ for the winding and $l = 2$ for the fold, also sets how fast the emission grows with the motion. On an orbit the power is $P \propto r^{-(l+3)}$ at the binding $E \propto 1/r$. Until the fall passes $t = E/((l + 2)\cdot P)$: left open, the winding falls after a third, the fold after a quarter of the time its initial loss per turn would need for the whole binding. The Earth would so fall onto the Sun in $1.1\cdot 10^{23}$ years. Two equal masses with $\delta = 3\Delta s/a$, Δs that of one, merge after

$t = (5/512)\cdot(3\Delta s/c)\cdot\delta^{-4}$ in $N = \delta^{-5/2}/(64\sqrt{2}\cdot\pi)$ turns. The number of turns depends on the spread alone, the time counts in the light time of the depth: from $\delta = 0.001$ on $1.1\cdot10^5$ turns, for one solar mass each in 13 hours. A rotating body emits with $P \propto \Omega^{2l+2}$ and brakes with $\Omega \propto \Omega^n$, braking index $n = 2l + 1$: $n = 3$ through the winding of locked currents whose axis stands oblique to the axis of rotation, $n = 5$ through a deformation of its fold across the axis. As a dipole the twist of the fold emits nothing, for it is the rotation itself, $J = Mc\cdot a_J$, and stands fixed like its centre. Orbit and rotation are one form: if the emission grows with the energy as $P \propto E^k$, the end or the start lies $t = E/((k - 1)\cdot P)$ away. The orbit gains binding, $k = l + 3$, and falls in finite time. The rotation loses energy, $k = l + 1$, never stands still and has turned with Ω for at most $t = \Omega/((n - 1)\cdot|\dot\Omega|) = E/(l\cdot P)$, through the winding E/P, through the fold $E/(2P)$.

Mirror of emission. With $|k - 1|$, $t = E/(|k - 1|\cdot P)$ also holds for k below one and for an energy that grows. Over ln E, ln P is a straight line of slope k. Slope 1 keeps the time E/P fixed, and t is E/P divided by the distance of the slope from 1. The fold radiates with $P \propto M^{-2}$, $k = -2$, and lives $E/(3P)$. At the limit it grows with $k = 1$ by the factor e per time E/P, without end and without start in finite time. The mirror $M \leftrightarrow \tilde m$ turns $P \propto E^k$ into the rate of $\tilde E = \tilde m c^2$, $(\tilde E/E)\cdot P \propto \tilde E^{2-k}$. It mirrors the slope about 1, reverses $k - 1$ and keeps the time fixed. Only growth at the limit stays fixed: if the fold grows there, its mirror mass decreases by the factor e in the same time. When the fold evaporates, $k = -2$ becomes mirrored $2 - k = 4$: its temperature rises with $\dot T \propto T^4$ and reaches the end after the same time $E/(3P)$.

Fall in every field. On a circular orbit in a field of flux dimension d, $g \propto r^{1-d}$, $v^2 \propto r^{2-d}$, and per e-fold of the radius the binding changes by $(4 - d)\cdot E_{kin}$, with the same $4 - d$ as in $\kappa^2 = (4 - d)\cdot\Omega^2$. The emission of order l grows as $r^{2l}\cdot\Omega^{2l+2}$, and until the fall passes

$$t = (4 - d)\cdot E_{kin}/((l\cdot(d - 2) + 2)\cdot P)$$

For Newton, $d = 3$, that is $E/((l + 2)\cdot P)$. In the deep regime, $d = 2$, the speed is fixed: every order emits with $P \propto v^{2l+2}/r^2$, the binding counts in e-folds of the radius, per factor $m\cdot v_{flat}^2$, and every orbit falls in $t = E_{kin}/P$, whether it emits through the winding or the fold. Over ln E, ln P is no longer a straight line there, over E itself it is one, and r^2 decreases uniformly with time. At $d = 4$ the binding does not depend on the radius, and every loss is the fall.

22 Galaxies: the Rule in the Data

A galaxy is a collective funnel: stars are **single funnels**, while the stellar system as a whole follows the rule in its total field. Compact funnels, denser than the tidal density of their surroundings, remain Newtonian up to their smoothing length. Only the collective field, smoothed over them, is enhanced. The tidal density is $\rho_J = 3\mathscr{T}/(4\pi G)$ with the tide $\mathscr{T}$ of the smoothed parent field, and the smoothing length ℓ follows from $\ell^3 = GM/\mathscr{T}$. Because the rule acts at the place on the smoothed field, in which the body's own field vanishes at its centre of mass, bodies at the same place fall alike. Compact globular clusters ($g = 25\cdot a_0$) show hardly any addition ($\nu = 1.007$), thin systems ($g = 0.13\cdot a_0$) the full one ($\nu = 3.3$). Pressure-supported spheres have, isotropically and for every profile, $\sigma_v^4 = (4/81)\cdot GMa_0$ (σ_v the speed of their thermal motion): 9.1 km/s at 10^7 M⊙, 19.3 km/s at $2\cdot10^8$ M⊙. For an isotropic swarm of test bodies with $\rho \propto r^{-\gamma}$ in the deep regime of a source, the Jeans equation gives $\sigma^2 = v_{flat}^2/\gamma$ at every radius: without pressure from outside every such swarm is isothermal there, its dispersion reads the source only through v_{flat} and itself only through its profile. Its mass outwards is finite only for $\gamma > 3$, the dimension of space: gas that does not carry the field itself has a finite mass only with $\sigma < v_{flat}/\sqrt{3} = 0.58\cdot v_{flat}$. The isothermal halo the observer reads, $\rho_X \propto r^{-2}$, is the swarm with $\gamma = 2$, its mass grows without end. If an isothermal sphere entirely in the deep regime carries itself, $\sigma^4 = v_{flat}^4/\gamma^2 = (4/81)\cdot GMa_0$, so $\gamma = 9/2$: it falls off outside like $r^{-9/2}$, virial theorem and profile read the same number. The rule lies everywhere above its deep limit, $\nu > 1$/s: therefore 9/2 is the upper bound, with a Newtonian core the sphere falls off less steeply. Around a region that follows the rule as a whole, the Bondi radius GM/c_s^2 lies where the spread of the funnel reaches that of the gas, $\delta = (c_s/c)^2$, and the speed after Newton is c_s. Its mirror image about r* is $c_s^2/a_0 = \ell_0\cdot(c_s/c)^2$, where the speed of the threshold is c_s, as ℓ_0 is the mirror image of $3\Delta s$ at $c_s = c$. If $c_s < v_{flat}$, it moves beyond r* into the deep regime, and there it does not exist: the orbital speed of the rule never falls below v_{flat}, acceleration of the orbit and pressure force per mass both fall like 1/r, the scale-invariant deep regime has no radius. Only v_{flat}/c_s decides: isothermal gas is there the swarm with $\gamma = (v_{flat}/c_s)^2$. As a whole, a system lies below the threshold with surface density $\Sigma < a_0/(\pi G) = 242$ M⊙/pc².

Surroundings and dimension. The rule acts on the total field of a region's own masses, not particle by particle. If the region lies in the field of a larger one, it falls with it: the uniform field cancels in the fall, and inside, the region reads only its own folding. What stays effective is the tide, and it decides whether the region remains collective. As a whole it moves in the enhanced field of its host. At the same time the rule lowers the dimension of every form: Newton is flux through the spherical surface, the

deep regime flux through a circumference, $d_{eff} = 1 + (d - 1)(1 - \chi/2)$. The sphere goes from 3 to 2, a filament from 2 to 1.5, a wall stays at 1. A filament of line mass μ has the threshold $r^*_{fil} = 2G\mu/a_0$ (at 10^{12} M☉/Mpc: 2.6 kpc). Deep inside it the circular speed grows like $r^{1/4}$. The epicycles change as well: instead of $\kappa^2 + \nu_z^2 = 2\Omega^2$ (ν_z the frequency of the oscillation perpendicular to the disc), $(2 + \chi)\Omega^2$ holds. The observer who assumes unchanged rules reads the missing part as a density, $4\pi G\rho_X = \chi\Omega^2$, outside an isothermal halo.

Surroundings in the data. The data carry this. Around isolated galaxies the lensing follows the rule without external field (KiDS-1000, Brouwer 2021): for the blue ones, which still form stars, $\chi^2 = 21$ in 15 bins, with $0.003 \cdot a_0$ in magnitude already 534. The 22 classical satellites of the Local Group, whose host field exceeds their own, lie $+0.15 \pm 0.04$ dex above the rule without the host field, with it in magnitude $+0.46 \pm 0.05$.

In the data (SPARC). On the rotation curves of 141 galaxies with 2688 points, with distance, inclination and mass-to-light ratio estimated for each galaxy, the threshold reads $a_0 = 1.11 \cdot 10^{-10}$ m/s² (16 to 84 %: 1.00 to 1.21). The mass-to-light ratio of the discs follows from the requirement that stars and gas obey the same rule, because the rule acts on the fold, not on the state of matter: $Y_d = 0.67$. With the value of the stellar models, $Y_d = 0.5$, one would read a higher threshold and an apparent trend from stars to gas, two consequences of the same foreign conversion. The central surface density from the rotation curve lies, galaxy by galaxy with its own baryons, on average 0.05 dex below the rule, compatible. The Bose form fits best. The simple form, $\nu = \frac{1}{2} + \sqrt{(\frac{1}{4} + a_0/a_N)}$, and the standard form, $\nu = \sqrt{(\frac{1}{2} + \sqrt{(\frac{1}{4} + (a_0/a_N)^2)})}$, are worse by $\Delta\chi^2 = 53$ and 41 (χ^2 the error measure of the fit, not the switch χ). The return has dimension 0.92 ± 0.05, one-dimensional as the square root in q requires. More precisely it is the first passage of a random walk in one dimension: the time T in which it covers the distance of the return is distributed as $T^{-3/2}$, and its mean $\langle e^{-x \cdot T} \rangle$ is $e^{-\sqrt{x}}$ with $x = a_N/a_0$, equal to 10^{-15} when integrated over the distribution. In three dimensions a random walk would return only with 0.341, the sum of returns would stay at 1.516, and there would be no flat curves. The residual per galaxy depends neither on mass nor on gas fraction, neither on shape (Hubble type) nor on surface brightness: the rule reads only the fold. Stacked by the fold of the baryons, at most 0.02 dex of the rule remains in every half decade.

Against the halo. For the same curves the standard model needs an NFW halo with two free numbers per galaxy. The rule has $\chi^2 = 2918$, the halos 1434 with 282 more free numbers. By the information measure BIC (χ^2 plus the number of free numbers times ln 2688) the rule leads by 736, 2926 against 3661. For 31 of the 141 galaxies the halos lie at the edge of their range, and their mass scatters against that of the baryons by 0.58 dex: the halo is bookkeeping, not an object of its own.

Discs at the threshold. The disc scale length R_d is, in the median, $0.71 \cdot r^*$ (16 to 84 %: 0.37 to 1.28) across four decades of mass. That is $r^*/\sqrt{2}$: the centre of the disc lies exactly at the threshold of surface density, $\Sigma_0 = M/(2\pi R_d^2) = a_0/(\pi G)$. It grows with the baryonic mass as $M_b^{0.31 \pm 0.02}$: the envelope as at fixed density of the whole with $M^{1/3}$, the threshold from the galaxy's own fold with $M^{1/2}$, hence $R/r^* \propto M^{-1/6}$, like R_{ta}/r^* in the inactive measure (Section 26).

Response to a perturbation. From $a = \nu(|g_N|) \cdot g_N$ at the place of the orbit a perturbation gives

$$\delta a = \nu \cdot [\delta g_N - (\chi/2) \cdot \hat{g}(\hat{g} \cdot \delta g_N)]$$

Along the slope the rule answers with its derivative, $da/da_N = \nu(1 - \chi/2)$, across it with its enhancement ν, deep along it with $\nu/2$, because in the mean $\sqrt{(a_N \cdot a_0)}$ the perturbation moves only the slope. A radial wave in the midplane of a thin disc lies along the slope and rearranges the bound baryons themselves. So the return answers as well, and locally and axisymmetrically Toomre's criterion carries $G \cdot \nu(1 - \chi/2)$ instead of G:

$$\omega^2 = \kappa^2 - 2\pi G \cdot \nu(1 - \chi/2) \cdot \Sigma \cdot |k| + \sigma^2 k^2 \qquad Q_T = \kappa\sigma/(\pi G \cdot \nu(1 - \chi/2) \cdot \Sigma) \qquad \kappa^2 = R \cdot d\Omega^2/dR + 4\Omega^2,\ \Omega^2 = a/R$$

For stars $2\pi \cdot \max_x (1 - e^{-x} I_0(x))/\sqrt{x} = 3.36$ replaces π. At the threshold of surface density the mass drops out, $a_N/a_0 = (R/R_d) \cdot [I_0 K_0 - I_1 K_1](R/(2R_d))$ (Freeman). The dispersion at the limit $Q_T = 1$ is therefore the same function of R/R_d for every mass. In units of the orbital speed ν at the same place it is, for a disc of gas, $0.259 \cdot \nu$ at R_d and $0.144 \cdot \nu$ at $2R_d$ (there $\nu = 2.16$, along the slope $\nu(1 - \chi/2) = 1.382$). It falls outward: if the gas disperses with more than $0.224 \cdot v_{flat}$, every disc at the threshold stays stable outside R_d. At $6 \cdot 10^{10}$ M☉ the gas needs at least 26.7 km/s at $2R_d$ for this. Perpendicular to the disc, in its midplane, the rule answers across, with ν like the orbit: ν_z/Ω is the Newtonian ratio. The observer with a halo reads the same curve with a rigid mass that does not answer the wave. The observer computes $0.104 \cdot \nu$ (19.3 km/s) at $2R_d$ and reads Q_T too large by the factor $\nu(1 - \chi/2)$. From the oscillation perpendicular to the thin disc the observer reads ν times its density in the midplane, from its stability $\nu(1 - \chi/2)$ times. Read as matter, the disc would have two densities, their ratio is $1 - \chi/2 = d \ln a / d \ln a_N$. This is a prediction for the local dispersions.

Weight of a layer. In a region whose mode has been handed over, the Newtonian field perpendicular to a thin layer of surface density Σ grows with the mass between $-z$ and z, $u = 4\pi G\cdot\int_0^z \rho\, dz'$, up to $2\pi G\Sigma$ at its edge. The rule reads the magnitude together with the field g_R along the layer, $a_z = \nu(\sqrt{(g_R^2 + u^2)})\cdot u$. The midplane carries the weight of the layer, for every profile and every dispersion:

$$P_0 = \int \rho\cdot a_z\, dz = (a_0^2/2\pi G)\cdot\int_{s_R}^{s_o} x^3\cdot\nu(x)\, dx \qquad s_R^2 = g_R/a_0 \qquad s_o^4 = s_R^4 + (2\pi G\Sigma/a_0)^2$$

The column of the layer runs through the ladder from s_R in the midplane to s_o at the edge. The spontaneous part x^3 is Spitzer's layer, $P_0 = \pi G\Sigma^2/2$, isothermal $\rho = \rho_0\cdot\mathrm{sech}^2(z/z_0)$ with $z_0 = \sigma^2/(\pi G\Sigma)$. The return adds $x^3\cdot n_B$, the stage $d = 3$ of the ladder, the structure action. If the layer is thin against the field along it, $2\pi G\Sigma \ll g_R$, it is Spitzer's layer with $G\cdot\nu(g_R)$, the answer across the slope, $z_0 = \sigma^2/(\pi G\cdot\nu\cdot\Sigma)$. In a disc at the threshold the midplane at $2R_d$ carries 2.08 times Spitzer's weight, just below ν. Without a field along the layer the field vanishes in the middle of every layer, and there it is deep. Below the threshold, $2\pi G\Sigma \ll a_0$, $P_0 = (\Sigma/3)\cdot\sqrt{(2\pi G\Sigma a_0)} = (2/3)\cdot\sqrt{((\pi G\Sigma^2/2)\cdot\Sigma a_0)}$, the mean of Spitzer's weight and that of the threshold, $\Sigma\cdot a_0$, as the orbit in the deep regime is the mean of slope and threshold. Isothermally half of the mass then lies within $0.888\cdot\sigma^2/\sqrt{(2\pi G\Sigma a_0)}$ instead of $0.549\cdot\sigma^2/(\pi G\Sigma)$. Far above it the column runs through the whole stage, $P_0 = \pi G\Sigma^2/2 + \pi^3 a_0^2/(30G)$. The return gives no more to the midplane of any layer: at most $\pi^3 a_0^2/(30G) = \rho_s\cdot c^2 = 1.74\cdot10^{-10}$ Pa, independent of the mass. The same constant $\pi^4/15$ gives, over all s of the whole, the restoring of one direction, the vacuum part, and over the column of a layer the largest weight of the return. A layer of gas that never cooled reads $(1 + Q)\cdot$Newton, and its addition $Q\cdot\pi G\Sigma^2/2$ has no bound. This is a prediction.

Friction. A body of mass m that moves with ν through the bound baryons of a region whose mode has been handed over (density ρ_b, dispersion σ) gathers a wake behind it that brakes it (Chandrasekhar). Raising the wake and being pulled by it are both responses to a disturbance: a wave k of the wake is raised with $G\cdot\nu[1 - (\chi/2)(\hat{k}\cdot\hat{g})^2]$, and across the slope it pulls with $G\cdot\nu$. Up to its smoothing length ℓ, $\ell^3 = Gm/\mathscr{T}$ with the tide $\mathscr{T} = -dg/dr$, a compact body is a single funnel, and there Newton counts for both. On a circular orbit the path lies across the slope, and for an isotropic dispersion

$$dv/dt = -(4\pi G^2\cdot m\cdot\rho_b/v^2)\cdot F(X)\cdot[\ln(\ell/b_{min}) + v^2\cdot(1 - (\chi/4)\cdot(1 - \bar{\mu}^2))\cdot\ln(b_{max}/\ell)] \qquad F(X) = \mathrm{erf}\, X - (2X/\sqrt{\pi})\cdot e^{-X^2},\ X = v/(\sqrt{2}\cdot\sigma)$$

$F(X)$ is the share of stars slower than the body, $\bar{\mu}^2$ the mean square of the cosine μ between wave and path, weighted like the braking with $\mu^2\cdot e^{-X^2\mu^2}$: 0 for fast bodies and 3/5 for slow ones. b_{max} is the distance from the centre, b_{min} the larger of the radius of the body and Gm/v^2, where its spread reaches that of the motion, $3\Delta s/b = (v/c)^2$, as at the Bondi radius. In the Newtonian range this is Chandrasekhar with the baryons alone. The slope reads the rule once, friction reads it twice, $v^2 = \Sigma(n + 1)\cdot q^n$: n returns split in $n + 1$ ways between raising and pulling. Beyond ℓ the rule brakes a fast body at the threshold 2.14 times, at $a_N = 0.1\cdot a_0$ 10.7 times as strongly as Newton with the same baryons. Read as matter, friction would have a density of its own, $v^2\cdot(1 - (\chi/4)\cdot(1 - \bar{\mu}^2))\cdot\rho_b$, quadratic in the response. The observer with a halo computes friction at the same dispersion with the density read from the slope, in a sphere $\nu(1 - \chi/2)\cdot\rho_b + \chi\Omega^2/(4\pi G)$. The two part outside the baryons: there no wake forms, while the observer computes friction with $\chi\Omega^2/(4\pi G)$. This is a prediction.

Relaxation. In a collective region of N stars of mass m, with $M = N\cdot m$ and radius R, whose mode has been handed over, every encounter at distance b gives a star moving with speed v the kick $\delta v = 2Gm/(bv)$. Up to its smoothing length ℓ the kicking star is a single funnel and kicks after Newton. Beyond it, its field is a disturbance of the smoothed field, and the rule answers along the slope with $\nu(1 - \chi/2)$, across it with ν. In the spherical mean, one direction along and two across, the square of the kick grows by the factor $v^2\cdot f$ with $f = (2 + (1 - \chi/2)^2)/3$, deep ¾. In the time R/v

$$\Delta v^2/v^2 = (8/N)\cdot(GM/(v^2R))^2\cdot[\ln(\ell/b_{min}) + v^2\cdot f\cdot\ln(R/\ell)] \qquad b_{min} = Gm/v^2 \qquad \ell^3 = Gm/\mathscr{T}$$

b_{min} is the distance at which the spread of the star reaches that of the motion, as for friction. In the deep regime the tide at the edge is $\mathscr{T} = \Omega^2 = v^2/R^2$, so $\ell^3 = b_{min}\cdot R^2$: the smoothing length divides $\ln(R/b_{min})$ at one third, two thirds near, one third far. Without memory $v^2 = \nu\cdot GM/R$. The field of a far star and the square of the speed then both grow by the factor ν: in $\Delta v^2/v^2$ the far part stays as after Newton up to f, the near part falls by $1/v^2$. In the time R/v there remains $(8/N)\cdot\ln(R/b_{min})\cdot(2/(3v^2) + f/3)$, deep $2\cdot\ln(R/b_{min})/N$, a quarter of the Newtonian value at equal logarithm. The observer with a halo computes at the same speed only with the kicks of the stars after Newton, $(8/N)\cdot(GM/(v^2R))^2\cdot\ln(R/b_{min})$. The region thus relaxes $(2 + v^2\cdot f)/3$ times as fast as this observer expects, with ν and χ of the return, which reads the baryons once bound: at $a_N = 0.13\cdot a_0$ ($\nu = 3.3$) 3.5 times, at $a_N = 0.01\cdot a_0$ ($\nu = 10.5$) 29 times. Read as matter, the halo would be grainy: without memory it would consist of bodies of $(v^2\cdot f - 1)/(3(\nu - 1))$ stellar masses, there 1.1 and 2.9. This is a prediction for the relaxation times.

Arms. A wave in the disc turns with the shear from leading to trailing and is amplified most when its crests stand radially. Then k lies azimuthally, across the slope, and the wave arises with $G\cdot\nu$. Its critical length there is $\lambda_T = 4\pi^2 G\cdot\nu\cdot\Sigma/\kappa^2$, and around the circumference fit

$$m_T = 2\pi R/\lambda_T = \kappa^2 R/(2\pi G\cdot\nu\cdot\Sigma) = (4 - d_{eff})\cdot g_N/(2\pi G\Sigma) \qquad -d \ln \Omega/d \ln R = d_{eff}/2$$

Orbit and wave both read the full enhancement ν, and it cancels. So m_T reads only the baryons, the ratio of their field in the plane to their field perpendicular at the disc, and through the flux dimension the shear. The number of arms is m_T/X with the measure X of the swing amplification. At the threshold of surface density m_T is the same function of R/R_d for every mass, 2.16 at R_d, 3.45 at $2R_d$ (there $d_{eff} = 1.58$). Newton with the baryons alone would give 2.98 there: only the shear differs. The observer with a halo computes with a rigid mass that does not answer the wave and expects, at the same X, ν times as many, 7.45 at $2R_d$. From the number of arms the observer reads, as from the oscillation perpendicular to the disc, ν times the surface density. This is a prediction.

Memory of binding. The return of a bound region reads the baryons it has bound, the slope at the place only those in it today: $a = a_N + (\nu(a_N,B) - 1)\cdot a_N,B$, with a_N,B from the fold of those once bound. This is the decomposition of the rule after Einstein, $\nu = 1 + n_B$: Newton, the spontaneous part, reads at the place what is there today, the return n_B, the stimulated part, in the inner direction what was bound. Discs that still form stars still carry their gas: in SPARC the residual depends neither on the gas fraction nor on the type.

Tidal dwarfs. In an encounter the tide tears gas out of the disc of a galaxy whose mode has been handed over, and in the tail it gathers into dwarfs. Their gas has cooled and is bound: such a dwarf no longer breathes with the whole and, by the state of its source, reads the rule, not (1 + Q)·Newton. As a system of gas and stars it is collective and a region of its own out to its Jacobi radius, and also close to the host it reads only its own fold. It has bound only what is in it today: the memory adds nothing, and young and gas-rich it hardly depends on the mass-to-light ratio. It therefore lies on the rule with its present baryons like an isolated gas-rich dwarf, not above it like the classical satellites. The host keeps reading the gas that went into the tail in its return, the dwarf in its own: two returns read the same gas, for the return is rule, not matter. A dwarf of 10^8 M☉ of gas has $r^* = 0.36$ kpc and, on a circular orbit at a distance of 50 kpc from the Milky Way, binds out to 6.8 kpc. At 2 kpc it rotates by the rule with 36.0 km/s, $\nu = 6.03$. The observer with a halo expects Newton there, 14.7 km/s. The halo of that observer around the Milky Way is the swarm with $\gamma = 2$ and $\sigma = v_{flat}/\sqrt{2} = 120.5$ km/s, and by Newton the dwarf holds at 2 kpc only what moves slower than 20.7 km/s. Even fully captured, that halo at the place, $4\pi G\rho_X = \chi\Omega^2$, would carry only $7.2\cdot10^6$ M☉ within 2 kpc. Reading the rule in the dwarf as matter, the observer needs $5.0\cdot10^8$ M☉ of dark matter in it, which cannot get there. This is a prediction for the tidal dwarfs.

Galaxies in clusters. A galaxy whose mode has been handed over reads its own rule inside a cluster, while the tide of the cluster reads the state of its source, (1 + Q)·Newton around the gas that never cooled, $\mathscr{T} = d_{eff}\cdot\Omega^2$ with the orbital rate Ω in the cluster. Its edge is the Jacobi radius, where its field reaches the tide, deep at the edge $r_J = v_{flat}/(\sqrt{d_{eff}}\cdot\Omega)$: the same form as $R_{max} = v_{flat}/H_\infty$ on its own, with the orbital rate of the cluster instead of the rate of the end state. Against its threshold $r_J/r^* = a_0/(\sqrt{d_{eff}}\cdot\Omega\cdot v_{flat})$. It keeps its deep regime as long as the product of its speed and the orbital rate of the cluster stays below the threshold. The Milky Way ($6\cdot10^{10}$ M☉) in a cluster of $(1 + Q)\cdot M_b = 10^{15}$ M☉, taken as a point mass with $d_{eff} = 3$, thus reaches at a distance of 1 Mpc out to 50 kpc instead of 3.1 Mpc and loses its deep regime within 0.33 Mpc. This is a prediction.

What the observer sees. Whoever assumes unchanged rules converts the addition into a mass:

$$M_{apparent}/M = 1/(1 - e^{-s}) = r/r^* + \tfrac{1}{2} + r^*/(12r) - (r^*/r)^3/720 + \ldots$$

The fixed half ½ and the coefficients 1/12 and 1/720 are exactly the local coefficients of the ladder of steps. They are Bernoulli numbers: the error of the single returns against the continuum, as of the polygon against the circle. The additional potential that the observer ascribes to an invisible mass is $\Phi_X = v_{flat}^2\cdot\ln\nu(s)$, the logarithm of the enhancement. The density that is read off is an output of the observer's calculation, not a content of space. Read as mass, the share of the baryons in it changes sign at the largest additional acceleration: thin matter enhances it, dense matter reduces it.

Part V: The Whole

23 The Threshold from the Whole

The threshold $a_0 = c^2/(2\Lambda)$ is the same for every funnel. Where does Λ come from? If one reads the closure length as the entire fold of the cell, counted with its returns,

$$\Lambda = (1 + Q)\cdot\Sigma\Delta s_b(\text{cell}) \quad \Rightarrow \quad a_0 = 3c^4/(2G\cdot\rho_m\cdot L^3) \quad L = 27.9\ \text{Gpc}$$

with ρ_m the mean density of matter, baryons and mode, then the threshold is tied to the spatial cell L of the torus. Today this is the diameter of the last scattering surface. Read this way, Λ is the fold of the whole per direction, and the threshold becomes a mirror between part and whole: in $r^{*2} = r_0\cdot\Lambda$, r^* is the geometric mean of the fold length $r_0 = 2GM/c^2$ of the single funnel and the fold of the whole, just as ℓ_P is the geometric mean of $\bar{\lambda}$ and $3\Delta s$. **Every funnel reads its threshold from the whole.** This reading of Λ is posited. The mirror between part and whole supports it.

24 The Closure Rule and Q

In the background radiation and in the build-up of clusters, the local rule is not enough. There a restriction of the degrees of freedom of the whole acts, the **closure rule**, not a substance:

$(\partial\xi)^2 = -1$	the pressure direction of the conformal mode is closed
$\dot{z} = (I - w\cdot\Pi\perp)F$	$\Pi\perp$ projects onto the escape direction (pressure, scattering)
$w = 1 - e^{-\sqrt{(\tilde{\theta}\tau_D)}}$	weight: 1 co-breathing, 0 bound

F is the vector field of the motion, I the identity. The **mode** ξ is an oscillation form of the whole whose pressure direction is closed. The **weight** w says how strongly it acts, and it depends on the expansion rate $\tilde{\theta}$ of the region: if the region breathes along, $w \approx 1$. If it is bound, w goes to zero. As long as a region breathes along ($w \approx 1$), the mode is read like pressureless dust with the weight

$$Q = q/(1 - q) = \Omega_c/\Omega_b$$

With the measured $Q = 5.36$, $q = 0.843$. By Kac, $1 - q = \Omega_b/\Omega_m$ is the measure of the open region and $1 + Q$ the mean return time of the whole. What cosmology reads as dark matter Ω_c is this mode. Linearly this is identical with the standard model. The mode has no initial data of its own, it starts in the state of the baryons. From the turnaround point of a region onward, it passes into the local rule, about one radian of the orbit after the turnaround. The balance of whole and place closes at the **seam** $R_Q = r^*/(-\ln q)$, where the local q reaches the cosmic one ($a_N = 0.029\cdot a_0$, Milky Way $R_Q = 52.0$ kpc). There $\chi = Q\cdot(-\ln q) = 0.917$, almost the deep limit, inside the local rule acts, outside $1 + Q$ counts as the measure of the whole. In SPARC 131 points lie beyond the seam, their residual against the local rule is $+0.013 \pm 0.006$ dex, capped at $1 + Q$ they would lie 0.094 dex above. Likewise the lensing around isolated galaxies follows the local rule up to 2.6 Mpc. In clusters the mode is not handed over: their acceleration follows the measure of the whole, $(1 + Q)\cdot g_{bar}$, down into the core, not the local rule. It is the same formula with the return number at the nome of the whole, $n_B(q) = q/(1 - q) = Q$, instead of at the nome of the place, and at the seam both are equal. The mode passes over only where the phase-space sheet folds, and gas folds only once it has given off its heat: cold gas breaks up into clouds whose orbits cross, hot plasma holds itself up by its pressure, and the sheet does not fold. The plasma of clusters has never given off its heat, in the core its cooling takes 13.6 to 38 Gyr (X-COP). Because the folding cannot be undone, the return of extinct systems keeps reading what they once bound. In HIFLUGCS (10 clusters, masses from the motion of the cluster galaxies, Li et al. 2023) $(1 + Q)\cdot g_{bar}$ lies +0.03 dex on the data over all 108 points ($\chi^2 = 601$), the local rule 0.17 dex below ($\chi^2 = 4173$). In X-COP (12 clusters, hydrostatic, baryons only as gas) R_{200} lies at the seam of the clusters: the gas alone gives there $a_N = 0.036\cdot a_0$ (0.031 to 0.065) against $0.029\cdot a_0$. There $M/M_{gas} = 7.9$ (4.7 to 10.5) against $1 + Q = 6.28$, the balance holds with stars of a quarter of the gas, which are missing in the data. The local rule demands in R_{200} 1.7 times the gas in baryons, in the core ($0.1\cdot R_{500}$) 12 times, the measure $1 + Q$ there only 2.9 times. In the form: with $g = (1 + Q)\cdot g_{bar}$, at the edge R_Δ of every cluster $a_N = \Delta\cdot H^2\cdot R/(2(1 + Q))$. At the seam R_Δ therefore lies at one radius for all clusters, $R = 2(1 + Q)\cdot(\ln q)^2\cdot a_0/(\Delta\cdot H^2)$, with $1 + Q = 2\pi$ thus $R = 2\Theta\cdot(\ln q)^2\cdot(c/H)/\Delta$, for $\Delta = 200$ today 1.3 Mpc. There lies the mass $M = 8\cdot(\ln q)^6\cdot\Theta^3\cdot M_H/\Delta^2 = 2.5\cdot10^{14}\ M\odot$, a fixed part of the Hubble mass $M_H = c^3/(2GH)$. Heavier clusters have R_{200} inside the seam, lighter ones outside. Co-breathing, the mode feels no pressure: it follows the motion of pressureless dust and mixes its orbits

during infall, while the gas keeps its energy as heat. That is why it lies more concentrated than the gas. HIFLUGCS reads g/g_{bar} within 300 kpc 9.8, outside 1 Mpc 6.5. In the shape after infall (NFW) it carries the core without the stars of the central galaxy: X-COP reads the concentration $c = 4.2 \pm 0.8$ (residual in the core −0.04 dex, in the shape of the gas +0.41), HIFLUGCS $c = 5.5 \pm 2.5$ ($\chi^2 = 447$ instead of 601, within 300 kpc −0.04 instead of +0.21 dex, errors from leaving out one cluster at a time). Computed from the infall itself, dust in the breathing of the model falling onto a peak of its spectrum, it is $c = 4.0$ (3.3 to 5.2), and with it the residual in X-COP in the core is −0.02 dex. In spherical shells the measure lies between two limits: around the peak the mean linear fold is flat inside, and the inner shells turn around together. Falling radially, the profile becomes steeper than NFW inside, mixed in all directions at the first fold of phase space it becomes flatter. In between, the deformation of the peak sets the measure: trace and shear are independent, at a peak of height ν above the fluctuation of the field the trace carries $\nu^2\sigma^2/3$, the shear $2\sigma^2/3$. If at the first fold every shell gives the share of the shear, $2/(\nu^2 + 2)$, of its motion to the transverse direction, the profile reads $c = 4.4$ and 5.3 for the two peaks of the runs ($\nu = 2.9$ and 3.8), like the runs of many bodies and the clusters. Every shell carries the share of the whole peak because the shells around a peak are similar ellipsoids: an outer shell of this form pulls nothing inside (Newton), every shell reads only the ellipsoid it encloses, and inside a uniformly held ellipsoid the slope is linear, its shear stands to its trace in a ratio that depends only on the axes. The peak sets the axes once on its own length. Where the gas never cooled, the mode computes like cold dark matter, and the observer reads the halo of the standard model. At recombination, linear perturbations lie above the threshold but breathe along ($w \approx 1$) and are read like dust with $1 + Q$. The local rule sets in for each region only at its turnaround.

What gas, galaxies and light read. Gas and galaxies of a cluster read the same field, that of the source in its state. In equilibrium the pressure per density carries the circular orbit, $(P/\rho)\cdot\gamma_P = v_c^2 = g\cdot r$ with $\gamma_P = -d \ln P/d \ln r$, for the gas $P/\rho = kT/(\mu m_p)$, for the galaxies, isotropic, σ^2. In the ratio $\beta_{spec} = \mu m_p\cdot\sigma^2/kT = \gamma_{P,gas}/\gamma_{P,gal}$ the rule drops out: it reads only how the two lie. Around a cluster whose gas never cooled, the temperature reads the baryons only with an edge. At R_{200}, $GM_{200} = v_c^3/(10\cdot H)$, in both forms of the mode $M_{200} = (1 + Q)\cdot M_b(R_{200})$, so $M_b(R_{200}) = (\gamma_P\cdot kT/(\mu m_p))^{3/2}/(10\cdot(1 + Q)\cdot G\cdot H)$: the exponent 3/2 and, through the edge at $200\cdot\rho_{crit}$, H, the epoch, instead of 2 and a_0 as around a galaxy with handed-over mode. At the seam both states give the same field, $v_c^2 = (1 + Q)\cdot(-\ln q)\cdot\sqrt{(GM_b\cdot a_0)} = 1.09\cdot\sqrt{(GM_b\cdot a_0)}$. If R_{200} lies there, the temperature reads the same baryons in both states. To the light the gas hardly gives off its heat: per scattering the photon gains on average $4kT/(m_e c^2)$ of its energy, so the heat passes to the light in $t = 3\mu_e\cdot m_e\cdot c/(8\mu\cdot\sigma_T\cdot u_\gamma)$, independent of density and temperature, with $\sigma_T = (8\pi/3)\cdot(\alpha\cdot\lambda\!\!\!^{-}_e)^2$ and u_γ the energy of the background radiation per volume. For ionised gas of the primordial mixture this is $2.3\cdot10^3$ Gyr today, shorter by $(1 + z)^4$, at $z = 1$ still $1.4\cdot10^2$ Gyr. This is a prediction for y and the dispersion of cluster galaxies.

Friction in clusters. The rule of a cluster is the sum of returns at the nome of the whole, $q = Q/(1 + Q)$. It does not depend on a_N and therefore responds equally in every direction. Both the forming and the pulling of the wake read it, and the friction counts, as in a galaxy, $\Sigma(n + 1)\cdot q^n$, here $(1 + Q)^2 = 40.5$ times Chandrasekhar with the baryons alone. At the nome of the whole the rule is a fixed factor: what reads it k times carries $(1 + Q)^k$, and in the ratio of two effects of the same order it drops out. The wake forms in the gas and in the galaxies of the cluster, and since the gas that never cooled fills the cluster, it forms everywhere, unlike between two galaxies whose mode has been handed over. The same separates groups: where their gas never cooled and fills the space between the galaxies, the galaxies sink to the centre as in a cluster and merge into one, where it cooled and lies in the galaxies, an encounter brakes only within the baryons, and the group stays loose. A galaxy whose mode has been handed over disturbs the field of the cluster beyond its Jacobi radius r_J with its baryons m_b. The observer with a halo computes with dust of density $(1 + Q)\cdot\rho_b$ and with the mass $\nu(r_J)\cdot m_b$ that the observer reads from its motion within r_J. The friction is therefore $(1 + Q)/\nu(r_J)$ times the observer's. Both are equal where r_J lies at the seam of the galaxy: there its field passes continuously into the one the cluster reads from its baryons. The Milky Way ($6\cdot10^{10}$ M☉) at a distance of 1 Mpc in a cluster of 10^{15} M☉ reaches out to 49.6 kpc, almost to its seam at 52.0 kpc, and the ratio is 1.04. If r_J moves in to the threshold, it is $(1 + Q)\cdot(1 - 1/e) = 4.0$: near the centre of the cluster the rule brakes a galaxy four times as strongly as the observer expects, and it sinks faster to the central galaxy. This is a prediction.

Waves in the gas of a cluster. A compression of the gas that never cooled pulls in through the rule of the cluster, with $(1 + Q)\cdot G$ in every direction, and runs by the trace of a wave with $\omega^2 = c_s^2\cdot k^2\cdot(1 - q)$, $q = 4\pi G\cdot(1 + Q)\cdot\rho_{gas}/(c_s^2\cdot k^2)$ with $c_s^2 = \gamma\cdot kT/(\mu m_p)$. The same slope holds the gas in equilibrium, $4\pi G\cdot(1 + Q)\cdot\rho_b\cdot r^2 = v_c^2\cdot\alpha_M$ with $\alpha_M = d \ln M/d \ln r$, and the pressure carries the circular orbit, $v_c^2 = \gamma_P\cdot kT/(\mu m_p)$. Therefore the rule drops out: for the longest wave that reaches out to the radius, $k = 2\pi/r$, $q = f_{gas}\cdot\alpha_M\cdot\gamma_P/(4\pi^2\cdot\gamma)$, with the share f_{gas} of the gas in the baryons. Isothermal ($\alpha_M = 1$, $\gamma_P = 2$), with $\gamma = 5/3$ and stars of a quarter of the gas, $q = 0.024$: the wave turns, and the gas amplifies it by 2.5 %. Even made entirely of gas and with $\alpha_M = 3$ it builds up only at $\gamma_P = 4\pi^2\cdot\gamma/3 = 22$. A cluster in equilibrium therefore does not fold its gas through its own slope, only when it gives off its heat, and only then is the mode handed over. This is a prediction.

Voids. A void never turns around and is not a region. Its $\tilde{\theta}$ always lies above that of the mean, and with it its weight w. Its mode is never handed over: as a co-breathing perturbation it is read like dust with 1 + Q, nonlinearly too, as in the standard model. Its galaxies and those of the walls are regions of their own, bound at most out to R_{max}. The void is the rolling circle with an imaginary angle, $\theta = i\vartheta$: the mirror $\Phi \to -\Phi$ reverses the sign of the energy, cos becomes cosh and the circle a hyperbola, $r = A\cdot(\cosh \vartheta - 1)$, $t = B\cdot(\sinh \vartheta - \vartheta)$, $1 + \delta = (9/2)\cdot(\sinh \vartheta - \vartheta)^2/(\cosh \vartheta - 1)^3$ and linearly $\delta_L = -(3/20)\cdot(6\cdot(\sinh \vartheta - \vartheta))^{2/3}$. Linearly the mirror is exact, $\delta_L \to -\delta_L$. Nonlinearly the turnaround breaks it: the bound region closes after the full turn, the void never does. Inside it expands faster than outside, and at a sharp edge its shells overtake the outer ones. There the wall forms. Outside the edge the mean perturbation within a shell falls as 1/M, and the shells cross where the comoving volume $M/(1 + \delta)$ stops growing with M, at $d \ln(1 + \delta)/d \ln|\delta_L| = -1$. With $d \ln(1 + \delta)/dt = 3\cdot(H - H_v)$ and $d \ln|\delta_L|/dt = f\cdot H$, $f = d \ln D/d \ln a$ the growth rate of the co-breathing perturbation, the edge crosses at $H_v = (1 + f/3)\cdot H$, and the wall flows outward at $(f/3)\cdot H\cdot R$. Without restoring (f = 1) this is exact, $H_v = (4/3)\cdot H$ at $8\cdot(\cosh \vartheta - 1)^2 = 9\cdot\sinh \vartheta\cdot(\sinh \vartheta - \vartheta)$, $\vartheta = 3.49$: linearly $\delta_v = -2.72$, larger in magnitude than δ_c of the bound region, with $1 + \delta = 0.20$ at 1.70 times the radius its mass would fill at the mean density. A void crossing today in the breathing of the form part ($\Theta = 1.020$) crosses with the restoring at the same δ_v and $1 + \delta$, and $H_v = (1 + f/3)\cdot H$ holds to 0.1 %. This is a prediction for void profiles and outflows.

Web of the axes. Co-breathing, every baryon follows the deformation of the linear folding, $x = q - D\cdot\nabla\psi$ (Zeldovich). The eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3$ of $T_{ij} = \partial_i\partial_j\psi$ are three axes, each by its sign one of the three kinds of the shell: $\lambda > 0$ rotates and falls at $D\cdot\lambda = 1$, $\lambda = 0$ shears, $\lambda < 0$ dilates. The axes fall one after another, the largest first, and the number of positive eigenvalues is the form: one the wall, two the filament, three the knot, none the void. A plane wave of the folding has only one eigenvalue, $T_{ij} = \delta\cdot\hat{k}_i\cdot\hat{k}_j$: its round part, the trace, carries 1/3 of $\Sigma\lambda^2$, the shear 2/3, the same third as the folding per direction. Superposed Gaussian and isotropic, $\langle T_{ij}\cdot T_{kl}\rangle = (\sigma^2/15)\cdot(\delta_{ij}\cdot\delta_{kl} + \delta_{ik}\cdot\delta_{jl} + \delta_{il}\cdot\delta_{jk})$ with σ^2 the variance of the co-breathing linear perturbation, and trace and shear are independent. By mass, all three eigenvalues are positive for the fraction $P_3 = 1/2 - \arctan(\sqrt{5})/(2\pi) - \sqrt{5}/(3\pi) = 0.080$, two for 0.420, one for 0.420, none for 0.080, for every spectrum and at every smoothing scale. Independent signs would give the pure forms one eighth each. The trace gives all axes the same sign, the shear with zero sum different ones. The shear outweighs and shifts mass from knot and void into filament and wall. The mirror $\Phi \to -\Phi$ is $\lambda_k \to -\lambda_{4-k}$, its fixed point is the middle axis: on average $\lambda_2 = 0$, it shears like the middle shell, and exactly half of the mass has at least two positive eigenvalues. On average $\lambda_1 = -\lambda_3 = 3\sigma/\sqrt{(10\pi)} = 0.535\cdot\sigma$. An axis can fall only while the linear perturbation grows, up to D_∞: computed linearly to today, it falls only with $\lambda > 0.71$. The fractions by sign are the form, the fall reaches them only on scales with σ well above that. Along the falling axes the form turns around, along the open ones it dilates faster than the mean. After the fall the volume of wall and filament therefore keeps growing, $\tilde{\theta} > 0$: as a whole they do not turn around, like the void they are not regions, their mode is not handed over, it acts like dust with 1 + Q. Only the third axis closes the volume, only the knot turns around as a whole and stays, only the knot is a region. If its gas cools, it reads the rule with $d_{eff} = 3 - \chi$, deep 2, if the gas stays hot as in clusters, the mode acts like dust there as well. The rule thus acts only at the end of the sequence, and the dimensions 1, 2, 3 become 1, 2, 2 in the deep, not 1, 1.5, 2 as for forms whose mode has been handed over: around the gas of a filament that does not lie in its own knots, the field falls as 1/r by $(1 + Q)\cdot$Newton, $v^2 = 2(1 + Q)\cdot G\mu_b$ with its line mass μ_b, as around the cooled knot with $v^4 = GM_b\cdot a_0$. What lies in a wall or filament on one scale falls into its own knots on a smaller one. This is a prediction for lensing and orbits around filaments.

Fold per photon. Light and baryons dilute equally with the cell, fixed in time is the fold of the baryons per photon, $\varepsilon_b = \rho_b\cdot c^2/n_\gamma = 0.574$ eV (from ω_b and T_{CMB}). The number of photons is the stage d = 2 of the Bose form, $n_\gamma = (2\zeta(3)/\pi^2)\cdot(kT/\hbar c)^3$, their energy the stage d = 3, per photon $(\pi^4/15)/(2\zeta(3))\cdot kT = 2.70\cdot kT$, with the neutrinos ($N_{eff} = 3.046$, each 7/8 at $(4/11)^{1/3}$ of the temperature) 1.69 times that. Radiation and fold are equal where this energy per photon reaches the fold per photon with its returns, $(1 + Q)\cdot\varepsilon_b$, at $kT_{eq} = 0.80$ eV: a temperature of the whole, $1 + z_{eq} = kT_{eq}/kT_{CMB}$ reads it against today's. After recombination the gas holds itself with its heat. As long as it carries the heat of the light, the spacing of its particles, $\ell = n^{-1/3}$, is a fixed multiple of the thermal length $\lambda_H = \hbar c/kT$, $(\ell/\lambda_H)^3 = (\pi^2/2\zeta(3))\cdot n_\gamma/n$, $\ell = 2.0\cdot 10^3\cdot\lambda_H$. Breathing along, the slope reads $(1 + Q)\cdot$Newton, $\ell_J^2 = kT/(4\pi G\cdot(1 + Q)\cdot\rho_b\cdot m)$, and the Jeans mass is fixed in time, $M_J = (1 + Q)\cdot\rho_b\cdot\ell_J^3 = (\ell/\lambda_H)^{3/2}\cdot m_{Pl}^3/((4\pi)^{3/2}\cdot\sqrt{(1 + Q)}\cdot m^2)$: the mass scale of the stars, stretched by $(\ell/\lambda_H)^{3/2}$, for neutral gas of the primordial mixture ($m = \mu\cdot m_H$, $\mu = 1/(X + Y_p/4) = 1.225$) read with $1 + Q\ M_J = 1.0\cdot 10^3\ M\odot$. Before, the light holds, its pressure u/3 gives the speed $c/\sqrt{(3(1 + R))}$ with $R = 3\rho_b/(4\rho_\gamma)$, and at the last scattering surface $M_J = 2.8\cdot 10^{16}\ M\odot$, $\ell_J = 0.43\cdot c/H$, almost the Hubble length. With the change of holder the Jeans mass falls there by $(\mu\cdot m_H\cdot c^2/(3(1 + R)\cdot kT_*))^{3/2} = 2.8\cdot 10^{13}$.

Cooling by the light. The time in which ionised gas gives its heat to the light is $\propto T^{-4}$ with T the temperature of the light: the light scatters at the winding of the electron, $\sigma_T \propto \alpha^2$, and carries its energy in the step d = 3, $u_\gamma \propto T^4$. The time of the breathing 1/H is only $\propto T^{-3/2}$ as long as the folding carries the breathing, because the folding per photon is fixed in time, $H^2 =$

$(8\pi G/3)\cdot(1 + Q)\cdot\varepsilon_b\cdot n_\gamma/c^2$. Both are equal at a temperature of the whole, $(kT_{ab}/m_e c^2)^{5/2} = (135/(64\pi^3\alpha^2))\cdot(n/n_e)\cdot\sqrt{((16\zeta(3)/3\pi)\cdot(1 + Q)\cdot\alpha_G(m_e)\cdot\varepsilon_b/(m_e c^2))}$ with $\alpha_G(m_e) = (m_e/m_{Pl})^2$ and n/n_e the particles per free electron, 1.93 for the fully ionised primordial mixture. This gives $kT_{ab} = 1.4$ meV (16 K), $z_{ab} = 5.0$. The cooling is winding (α^2), the breathing is folding ($\alpha_G(m_e)$). Before z_{ab} any ionised gas gives its heat to the light within one time of the breathing, however thin and however hot it is: a region whose gas is ionised as it binds cools and can fold. Afterwards gas cools only by its own radiation, and the thin plasma of the clusters, which bind later, keeps its heat. If per hydrogen only the fraction x_e of an electron is free and helium is neutral, n/n_e grows to $(1 + f_{He} + x_e)/x_e$ with $f_{He} = n_{He}/n_H$, and the gas carries the heat of the light until $(1 + z_t)^{5/2} = (1 + z_{ab})^{5/2}\cdot(n/n_e)/1.93$. For fixed x_e it still has 0.68 of the temperature of the light there, afterwards the coupling against the breathing falls with $(1 + z)^{5/2}$. When gas and light part after recombination is thus read by the residual ionisation x_e alone. This is a prediction for the residual ionisation.

Gas and light apart. If the energy of a carrier is $\propto p^n$, its pressure is $n\cdot\langle\cos^2\theta\rangle\cdot u = (n/3)\cdot u$, with u the energy of its motion per volume, and on its own its heat falls with the breathing as a^{-n}. Light is linear in momentum, $p = u/3$ and $kT \propto 1/a$. Gas is quadratic, with the spreading $(v/c)^2$, $p = (2/3)\cdot u$ and $kT \propto 1/a^2$: the sphere average counted once and twice. The thermal length of the gas, $\lambda_{dB} = \hbar/\sqrt{(m\cdot kT)}$, is the mean of its quantum length $\bar{\lambda} = \hbar/(mc)$ and the thermal length $\lambda_H = \hbar c/kT$ of the same heat, $\lambda_{dB}^2 = \bar{\lambda}\cdot\lambda_H$. The ratio $\bar{\lambda}/\lambda_{dB} = \lambda_{dB}/\lambda_H = \sqrt{(kT/(mc^2))}$ is the distance from the meeting point $kT = mc^2$, where the motion reaches c. As long as the gas carries the heat of the light, λ_{dB} grows only as $\sqrt{a}$. What the light gives for this does not cool it, because the heat of the neutral gas is $2.8\cdot10^{-10}$ of the energy of the light. Once the gas no longer carries it, λ_{dB} breathes like the wave of the light, the spacing ℓ of the particles is now a fixed multiple of λ_{dB} instead of λ_H, and T_{gas}/T_{light} falls as 1/a. The Jeans mass is then no longer fixed in time, with $\ell/\lambda_H \propto 1/a$ it falls as $a^{-3/2}$. If the spin of hydrogen is as warm as the gas, $T_S = T_{gas}$, the colder gas absorbs light in its line. The line lies deep in the Rayleigh-Jeans part, $x = h\nu_{21}/kT_{light} = 0.025/(1 + z)$, where each mode carries $n_B \approx 1/x$ photons. With $x_S = h\nu_{21}/kT_S$ and the sum of returns $\nu = 1 + n_B = 1/(1 - e^{-x})$, the absorption is reduced by the stimulated part, $1 - e^{-x_S} = 1/\nu(x_S)$, and the signal against the light is $\propto (n_B(x_S) - n_B(x))/\nu(x_S) = 1 - \nu(x)/\nu(x_S)$, with the ratio of two sums of returns, in the deep regime $1 - T_{light}/T_S$. If gas and light part at z_t, afterwards $T_{light}/T_{gas} = (1 + z_t)/(1 + z)$, and the depth of the 21-cm signal reads z_t. This is a prediction for the signal.

Sunyaev-Zeldovich. Light of the background radiation that runs through the hot plasma of a cluster scatters off electrons with the spread per direction $\delta_e = kT_e/(m_e c^2)$. To first order in δ_e the occupation n of its modes follows the Kompaneets equation, $\partial n/\partial y = x^{-2}\cdot\partial_x[x^4\cdot(\partial_x n + (T_{CMB}/T_e)\cdot(n + n^2))]$ with $x = h\nu/kT_{CMB}$ and $y = \int\delta_e\cdot n_e\cdot\sigma_T\cdot dl$, the spread counted per scattering. The uptake from the heat of the electron is $\partial_x n$, the recoil on the electron $n + n^2 = n\cdot(1 + n)$, spontaneous and stimulated as in Einstein's equilibrium. At equal temperature and without flux, $-\partial_x n = n\cdot(1 + n)$: the same equation as the fluctuation of the returns, $\langle\Delta n^2\rangle = n_B\cdot\nu = -d\nu/ds$, and the Bose form solves it, $-\partial_x n_B = \langle\Delta n^2\rangle$. If the gas is hotter than the light, $T_e \gg T_{CMB}$, the recoil recedes, and the fluctuation pushes the light upwards in ln x, $\Delta n = -y\cdot x^{-2}\cdot\partial_x(x^4\cdot\langle\Delta n^2\rangle)$. The number of photons, the step d = 2, stays, their energy grows by the fraction 4y: integrated, the step d of the fluctuation is $\int s^d\cdot\langle\Delta n^2\rangle ds = d\cdot\Gamma(d)\zeta(d)$, at d = 4 four times the step d = 3 of the returns. In the Rayleigh-Jeans part light is missing, $\Delta T/T = -2y$, above the null it is added. The null is the peak of $x^4\cdot\langle\Delta n^2\rangle$. The step $d > 2$ of the fluctuation, $s^d\cdot\langle\Delta n^2\rangle$, peaks at $s\cdot\coth(s/2) = s\cdot(2\nu - 1) = d$, the step $d > 1$ of the returns, $s^d\cdot n_B$, at $s\cdot\nu = d$: the mean reads ν, the fluctuation, a square, reads the part of the oscillator twice, $2\nu - 1 = \coth(s/2)$. At d = 4, $x_0 = 3.830$, the null lies at 217.5 GHz. $x = 2\pi\cdot\lambda_H/\lambda$ is the ratio of two loops, the thermal loop of the light with $\lambda_H = \hbar c/kT_{CMB}$ and the wavelength λ, which both breathe along. Therefore x is time-invariant: for every redshift the null lies at the same frequency for the observer, and $\Delta T/T$ does not weaken with distance.

Mass function. Smoothed over the mass M around a baryon, the breathing-along linear perturbation at epoch t has the variance $S = \sigma^2(M, t)$. With every smaller mass independent waves are added, and with a sharp cut in k the perturbation performs a random walk in S in one dimension. Every baryon belongs to the largest region whose perturbation first reaches the linear threshold of the rolling circle δ_c (without restoring), as in the escape measure of binding: a later crossing at larger S is a subbinding. The first crossing is distributed like the time of return, over scale instead of over time: $f(S) = \delta_c/\sqrt{(2\pi)}\cdot S^{-3/2}\cdot e^{-\delta_c^2/(2S)}$ has the form $T^{-3/2}\cdot e^{-\tau_D/(4T)}$, with $S/(2\delta_c^2)$ instead of T/τ_D. In regions heavier than M lies the fraction $\mathrm{erfc}(\delta_c/\sqrt{(2S)})$, as by time T the fraction $\mathrm{erfc}(\sqrt{(\tau_D/(4T))})$ of the return has arrived. Counted per logarithmic step of mass, the peak lies for every slope of the spectrum at $\sigma = \delta_c$, the typically binding mass M_z, and there it is 0.317. Likewise the return per logarithmic step of time has its peak at $T = \tau_D/2$, where 0.317 has arrived. The factor 2 of Press and Schechter is the mirror at the threshold: every path that has crossed it and lies below again has its mirror image above. The same mirror stands in the arrived fraction of the return. Without it only half of the baryons over all scales would find a region, with it every one, as every return without loss arrives once.

Q per radian. The measured value is Q = 5.36 ± 0.06. The model counts per radian, as in ħ and $\bar{\lambda}$: the whole is one full turn, the open part, the baryons, one radian of it (the author's derivation):

$$\nu_{cosm} = 2\pi \qquad \Omega_b/\Omega_m = 1/2\pi \qquad Q = 2\pi - 1 = 5.283$$

This lies 1.3 σ from the measurement. For a given ω_b the geometry of the whole (sound angle, sound wave and distances) reads almost only the folding of the baryons counted with the mean number of their returns, $(1 + Q)\cdot\omega_b$, and Q follows only with the open part ω_b. Taken from nucleosynthesis (D/H and Y_p, $\omega_b = 0.0219 \pm 0.0002$), entirely without the height of the peaks, $(1 + Q)\cdot\omega_b = 0.1384 \pm 0.0009$ and $Q = 5.32 \pm 0.06$: $2\pi - 1$ then lies $\Delta\chi^2 = 0.3$ away, the measured ratio $\Delta\chi^2 = 0.6$. With the state part $Q = 5.40 \pm 0.08$: the state part raises Q by as much as $2\pi - 1$ and the measured ratio lie apart. In the model itself, with H_0 from a_0 and the maximum of the dark energy at the saturation of the loop, all precision data read $Q = 5.36 \pm 0.05$ and $a_0 = 1.076 \pm 0.006$: $2\pi - 1$ lies $\Delta\chi^2 = 2.4$ away. Independently the clusters read the same measure: with the mode in the shape after infall HIFLUGCS reads $Q = 5.5 \pm 0.3$, with the concentration from the infall 5.6 ± 0.2. Geometry and clusters read Q without contradiction to each other, to $2\pi - 1$ or to the measured ratio. The measurement carries both. From below, the form closes: the return depth of all baryons of the cell must not reach beyond Λ, $r^*/\Lambda = \sqrt{(6/(1 + Q))} \leq 1$, hence $Q \geq 5$. The measured value is $r^*/\Lambda = 0.971$, so the whole lies just below its own meeting point. The same counting per radian closes the atom: the wave of the electron closes after one turn, $2\pi\cdot a_B = h/(m_e\cdot v)$. The roles are mirrored: for the whole, closure sets the ratio (Q) and the content sets the length (Λ). For the atom, closure sets the length (a_B) and the content, the charge, sets the ratio α. The coupling α is content, not form, like the amount of baryons.

25 The Breathing and the Dark Energy

The growth of distances is the **breathing** of the medium: the loops stretch, $L(t) = \bar{L}\cdot a(t)$, with a(t) the measure of the stretching and $\bar{L}$ the length it refers to. The wavelength of light is a loop length and breathes along, $1 + z = L(t_{ob})/L(t_{em})$. The loops breathe alike: beyond z = 0.023, Pantheon+ shows no preferred direction, the dipole is ΔH/H = 0.9 % (0.7 σ). A quadrupole appears only at z > 0.1, where the supernovae come from a few fields of the sky, and falls to 1.8 σ with a zero point of its own per subcatalogue.

The vacuum part, derived. The action of the field carries the same function as the rule, on the third step:

$$\tilde{Q}(s) = -4\int_s^\infty x^3\cdot n_B(x)\,dx \qquad \tilde{Q}(0) = -4\pi^4/15$$

The constant $-4\pi^4/15$ depends neither on single funnels nor on the smoothing: it is the restoring of the closed whole. In the action it appears with the measure of the inner direction, $1/\ell_0^2$ with $\ell_0 = 2\Lambda$, doubled for the outward and return path of the return and as a restoring with reversed sign: $(2/\ell_0^2)\cdot 4\pi^4/15$, twice a constant per closure length, $\Lambda_s = \pi^4/(15\Lambda^2)$. Read as an energy density, with $\Lambda = c^2/(2a_0)$:

$$\rho_s = \Lambda_s\cdot c^2/(8\pi G) = \pi^3 c^2/(120\cdot G\cdot\Lambda^2) = \pi^3 a_0^2/(30Gc^2)$$
$$\Omega_s = \rho_s/\rho_{crit} = (\pi^2/45)\cdot\Theta^2 \qquad \Theta = 2\pi a_0/(cH_0) = 1.020 \qquad \Omega_s = 0.228$$

Θ is the ratio of two loops, the thermal length of the horizon $2\pi c/H_0$ to the loop of the inner direction $\ell_0 = 2\Lambda$. They are equally long when three vacuum parts carry $\Omega_{DE} = \pi^2/15 = 0.658$, today nearly so. This is a vacuum part with no number other than a_0, with the equation of state $p = -\rho c^2$ ($w_{DE} = -1$, this w_{DE} is not the weight w of the mode). Because Λ has been fixed since z ≈ 10, it is constant. Early on, when radiation contributed more to the fold, a_0 was smaller and the vacuum part is absent, as the background radiation requires. If it alone carried the expansion, this would run into the final state $H_s = H_0\cdot\sqrt{\Omega_s} = 32.1$ km/s/Mpc, with three parts into $H_\infty = c\cdot\sqrt{\Lambda_s} = 55.6$ km/s/Mpc. Because the fold is fixed, Λ, a_0 and ρ_s remain fixed while the loops stretch. Θ runs from 0 to the final value 1.23 and stands at 1.020 today. The two coincidences of today, $a_0 \approx cH_0/2\pi$ and L at the diameter of the last scattering surface, are one: from $L^3\rho_m = 3c^4/(2Ga_0)$ follows $LH/c = (8\pi^2/(\Theta\cdot\Omega_m))^{1/3}$, and with three vacuum parts $\Omega_m = 1 - (\pi^2/15)\cdot\Theta^2$. Without radiation L/D_S thus depends only on Θ, and the cell reaches the last scattering surface at Θ = 1.020, for every a_0. With radiation it is Θ = 1.024 to 1.030 (a_0 = 0.9 to 1.2), shortly after the loop of the horizon has caught up with the inner loop. That is the present inflation state of the balloon, not a number to be hit. The whole has three loops, one per direction: the cell L (tangential, the circuit of light), the thermal loop of the horizon $2\pi c/H$ (parametric) and the inner loop ℓ_0 (transversal). Each two become equal at Θ = 1 (horizon and inner), 1.012 (cell and inner, $L/\ell_0 = \Theta^{2/3}/(\pi\cdot\Omega_m)^{1/3}$) and 1.023 (cell and horizon, $LH/2\pi c = (\pi\cdot\Theta\cdot\Omega_m)^{-1/3}$). Today, at Θ = 1.020, the whole lies inside this window of 2 %, its three directions almost coincide. In the breathing of the form part the window lasts 0.68 billion years: it opened 0.58 billion years ago

($z = 0.041$) and closes in 0.10 billion years. At a fixed threshold $\Theta \propto 1/H$ is a clock of the whole for the observer breathing along. Without radiation the form part breathes by $\dot{H} = -(3/2)\cdot(H^2 - H_\infty^2)$, so $H = H_\infty\cdot\coth(3H_\infty t/2)$ and $\Theta = \Theta_\infty\cdot\tanh(3H_\infty t/2)$ with $\Theta_\infty = \sqrt{15}/\pi$. With $H_\infty\cdot\tau_D = 2\pi/\Theta_\infty$ the time, counted in return times, is a function of Θ alone: $t/\tau_D = (\sqrt{15}/3\pi^2)\cdot\operatorname{artanh}(\Theta/\Theta_\infty)$, at $\Theta = 1.020$ today 13.8 billion years, $H_0\cdot t_0 = 0.95$. The deceleration is the rate of the clock with the shell, $-\ddot{a}\cdot a/\dot{a}^2 = -1 + d\ln\Theta/d\ln a$, in the form part $\frac{1}{2} - (3/2)\cdot(\Theta/\Theta_\infty)^2$, today −0.53. The stretching accelerates as soon as the form part carries one third ($\rho + 3p/c^2 = 0$, the pressure counts in three directions), at $\Theta = \Theta_\infty/\sqrt{3} = 0.712$. Form part and fold are equal at $\Theta_\infty/\sqrt{2} = 0.872$. Both stages depend only on Θ, not on a_0. It is itself a mirror: its energy density is the geometric mean of the Planck density and the quantum density of the closure length, $\rho_s\cdot c^2 = (\pi^3/30)\cdot\sqrt{(c^7/(\hbar G^2)\cdot\hbar c/\ell_0^4)}$. $\hbar$ drops out. In Planck units it therefore lies at ε_L^2, $\Lambda_s\cdot\ell_P^2 = (4\pi^4/15)\cdot\varepsilon_L^2 = 0.94\cdot10^{-122}$, exactly midway between one and ε_L^4. The often-quoted vacuum energy that is "10^{122} times too small" is this midpoint.

Three loops. The whole too splits into the triad: the centre is the flux out of the sources, the fold of the baryons. The vortex is the return at the place, the dark matter. The background is the winding classes of the $b_1 = 3$ loops of the torus, the restoring: the **dark energy**. Each loop restores the share of the fold in its direction, because $\Delta s = GM/(3c^2)$ is the share of one of the three spatial directions, and together this is the whole fold, $3\cdot\Sigma\Delta s_b = \Sigma GM_b/c^2$, nothing twice, nothing missing:

$$\rho_{DE} = b_1\cdot\rho_s = 3\cdot\pi^3 a_0^2/(30Gc^2) \qquad \Omega_{DE} = b_1\cdot(\pi^2/45)\cdot\Theta^2$$

What counts is the length of each loop (magnitude), not its winding (phase). The weight is a quantity of the inner direction and holds for every shape of the cell. The 3 of the fold per direction ($d = 3$, at the place) and the 3 of the loops ($b_1 = 3$, in the whole) are two roles of the same dimension. Confusing them would be an observer's fallacy.

One threshold for galaxies and cosmos. Like dark matter, dark energy thus depends only on a_0. From ω_b, ω_c, T_{CMB} and a_0 follows the breathing: with $a_0 = 1.06\cdot10^{-10}$ m/s²: $H_0 = 67.2$ km/s/Mpc and $\Omega_{DE} = 0.685$, with $1.1\cdot10^{-10}$: $H_0 = 68.9$ and 0.700. Conversely, reading a_0 from the measured distances alone (Pantheon+, 1590 supernovae, DESI DR2 BAO, with ω_b and ω_c from Planck) gives $a_0 = 1.09\cdot10^{-10}$ m/s² (1.07 to 1.11), with $H_0 = 68.4$. Rotation curves (1.11), sound wave (1.10) and distances (1.09) read the same threshold: the threshold of the single funnel is that of the whole. The breathing, read after Planck, lies below at 1.065. That is at the same time the upper limit for a positive state part and lies within the band of the galaxies. Like Q, only a ratio without a clock is fixed in time: $k_{DE} = \rho_{DE}\cdot Gc^2/a_0^2$, the vacuum part has $k_s = \pi^3/30 = 1.034$. Quantities with H_0 or the age of the universe hold only in one epoch. Today $\Omega_{DE}/\Omega_s = 3.03$ (read after Planck).

In the distances. The form part, the restoring alone with $w_{DE} = -1$ (the state part, which depends on the state of binding, follows in Section 26), with no free number ($\Omega_m = 0.315$), lies only $\Delta\chi^2 = 2.0$ above the fitted standard model ($\Omega_m = 0.304$), 1.4 σ. With the sound angle and $Q = 2\pi - 1$ the data require a dark energy that weakens late (equation of state $w_{DE} = w_0 + w_a\cdot z/(1 + z)$ with $w_0 = -0.85$, $w_a = -0.49$), $\Delta\chi^2 = 8.5$ against the form part alone: the direction of the state part. The maximum of the dark energy then lies at $z = 0.43$ (0.34 to 0.55), between the onset of the accelerated stretching at $z = 0.67$ and the equality of form part and fold at $z = 0.32$. Without any calibration, because r_d and H_0 cancel, DESI reads the shape of the expansion in the ratio of transverse to radial distance, D_M/D_H, at six redshifts ($z = 0.51$ to 2.33). The form part meets it with $\chi^2 = 6.8$. The absolute scale has no free number either: early on the mode acts linearly like dust, the **sound horizon**, the distance the sound wave travels in the early plasma until it ends, follows from ω_b and ω_c alone ($r_d = 147.1$ Mpc), and H_0 and Ω_m follow from a_0. DESI then reads $a_0 = 1.10\cdot10^{-10}$ m/s² (1.09 to 1.11), with $H_0 = 68.9$ at $\chi^2 = 10.4$ for 13 values. SPARC reads the threshold the same way when the distances of 79 of the 141 galaxies come from the breathing of the whole instead of from the Hubble flow with $H_0 = 73$: $a_0 = 1.085$ (0.98 to 1.19), at $Y_d = 0.66$. If the sound angle reads with $Q = 2\pi - 1$, it gives $a_0 = 1.08$ ($H_0 = 68.0$), and the sound wave lies $\Delta\chi^2 = 5.9$ from it instead of 21.4 with the measured ω_c: the gap between breathing and sound wave comes from the inserted Q, not from the threshold. Sound angle, sound wave and distances together read $a_0 = 1.088$ (1.084 to 1.092), at $H_0 = 68.2$ and $\Omega_m = 0.30$. The standard value 1.06 is the breathing with the measured Q.

At the background radiation (Planck 2018 $\text{plik}_{\text{lite}}$ with TT below $\ell = 30$, 615 bins, $\tau = 0.054 \pm 0.007$ and calibration after Planck, lensing Planck PR4, together with DESI DR2, Pantheon+, growth $f\cdot\sigma_8$ from SDSS DR12 and eBOSS DR16, D/H and Y_p): early on the mode acts linearly like dust. The spectra of the background radiation (Planck), its lensing, DESI, Pantheon+, the growth and the light nuclei are therefore computed by the standard calculation with $\omega_c = (2\pi - 1)\cdot\omega_b$ and H_0 from a_0, with one free number fewer than the standard model. On the spectra alone the model lies $\Delta\chi^2 = 1.0$ above the standard model, on all data 0.01. From Planck's spectra alone it meets DESI with $\chi^2 = 18.6$ (13 values), the standard model with 31.4. With the state part (w_0 and w_a free) it gains $\Delta\chi^2 = 9.6$: $w_0 = -0.84 \pm 0.05$, $w_a = -0.56 \pm 0.18$, the maximum of the dark energy lies at $z = 0.40 \pm 0.07$, and the threshold reads 1.070 ± 0.013, the standard value only $\Delta\chi^2 = 0.8$ away. If the maximum lies where the

loop saturates, where three parts catch up with the fold (w_0 then follows from w_a, one number fewer), the data meet this just as well and read the dark energy today as $a_0 = 1.085 \pm 0.003$, like the geometry of the whole. The maximum then lies at $z = 0.32$. What is read is form and state together: as the form part with the saturation at the standard value the data meet it just as well, and the state part today is 5.4 ± 0.7 % of the form part. So the model with free a_0 has as many free numbers as the standard model and lies $\Delta\chi^2 = 8.7$ below it.

H_0 of the ladder. Calibrated with Cepheids, Pantheon+ instead reads $H_0 = 73.5 \pm 1.0$, in every interval up to $z = 1$ between 73.0 and 74.5. This is no state of the stretching. The restoring is the same at every place, locally only a void stretches faster. Against the breathing of all precision data ($H_0 = 68.25$) it would have to be, read linearly, at least $\delta_L = -0.41$ deep around the observer and measure 0.59 Gpc in radius, 43 times the fluctuation of the spectrum on the radius its mass would fill at the mean density. Beyond its edge space would stretch with the breathing again. Pantheon+ shows this jump for no edge up to $z = 0.3$. If the edge lies further out, the void no longer helps: at the same z, from 0.25 to 1, sound wave and supernovae measure the same path, since light breathes along and its number is kept. Still the sound wave ($r_d = 147.4$ Mpc) reads $H_0 = 68.3 \pm 0.2$, the Cepheids 73.2 ± 1.1. What differs is the zero point of the supernovae, which the ladder places 0.15 mag fainter. The rule does not change it, every rung of the ladder lies far above the threshold.

26 The State Part

Besides the fixed form part there is a part that depends on the state: the **escape measure of binding**. Binding makes escape directions of the state space inactive. The inactive directions pass into the return and act as a stretching of the loops. The same measure acts at once as binding in the region and as stretching in the whole, two sides of the same fold. The state part is carried by the baryons with orbits of their own, the bound gas whose mode has been handed over, not the stars (a star is a single funnel) and not the plasma of the clusters, in which the mode has not been handed over. Both parts hang on the baryons, on different sides: the vacuum part on their fold, with $\Lambda = (1 + Q)\cdot\Sigma\Delta s_b$ one has $\rho_s \propto 1/((1 + Q)\cdot\Sigma\Delta s_b)^2$, more fold means a longer return and a weaker restoring, whether gas or star. The state part on their binding. In SPARC ($Y_d = 0.67$), below $v_{flat} = 40$ km/s 72 % of the baryons are gas, above 150 km/s 12 % (median): the small, gas-rich regions carry the state part. What is counted is the accessible state space, not the place: per baryon the largest region counts, a sub-binding within it adds no measure. Hence only regions that are themselves the largest can lose, single galaxies, not galaxies in groups.

Measure up to the turnaround and lower limit. Counted up to the edge of the turnaround, the inactive measure of a region in the deep regime is $(R_{ta}/r^*)^{3/2}$, with the turnaround radius $R_{ta} \propto M^{1/3}$ and $r^* \propto M^{1/2}$, per baryon $\propto M^{-1/4}$: small and late-bound regions carry more. In the quantities of the model it is $a_0/(\sqrt{\kappa}\cdot H\cdot v_{flat})$, with $R_{ta}^3 = GM/(\kappa\cdot H^2)$: the orbital rate at the threshold, $v_{flat}/r^* = a_0/v_{flat}$, against the rate of the breathing, hence $(c/v_{flat})\cdot\Theta/(2\pi\cdot\sqrt{\kappa})$, tangential against parametric. Until the turnaround the region breathes along and carries (1 + Q) times its baryons, hence $\kappa = \Delta/(2(1 + Q))$ with the compression Δ of the shell against the critical density. If it turns around today, with restoring $\kappa = 0.3$, without it, with $\Delta = (3\pi/4)^2$ after half a turn, $\kappa = 0.44$. It grows with Θ like the form part and with the inverse of the speed, summed over all regions the state part follows $\Sigma\, M_{gas}/v_{flat}$. The lower limit is set by the gas itself: only what thermal motion holds becomes bound, in the deep regime $M_{Gas} = (81/4)\cdot\sigma^4/(Ga_0)$, independent of density. With the gas pressure of the slow dwarfs (SPARC, 8 to 10 km/s), $M_{Gas} = 0.6$ to $1.4\cdot10^7$ M☉. Written without mass, gas that carries its field itself holds together only with $\sigma < (4/81)^{1/4}\cdot v_{flat} = 0.47\cdot v_{flat}$. With 8 to 10 km/s there is no disc whose gas carries itself below $v_{flat} = 17$ to 21 km/s. Thus the heat of the gas bounds the measure per baryon from above, $c/v_{flat} \le 0.47\cdot c/\sigma_{gas}$, in the field of the stars $c/v_{flat} \le 0.58\cdot c/\sigma_{gas}$: the smallest regions that still hold gas carry the largest measure. In SPARC the smallest measured rotation is 17.8 km/s, the smallest flat one 18.6 km/s.

Size from the counted gas. Counted within the seam, like the escape measure of binding, every region lies at the same threshold $s = -\ln q$ and makes the same measure inactive per bound baryon, independent of its mass: under the rule the tori that stay within the seam have ν_G times their measure under Newton, and $\nu_G - 1 = 16.8$ becomes inactive. With the conversion of the whole, restoring times return number, the state part is $\rho_Z = 16.8\cdot F\cdot\rho_s$, with F the share of the baryons bound as thin gas whose mode has been handed over, in the cycle zero early, largest at the saturation and falling after it. Counted, 1.21 % of the baryons lie today as neutral gas in galaxies (ALFALFA), independently 1.17 % (HIPASS), at $z = 0.34$ about as much (Bera et al. 2019). For the neutral gas this is the upper limit, since gas denser than the tidal density stays a single funnel. The limit is the tidal density of the disc, $3\mathscr{T}/(4\pi G)$ with $\mathscr{T} = -dg/dR = (d_{eff} - 1)\cdot\Omega^2$: in the Milky Way ($6\cdot10^{10}$ M☉ as a point mass) this is 1.7 hydrogen atoms per cm^3 at the position of the Sun (8.25 kpc), 0.57 at $2R_d$, 0.18 at 20 kpc. Thinner gas is smoothed and carries, the denser clouds do not. Bound ionized gas is not counted in our data. With the gas in this cycle and the standard value 1.06 as form part, all precision data require $F = 0.73 \pm 0.11$ % today, six tenths of the counted gas. If all counted gas carries, the spectra of the

background radiation read the form part $a_0 = 1.043 \pm 0.003$, the distances independently 1.05 ± 0.01, both within the band of the rotation curves, and all data together 1.047 ± 0.003, $\Delta\chi^2 = 4.1$ below the standard model with as many free numbers. The standard value with all counted gas lies $\Delta\chi^2 = 17$ above the standard model. Counted within the seam, the counted gas thus suffices for the size. If the form part is the standard value, six tenths of the neutral gas lie below the tidal density of their disc. This limit is the same as that of the stability of a wave. The middle of a Spitzer layer with $G{\cdot}\nu$ has $\rho_0 = \pi G{\cdot}\nu{\cdot}\Sigma^2/(2\sigma^2)$, with Toomre's Q_T and $\kappa^2 = (4 - d_{eff}){\cdot}\Omega^2$ it is $\rho_0/\rho_J = (2/3){\cdot}(4 - d_{eff})/((d_{eff} - 1){\cdot}Q_T^2{\cdot}\nu{\cdot}(1 - \chi/2)^2)$: whether gas stays a single funnel as a cloud or is smoothed and carries is read by its Q_T. In a disc at the threshold, gas at the limit $Q_T = 1$ lies above the tidal density inside $4.2{\cdot}R_d$ and below it outside. In SPARC four tenths of the gas lie there, from V_{gas} 0.37 in the median and 0.40 by mass, from the surface density that V_{gas} requires as a thin disc likewise 0.37 and 0.40, independently from M_{HI} and R_{HI} 0.36 and 0.40. Six tenths require gas a little above the limit, $Q_T = 1.2$ to 1.4, from the surface density 1.28. The dispersion is read from the two widths of the line: a disc at the threshold gives $W20 - W50 = 1.5{\cdot}\sigma$, the edge of a step $1.68{\cdot}\sigma$. In ALFALFA, W20 − W50 lies at 19 to 28 km/s in the median for W50 = 30 to 600 km/s, in HIPASS at 26 to 30 km/s: the dispersion does not grow with v_{flat}, so Q_T falls with v_{flat}. Up to $\lg M_{HI} = 9.5$ the gas disperses with 11 to 12 km/s, the heaviest with 19 to 22 km/s, as an upper limit, since resolution, warps and motion across the orbit only broaden. With this, in SPARC 0.6 of the gas lies below the tidal density in the median from the surface density, 0.5 by mass, 0.5 to 0.6 from V_{gas} and from M_{HI} and R_{HI}. For every dispersion between a third of and three times the one read it is 0.3 to 0.8. Weighted per mass with the density of the gas from ALFALFA, as the whole counts it, it is likewise 0.5 to 0.6, so $F = 0.6$ to 0.8 % of the baryons: as much as the standard value requires, not all gas. With this F_0 held, all precision data read $\chi^2 = 2025.2$ to 2026.3, $\Delta\chi^2 = 1.5$ to 2.6 below the standard model, with one free number fewer: of the dark energy only its course remains free. Left free, the form part itself then reads $a_0 = 1.058$ to 1.064 (± 0.003), the standard value that the breathing requires. Counted up to the edge of the turnaround, the same data would require only 0.003 to 0.03 % of the baryons as thin gas. The dispersion thus separates the two countings: counted within the seam, the gas carries the size.

The whole is thus not an equilibrium but a **feedback loop**:

fold (fixed) → restoring → stretching of the loops
stretching ⊣ binding (stretching keeps regions open)
binding → inactive escape directions → return → stretching

More stretching keeps regions open and so lowers the state part: the loop limits itself. Its lag is the time until the turnaround of a region, a time in space, not τ_D. Q and the vacuum part are form (nome of the cell, Λ), fixed in time. The escape measure is state and follows the bound measure of the largest region in each case. An amount "today" is a phase of the loop, not a property of space. The measurement reads form and state as one fixed dark energy: Ω_{DE} presupposes a fixed dark energy and $\Sigma\Omega = 1$, w_0 and w_a a smooth fluid, Ω_c reads the mode ξ as dust, and along the light cone the observer sees earlier phases of the same loop. The loop saturates where dark energy catches up with matter. The vacuum part alone would catch up with it only at $z = -0.10$, three parts do so at $z = 0.30$. The small state part hardly shifts this, the moment hangs on a_0 alone. How far structure grows is a form without a_0. In the breathing of the form part a linear perturbation breathing along grows with the stretching, $D = a$, as long as the fold dominates, and reaches in the end state $D_\infty = (5/2){\cdot}a_\times{\cdot}\int_0^\infty u^{3/2}{\cdot}(1 + u^3)^{-3/2}\,du = (5/6){\cdot}B(5/6, 2/3){\cdot}a_\times = 1.44{\cdot}a_\times$, with $a_\times$ the stretching at which three parts catch up with the fold. Its state D/D_∞ depends on Θ alone, since $(a_\times/a)^3 = (\Theta_\infty/\Theta)^2 - 1$ with $\Theta_\infty = \sqrt{15}/\pi$. At $a_\times$ it is 61 %, today, at $\Theta = 1.020$, 71 %: linearly, structure grows from today on only by 41 % in the form part. After that, the backflow dominates: gas is heated above the binding of its region and releases its inactive measure, regions merge, gas turns into stars. Dark energy has a maximum and weakens afterwards. This is the direction the distances weakly show.

Two paths of backflow. *Merging:* when a region falls into a larger one, its escape directions are already bound in the parent field and cease to be escape directions. The number of turnarounds remains. Counted within the seam, this changes the state part only when gas leaves the seam in the process. Counted up to the edge of the turnaround, whether it raises or lowers the state part depends on the slope n of the spectrum: typically, regions $M_z \propto (1 + z)^{-6/(n+3)}$ bind at time z, so $R_{ta}/r^* \propto (1 + z)^{-(n+2)/(n+3)}$. Below $n = -2$ merging lowers the state part, above it raises it. The boundary lies at 3 to $4{\cdot}10^{12}$ M⊙, set by the equality of radiation and fold at $z_{eq} = 3402$. *Heat:* when gas is heated above the binding of its region, it leaves the seam and its inactive measure is released. In the field of the stars the gas is a test body and bound only with $\sigma < v_{flat}/\sqrt{3}$, at $M > 9{\cdot}\sigma^4/(Ga_0)$, if it carries its field itself, at $M > (81/4){\cdot}\sigma^4/(Ga_0)$. Gas at 100 to 300 km/s is held in the field of the stars only by regions from $6.4{\cdot}10^{10}$ to $5.2{\cdot}10^{12}$

M⊙. If gas ΔN leaves the seam, the inactive measure drops by ΔN·(ln B − ¾), with B the coefficient of the region (the author's derivation). When the vacuum part halts the growth of structure, new binding ends but heating does not: then the backflow dominates.

27 Cycle, Memory and Entropy

On the largest scale the **cycle** closes: at the bottleneck the medium condenses into fold again. The total entropy grows in every round, and the cosmos runs a spiral, not into a heat death. The cycle closes in form, not in state.

Memory. Folded space does not forget: every vortex leaves a lasting crease, and new systems use the existing grooves. The crease is form, not a store beside space. In the cosmos it is a state of the closure rule: what lasts is its weight w, which depends on the smallest expansion rate in the history. A region once bound stays bound. Memory is the number of turnarounds of a region: its phase runs only forward under a restoring slope, the weight drops as a step at the first turnaround. What is protected is the count, not the single loop.

Entropy and heat. The bottleneck is, at the place, the fold $r \le 6\Delta s$ and, in the whole, the horizon. On both levels the same form of entropy holds, $S = 36\pi k\cdot(\Delta s/\ell_P)^2 = 4\pi k\cdot(M/m_{Pl})^2$, for the Sun $1.05\cdot10^{77}$ k. The temperature of the fold is $kT = \hbar c/(24\pi\Delta s)$, that of the horizon twice as high for the same fold. Heat is here a loop length: the thermal length $\hbar c/kT$ of every closure is its loop, for the fold $24\pi\Delta s = 4\pi\cdot r_0$, for the horizon $2\pi c/H$, for the inner direction $\ell_0 = 2\Lambda$ ($kT_I = \hbar a_0/c$). At the place the slope g has the thermal length $2\pi\cdot r^2/(3\Delta s)$, one turn of its mirror image about the place (Unruh, $kT = \hbar g/(2\pi c)$). At the horizon this is $4\pi\cdot r_0$: the heat of the fold is the heat of its slope at the horizon. By the three directions they are $\bar{\lambda}$ at the place ($kT = Mc^2$), $8\pi\cdot3\Delta s$ at the fold and ℓ_0 inside, and their ratios are the three couplings: $kT_I/(Mc^2) = \alpha_I$, at the fold $kT/(Mc^2) = 1/(8\pi\cdot\alpha_G)$ and $T/T_I = (c/v_{flat})^4/(8\pi)$. The fold reaches its rest energy at $M = m_{Pl}/\sqrt{(8\pi)} = 4.3\cdot10^{-9}$ kg. The entropy is the coupling, $S = 4\pi k\cdot(M/m_{Pl})^2$. A body of size R carries at most $S = 2\pi k\cdot R/\bar{\lambda}$, one turn of its position against the quantum length (Bekenstein). The fold reaches this bound, with $R = r_0$ it is $4\pi k\cdot\alpha_G$. For gravity the heat capacity reverses its sign. When a funnel held by heat gives off folding, it becomes tighter and hotter, because its temperature is its spread: what the contraction sets free is divided by the half spread, one half leaves, the other heats. With ½kT per direction $E = -(3/2)\cdot NkT$, T the mean over its N carriers, so $C = -(3/2)\cdot Nk$. If the blocking of the half turn of the electrons holds it, C changes sign through a pole at the maximum of its temperature, where the pressure of the blocking grows as fast as the pressure the binding demands, at the limiting mass at $\hbar k_F = 1.06\cdot m_e\cdot c$. Beyond it the funnel cools as it gives off. The fold becomes shallower and hotter as it gives off: its energy counts the mass, its temperature the mirror mass, and since $M\cdot\tilde{m} = m_{Pl}^2/2$ is fixed, the mirror reverses the direction, $C = -Mc^2/T = -4\pi k\cdot M/\tilde{m} = -2S$. For the horizon $T = \hbar c/(12\pi\cdot k\cdot\Delta s_H)$ with $\Delta s_H = c/(6H)$. Δs_H is the fold of the Hubble sphere at the critical density: Friedmann's $H^2 = 8\pi G\cdot\rho_{crit}/3$ says that this sphere is its own fold, $6\Delta s_H = c/H$. Order grows inside only as long as more entropy is exported than produced. This is syntropy, $Syn = S_{eq} - S$, with S_{eq} the entropy in equilibrium. It changes as $dSyn/dt = J - \dot{S}_i$, with the entropy export J and the production $\dot{S}_i \ge 0$. At the bottleneck of the whole the medium condenses into fold again. Λ grows and the vacuum part falls. The entropy belonging to the vacuum part depends, like that of a fold, only on the square of a fold length, $S_\Lambda = (45/\pi^3)\cdot k\cdot(\Lambda/\ell_P)^2$, and requires $\dot{\Lambda}/\Lambda \ge \dot{S}_i/(2S_\Lambda)$. This is the entropy of the horizon set by the vacuum part alone, $S_\Lambda\cdot\Lambda_s\cdot\ell_P^2 = 3\pi k$ as for de Sitter, and in the number of the ladder $S_\Lambda = (45/4\pi^3)\cdot k/\varepsilon_L^2 = 1.0\cdot10^{123}$ k. This minimum rate is tiny and does not act on the dark energy today. For gravity, compression increases entropy: the fold is, in the small, the place of greatest entropy and, in the large, of the simplest form. The thermal lengths of the slope and of the fold are counted in t from outside, with g as half the slope of $c^2\cdot(d\tau/dt)^2$ over r. A resting clock measures these loops with its own rate: it feels the slope as $g/(d\tau/dt)$, and a resting body in equilibrium has one thermal length, a loop of the whole body in t, read at a place as $T\cdot d\tau/dt = T_\infty$ (Tolman). In the depth the equilibrium is warmer, and at the horizon of the fold its temperature at the place grows without bound. The pressure carries the same rate: in a uniformly held body $T/T_R = 1 + P/(\rho c^2)$, and the temperature of equilibrium reads the pressure in rest energy. A neutron star of 1.4 M⊙ and 12 km, held equally in every direction and not denser outward, is in equilibrium at least 13 % warmer in the centre than at the hull.

28 The Web

The torus has Euler characteristic zero. Therefore every form of the slope obeys

$$\text{funnels} - \text{filament saddles} + \text{wall saddles} - \text{voids} = 0$$

The closed medium enforces the web of nodes, filaments, walls and voids. Between two funnels there is always a filament saddle, between two voids a wall saddle. Fewer than 1, 3, 3, 1 is impossible, the Betti numbers of the torus: the simplest form, cos x + cos y + cos z, has exactly one funnel, three filaments, three walls and one void. The three filaments of the smallest web are the three loops that carry the dark energy. Reversing the slope exchanges funnel with void and filament with wall: the web is a mirror that only the fold breaks, because only the funnels carry fold.

In the isotropic Gaussian field the mirror holds on average, and the form of the slope has the same ratios for every smoothed spectrum: at a point without slope the curvature $\partial_i\partial_j\Phi$ is independent of the slope, its eigenvalues repel, and every point counts with $|\lambda_1\lambda_2\lambda_3|$. Per funnel there are $(29 + 6\sqrt{6})/(29 - 6\sqrt{6}) = 3.055$ filament saddles, as many wall saddles and one void. The Euler sum does not fix this number, because the mirror makes it zero by itself. The number comes from the curvature. From every filament saddle two lines run down into funnels, so on average 6.11 filaments end at a funnel. In the smallest web there are $2b_1 = 6$, each of the three filaments ends with both ends in the one funnel. This is a prediction for the linear web, testable with counts of filaments.

Part VI: Reading and Testing

29 The Observer's Fallacy

When the rule of the whole changes, a measured value at a place deviates, and the observer invents a local thing as its carrier. In the model nothing behaves differently. The rule of the medium has changed. For every quantity one must therefore ask in which direction of view it holds: **at the place** (in space), **in the inner direction** (orthogonal to space: return, Λ, ϖ) or **for the whole** (the torus as a cell). And in which state: breathing along or bound, in which epoch, for which observer. A test body reads the field in the state of its source, not in its own.

Phenomenon	**read as a thing**	**as a state of the medium**
mass	substance in space	fold depth Δs
gravity	emitted force	slope g
time	a flow of its own	difference in unrolling δ
expansion	bodies fly apart	the loops stretch
dark matter	invisible particle	changed rule of motion below the threshold a_0
dark matter in the cosmos	dust with its own density Ω_c	mode ξ of the closure rule with weight Q
dark energy	a form of energy of its own	restoring of the three loops and escape measure of binding
vacuum energy	energy of empty space, 10^{122} times smaller than expected	mirror between Planck density and closure length
temperature of the return	heat	period of the return in the inner direction
field around a bound region	depends on the state of the test body	state of the source: around a galaxy with handed-over mode every shell reads the rule, around gas that never cooled $(1 + Q)\cdot$Newton
stability of a disc	rigid halo that does not answer a wave	response of the rule along the slope with its derivative $\nu(1 - \chi/2)$, across it with ν: read as matter, two densities
entanglement	signal through space	shared node in the inner direction
black hole	hole with a singularity	fold: shell at the centre, finite fold Δs

Rule instead of source. The difference between an additional substance and a changed rule can be stated precisely. Closure is an **operator** $\mathscr{P}$ acting on the generator of motion, $L \to \mathscr{P}L$, with $\mathscr{P}_q = (1 - q)\cdot\Sigma\, q^m T^m$, T the step by one return (one turn in the inner direction) and q^m its weight after m returns. It deletes no degrees of freedom and preserves the measure, while certain directions become inactive and pass into the return. In closed form $\mathscr{P}_q = (1 - q)(I - qT)^{-1}$: for $q \to 0$, $\mathscr{P} = I$, open and Newtonian. For $q \to 1$, space is closed. On the torus, applied to a power kernel, $\mathscr{P}$ is the lattice sum after Eisenstein. Its lowest series over the return steps of one direction is $\Sigma_{n\geq 1}\, q^n = q/(1 - q)$, the addition of the circular orbit. The bound measure grows only as long as a parameter wanders. A source, by contrast, adds density:

	Source (dark matter as substance)	Operator $\mathscr{P}$
where it acts	$\rho \to \rho + \rho_{DM}$	$L \to \mathscr{P}L$
potential	additive	multiplicative
localisable	yes	no
degrees of freedom	additional	existing ones become bound
time structure	without memory	memory (weight w, handover of the mode)

The density read off, $\rho_X = -\nabla\cdot(a - a_N)/(4\pi G)$, is an output of the calculation and not even unique: different distributions produce the same external field. If all tests run alike, source and operator are just two descriptions. If one test runs robustly apart, they can be told apart.

Entanglement. Two entangled particles may lie arbitrarily far apart in space. In the inner direction they share the same node. No signal travels. Both measurements act on the same node. Their correlation is the fundamental vibration of the winding, $E(\Delta) = -\cos\Delta$ with Δ the angle between the two measurement directions, and gives $S = 2\sqrt{2}$ (S here Bell's measure of correlation, not the entropy). As a property given along, $S \le 2$ would remain.

30 What Can Be Tested

- Wide binary stars remain Newtonian: stars are single funnels. At the place of the Sun a star of 1 M☉ has the smoothing length ℓ = 349 kAU, wider than the Jacobi radius of a pair of 1 M☉ (277 kAU). In general the heavier star of every pair reaches with $\ell^3 = Gm/\mathscr{T}$ at least to the Jacobi radius of the pair, $r_J^3 = G(m + m')/(2\mathscr{T})$ for a flat curve.
- The tide reads the flux dimension. On the circular orbit the tide of the host is $\mathscr{T} = -dg/dR = (d_{eff} - 1)\cdot\Omega^2$, with the revolution $d_{eff}\cdot\Omega^2$, and the epicycles are $\kappa^2 = (4 - d_{eff})\cdot\Omega^2$. The flux dimension is the slope of the field, $d_{eff} = 1 - d\ln g/d\ln R$, outside a sphere $3 - \chi$. It divides $4\Omega^2$ between tide with revolution and epicycles, after Newton 3 and 1, deep 2 and 2. For a flat curve $d_{eff} = 2$ and the tide with the revolution is $2\mathscr{T}$ as in the Jacobi radius of the pairs. A region binds up to its Jacobi radius r_J, where its own field reaches the tide of the host with the revolution, $d_{eff}\cdot\Omega^2\cdot r$ to first order in r/D. For a single funnel $r_J^3 = Gm/(d_{eff}\cdot\Omega^2)$. If a collective region is deep at its edge, its field there is v_{flat}^2/r, so $r_J = v_{flat}/(\sqrt{d_{eff}}\cdot\Omega)$, linear in the speed. If the host is deep as well, the flat speeds of both give $r_J/D = v_m/(\sqrt{2}\cdot v_M) = (m/M)^{1/4}/\sqrt{2}$ instead of $(m/3M)^{1/3}$: the edge reads, like $v^4 = GMa_0$, the fourth root of the folds. A region of 10^7 M☉ at a distance of 50 kpc around the Milky Way (point mass) binds up to 3.8 kpc, after Newton with the same baryons up to 1.9 kpc. Its return reads the baryons it has bound, the slope at the place today's: if it loses nine tenths of them as gas, the edge moves to 3.7 kpc, without this memory to 2.1 kpc.
- What the measurement reads as the dark matter fraction within a radius, $V_{DM}^2/V_c^2 = (a - a_N)/a$, is in the model the return share $q = e^{-\sqrt{(g_N/a_0)}}$, at every z with the same a_0.
- Galaxies in clusters follow the same relation inside as free ones. Only the tide disturbs them.
- When two clusters collide, pressure brakes the gas but not the mode: it runs on with the galaxies, and the lensing separates from the gas. At the galaxies it reads at least Q times the collided gas, at the gas only the gas's own fold. Measured at two cluster collisions, 1E0657−56 and MACS J0025.4−1222 (Bradač et al. 2006 and 2008): in all four subclusters the lensing peak lies closer to the galaxies than to the gas and 3.8 to 13.8 errors from the gas peak. In 1E0657−56, equal regions around the galaxies carry 16.3 and $4.3\cdot10^{13}$ M☉, while Q times the gas at the gas peaks is 10.6 and 2.2. At the gas, after the galaxies are subtracted, hardly any lensing remains, κ = 0.05 ± 0.06 and 0.02 ± 0.06, at the galaxies 0.36 ± 0.06 and 0.20 ± 0.05 (Clowe et al. 2006). Over the whole field the lensing carries 7.0 ± 0.9 times the gas: the measure $1 + Q = 2\pi$, with the stars 6.7.
- Counter-rotating stellar discs of the same galaxy follow the same relation: the one sense of circulation of the return is a sense of time, not of rotation.
- The largest additional acceleration lies at $a_N = 2.54\cdot a_0$ and amounts to $0.648\cdot a_0$. In SPARC the peak lies at $2.15\cdot a_0$ (1.56 to 2.65) with $0.75\cdot a_0$ (0.64 to 1.05), the simple form would have none.
- Dark energy weakens late ($w_0 > -1$, $w_a < 0$).
- Since $z \approx 10$ the dark energy never lies below the form part: the state part is the measure of bound regions and never negative.
- The next closure number of nuclei is 184.

- In disc galaxies the lensing mass inside is smaller than the dynamical mass, within $2R_d$ by 0.05 dex in the median, 0.08 for low surface brightness discs, 0.02 for high surface brightness discs. Outside $4R_d$ and for spheres both are equal.
- Gas just above the layer of a disc orbits more slowly than in its middle: there the orbit also reads the vertical field of the layer, $a = \nu(\sqrt{(g_R^2 + (2\pi G\Sigma)^2)}/a_0)\cdot g_R$, above a Kuzmin disc exactly the value that light reads in the plane. Within $2R_d$ of the SPARC discs this is 0.07 dex of the acceleration in the median, 8 % of the speed, 0.11 for low surface brightness discs, 0.02 for high surface brightness discs, outside $4R_d$ 0.01. Averaged over the whole layer the orbit reads the vertical field uniformly distributed between 0 and $2\pi G\Sigma$ for every profile, because the field at a height grows with the mass below it: the stars that fill the layer orbit on average more slowly than the cold gas in its middle, within $2R_d$ by 0.03 dex of the acceleration in the median, 4 % of the speed, 0.05 for low surface brightness discs, 0.01 for high surface brightness discs.
- Polar rings orbit more slowly than the disc of their galaxy: on the polar orbit the rule reads the slope at the place of the orbit. At 2 to $4\cdot R_d$ a polar ring around an exponential disc orbits with 0.90 to 0.92 of the speed in the plane, under Newton with the baryons alone with 0.86 to 0.88. At $8\cdot R_d$ it is 0.96 to 0.97.
- Hot gas in equilibrium shows the same enhancement as the stars of the same galaxy.
- At the saddle point between Sun and Earth there is no additional acceleration: both bodies remain Newtonian up to their smoothing length.
- The growth of structure departs from the standard model only below the turnaround scale, the size of the regions that detach from the expansion today (Section 26).
- No region binds beyond R_{max}. Around a galaxy whose mode is handed over, every shell at rest reads the rule of its bound baryons: there the slope of the deep regime v_{flat}^2/R balances the restoring $H_\infty^2\cdot R$, $R_{max} = v_{flat}/H_\infty$, with $H_\infty = c\cdot\sqrt{\Lambda_s} = 55.6$ km/s/Mpc, the rate of the end state of three vacuum parts. R_{max} is fixed in time, for the Milky Way ($6\cdot10^{10}$ M☉) it is 3.1 Mpc. Around a cluster whose gas never cooled only (1 + Q)·Newton acts, $R_{max}^3 = (1 + Q)\cdot GM_b/H_\infty^2$, for $M_{200} = 10^{15}$ M☉ 11 Mpc. Both edges lie at the same density. Whoever reads at the edge, from the slope $a = H_\infty^2\cdot R$, the dynamical mass $M_{dyn} = a\cdot R^2/G$ finds inside R_{max}, whichever rule gives the slope, the mean $3M_{dyn}/(4\pi R_{max}^3) = 3H_\infty^2/(4\pi G) = 2b_1\cdot\rho_s = \pi^3 a_0^2/(5Gc^2) = 1.7\cdot10^{11}$ M☉/Mpc^3, around the Milky Way as around a cluster, read fixed in time $\bar\rho\cdot Gc^2/a_0^2 = 2b_1\cdot k_s = \pi^3/5 = 6.20$. As M_{200} names the edge at 200 times the critical density, R_{max} is the edge at twice the critical density of the end state, $3H_\infty^2/(8\pi G) = b_1\cdot\rho_s$. The factor two joins two readings of the same restoring: in the expansion law as energy with w = −1, $H_\infty^2 = (8\pi G/3)\cdot b_1\rho_s$, as slope with −(1 + 3w) = 2 times that, $H_\infty^2\cdot R = (4\pi G/3)\cdot 2b_1\rho_s\cdot R$. The restoring is thus the tide of the whole, equal in every direction, $\mathscr{T} = H_\infty^2$, and $2b_1\cdot\rho_s$ is its tidal density $3\mathscr{T}/(4\pi G)$. $R_{max}^3 = GM_{dyn}/H_\infty^2$ has the form of the smoothing length $\ell^3 = GM/\mathscr{T}$: R_{max} is for a region in the tide of the whole what ℓ is for a compact funnel in the tide of its parent. A sphere at the mean of the folding runs like every shell, $\ddot r = -GM/r^2 + H_\infty^2\cdot r$, and for it $\ddot r = (\ddot a/a)\cdot r$ with $\ddot a/a$ of the whole. In the breathing of the form part the stretching therefore begins to accelerate when the mean falls below this tidal density and every sphere of the mean grows beyond the largest turnaround of its mass, at $\Theta = \Theta_\infty/\sqrt{3}$. Today ($\Theta = 1.020$) every sphere of the mean lies at $1.63\cdot R_{max}$. Only a sphere that, read with its dynamical mass, is compressed beyond 4.34 times the mean of the folding lies inside the largest turnaround of its mass, and only such a sphere can still turn around. The resting shell at R_{max} is the outermost circular orbit. With the restoring the orbit is $\Omega^2 = g/R - H_\infty^2$, and the epicycles become $\kappa^2 = (4 - d_{eff})\cdot g/R - 4H_\infty^2$: the restoring has $d_{eff} = 0$ and counts fourfold. The circular orbit stays stable as long as its angular momentum $R^2\Omega$ grows outwards, up to $(1 - d_{eff}/4)^{1/d_{eff}}\cdot R_{max}$ with $\Omega^2 = d_{eff}/(4 - d_{eff})\cdot H_\infty^2$. Around a galaxy whose mode is handed over ($d_{eff} = 2$) the last stable circular orbit lies at $R_{max}/\sqrt{2}$, for the Milky Way without neighbours 2.2 Mpc. For every mass it circulates with the rate of the end state, $\Omega = H_\infty$, once in $2\pi/H_\infty = \Theta_\infty\cdot\tau_D = 110$ billion years. Around a cluster ($d_{eff} = 3$) it lies at $R_{max}/\sqrt[3]{4}$, at $M_{200} = 10^{15}$ M☉ 7.0 Mpc, with $\Omega = \sqrt{3}\cdot H_\infty$. Between it and R_{max} the circular orbits are unstable.
- Around a cluster whose gas never cooled, every shell runs up to its turnaround with (1 + Q)·Newton and the restoring, $\ddot r = -(1 + Q)\cdot GM_b/r^2 + H_\infty^2\cdot r$ with the baryons M_b inside the shell. In the breathing of the form part the shell turning around today ($\Theta = 1.020$) therefore encloses the same compression for every mass: read with the dynamical mass $(1 + Q)\cdot M_b$, 11.8 times the mean of the fold, against $(3\pi/4)^2 = 5.55$ in the rolling circle without restoring. Today the turnaround of every cluster lies at $0.72\cdot R_{max}$ of the mass it encloses. The ratio depends on the clock Θ alone and tends to 1 as $\Theta \to \Theta_\infty$. After the turnaround the mode falls like dust through the centre and turns around a second time, at the edge of infall. If the mass inside stayed fixed, the edge would lie on the turnaround, because the restoring is fixed in time and the energy of the shell would be conserved. It moves inward because mass keeps falling in while the shell is on its way. When the growth of structure ends, the edge of infall and the turnaround move towards R_{max}. Around a galaxy whose gas cooled, the mode is handed over

already at the first crossing, at the latest on the passage through the centre, before the second turnaround. There the edge of infall is missing, the lens reads the rule of the bound baryons without a steep drop. Around a cluster it shows the edge as in the standard model. This is a prediction.

- From the deep regime of a galaxy whose mode has been handed over, a star escapes only through the restoring of the whole. Around its baryons as a point mass the potential of the rule is $\Phi = v_{flat}^2 \cdot \ln n_B(s) = v_{flat}^2 \cdot \ln(q/(1-q))$, since $-d(\ln n_B)/ds = \nu$ and hence $d\Phi/dr = \nu \cdot a_N$. Newton alone is $v_{flat}^2 \cdot \ln q$, the logarithm of the return fraction. The return adds the logarithm of the sum of the returns, $q \cdot \nu = n_B$. The binding between two places is thus v_{flat}^2 times the logarithm of the ratio of their return numbers. Outside, n_B grows like r/r^* without limit, and only the restoring sets the edge at R_{max}. From the distance r, $v_{esc}^2 = v_{flat}^2 \cdot [2 \cdot \ln(n_B(r^*/R_{max})/n_B(s)) - 1 + (r/R_{max})^2]$ with $n_B(r^*/R_{max}) \approx R_{max}/r^* = (\sqrt{15}/2\pi^2) \cdot c/v_{flat}$, so at the threshold $v_{esc}^2 \approx v_{flat}^2 \cdot [2 \cdot \ln((e-1) \cdot n_B(r^*/R_{max})) - 1]$: the escape reads the place on the ladder, c/v_{flat}, only in the logarithm. For the Milky Way as a point mass without neighbours this is $3.43 \cdot v_{flat} = 585$ km/s at the threshold and 591 km/s at the place of the Sun (8.25 kpc). After Newton with the same baryons v_{esc} would be finite at the threshold even without restoring, $\sqrt{2} \cdot v_{flat} = 241$ km/s. Around a cluster whose gas never cooled v_{esc} stays finite as after Newton, $v_{esc}^2 = 2(1+Q) \cdot GM_b/r$.

A star evaporates when relaxation lifts it above escape. Per relaxation a region loses the part of the Maxwell distribution beyond v_{esc}, $\mathrm{erfc}\, X + (2X/\sqrt{\pi}) \cdot e^{-X^2}$ with $X = v_{esc}/(\sqrt{2} \cdot \sigma)$. After Newton, with the mean escape $4\langle v^2 \rangle$ of a region in equilibrium ($X^2 = 6$), this is 0.0074. Around a region whose mode has been handed over, in the deep regime only the tide sets the edge R_e, in a host the Jacobi radius r_J, alone R_{max}, and if the region is deep at its edge, $d_{eff} \cdot \Omega^2 \cdot r_J^2 = H_\infty^2 \cdot R_{max}^2 = v_{flat}^2$. Up to there $v_{esc}^2 = v_{flat}^2 \cdot [2 \cdot \ln(n_B(r^*/R_e)/n_B(s)) - 1 + (r/R_e)^2]$. With $\sigma^2 = v_{flat}^2/\gamma$ the Boltzmann factor becomes the ratio of the numbers of returns to the power γ, $e^{-X^2} = (n_B(s)/n_B(r^*/R_e))^\gamma \cdot e^{(\gamma/2) \cdot (1 - (r/R_e)^2)}$, deep and for $r \ll R_e$ $(r/R_e)^\gamma \cdot e^{\gamma/2}$. Deep it reads no mass, only R_e/r, as the scale-invariant deep regime requires. A region that carries itself in the deep regime ($\gamma = 9/2$) thus loses per relaxation $1.0 \cdot 10^{-3}$ from $r = R_e/10$, alone at 10^7 M☉ from the threshold $1.2 \cdot 10^{-15}$. Alone a deep region hardly evaporates, only the tide of its host moves its edge close. This is a prediction.

- The form of the return can be rebuilt in the laboratory: an RC chain whose tap n_R is fed back to the input is the renewal $u(0) = a_N + u(\varpi_R)$ and gives the Bose form exactly, $\nu = 1/(1 - e^{-s})$ with $s = n_R \cdot \mathrm{arcosh}(1 + RG/2)$. At $R \cdot G = 0.01$ (R the resistance, G the conductance per link) and $n_R = 20$, $\nu = 1.1567$. This tests the equation, not gravity: in the laboratory $s \approx 3 \cdot 10^5$, there is no laboratory prediction that only the model makes.
- Docking: at one receptor k_{off} orders the modes of action, from full through partial agonists and antagonists to inverse ones. Cooperative receptors become steeper on cooling, after Curie and Weiss. Rebinding prolongs the effect strongly in a cleft or channel, hardly at dissolved targets. At equal occupancy a fast-changing drug habituates more.

31 What Is Set

Where the model takes a number from measurement, the standard model inserts a free number at the same place.

- **The reading of Λ** as the entire fold of the cell is posited, supported by the mirror between part and whole. The standard model sets Λ as a constant without origin.
- **Q** is, counted per radian, $2\pi - 1$ (the author's derivation), 1.3 σ from the measurement. The standard model inserts Ω_c/Ω_b as a measured number.
- **α and m_p/m_e** are measured numbers in the standard model of particles, and so in the model: content, not form. They follow from no equation of the model: a mirror sets only its meeting point, q_P for the winding, m_{Pl} for the fold, not the distance from it. Rule, threshold and dark energy carry neither ħ nor e, and computed on the same data with a different α and m_p/m_e, they and their readings of galaxies, clusters and distances stay the same. In the whole, α and m_e appear only where gas speaks with light, including the recombination of the early plasma, a standard calculation. Carter's relation and the distance of the electron from the closure, $\ell_0/\bar{\lambda}_e$ against $r_e/(3\Delta s_p)$, are near coincidences, not equations.
- **The course of dark energy:** the standard model reads it with two fit numbers w_0, w_a, without a cause. In the model, direction and time follow from the cycle of binding and backflow, and the data read its height.
- **The nome:** whether $e^{-r^*/r}$ or the core nome $\Delta s/r$ carries the imaginary part is the author's choice, on the data the two cannot be told apart.
- **Predictions that a measurement decides:** the ultra-diffuse galaxies, the red lenses, the shear, the motion of the Local Group, the galaxies per region and the zero point of the supernovae, which the Cepheid ladder sets 0.15 mag fainter than the sound wave.

- **Docking:** the ladder of clocks is a prediction, testable with k_{off} of known mode of action at the same receptor.

32 Conclusion

The treatise began with a funnel: a drawing in which more space within space appears as depth. From it followed slope, clock rate and energy. From the closedness of space, the return, the threshold and the rule. From the shape of the torus, the three generators mirror, threefold and full turn. The same forms recur on every level, from the Planck point through nucleus, atom, star and galaxy to the whole, tested on nuclei, atoms and substances. Galaxies, sound wave and distances read the same threshold. The whole sets the scale of the parts, and the parts fold the whole.

Quod est inferius est sicut quod est superius, et quod est superius est sicut quod est inferius, ad perpetranda miracula rei unius. (Tabula Smaragdina)

Appendix A: Symbols

Symbol	Name	Meaning
$\Delta s = GM/(3c^2)$	fold	extra length in the centre, measure of mass
$g = 3c^2\Delta s/r^2$	slope	steepness of the funnel wall, the weight we feel, $g_N = a_N$ the Newtonian one, g_m in $e{\cdot}g_m$ the magnetic charge, $g = 2$ the ratio of magnetic moment to angular momentum
$d\tau/dt = \sqrt{(1 - 6\Delta s/r)}$	clock rate	unrolling at the place against flat outer space, τ the proper time at the place
δ	spread	distance of the clock rate from one, on orbits v^2/c^2
$r_0 = 6\Delta s$	limit radius	where the clock stands still
fold	black hole	funnel whose shell lies at the centre, $R = r_0$
$\bar{\lambda} = \hbar/(Mc)$	quantum length	mirror partner of the fold
ℓ_P, m_{Pl}	Planck length, Planck mass	centre of the mirror, meeting point of fold and quantum length
ϖ	inner direction	its coordinate, the smoothing depth of the return
$\varpi_R = r^*$	return depth	depth at which the state returns, equal to the threshold radius
$\Gamma = a_N/c$, θ	losses	from the clock-rate gradient and from the expansion of the region, $\Gamma(k)$ the gamma function
$\tilde{\theta}$	expansion rate	of the region, the loss θ of the return equation
$D = GM/c$	spreading	in the inner direction
Λ	closure length	where the inner direction closes
$\tau_D = 2\Lambda/c$	return time	mean return time, the same for every mass
$a_0 = c^2/(2\Lambda)$	threshold	slope below which the return acts
$r^* = \sqrt{(GM/a_0)}$	threshold radius	transition to the background
$q = e^{-s}$, $s = \sqrt{(a_N/a_0)}$	return fraction	share that comes back
$\nu = 1/(1 - q)$	enhancement	sum of all returns
$\chi = s/(e^s - 1)$	switch	from Newton (0) to the deep regime (1)
$q_{\Delta s} = \Delta s/r$	core nome	second nome of the funnel, a third of the depth, cusp form $\Delta(q_{\Delta s}) = q_{\Delta s}{\cdot}\prod(1 - q_{\Delta s}^{\,n})^{24}$
ξ, w	mode, weight	closure rule of the whole, $w_{DE} = w_0 + w_a{\cdot}z/(1 + z)$ the equation of state of dark energy
$Q = \Omega_c/\Omega_b$	weight of the mode	mean return time of the whole minus one
$b_1 = 3$	number of loops	independent loops of the torus
ρ_s, Ω_s	vacuum part	restoring per loop
λ_H	holding length	distance at which a held funnel holds its building blocks
τ	modulus	shape of the torus, in $d\tau/dt$ the proper time
$n_B = 1/(e^s - 1)$	Bose occupation	number of stimulated returns
$\tilde{m} = m_{Pl}^2/(2M)$	mirror mass	mass whose quantum motion is the return
$kT_I = \hbar a_0/c$	period of the return	period of the inner direction, written as a temperature, not heat at the place
$\varepsilon_L = \ell_P/\ell_0$	ladder number	fixes the rungs of the ladder of lengths
$\Theta = 2\pi a_0/(cH)$	breathing number	threshold against the expansion rate, clock of the whole from 0 to $\Theta_\infty = \sqrt{15}/\pi$
α	fine-structure number	winding against the quantum length, $\alpha = r_e/\bar{\lambda}_e$, in hydrogen v/c
$\alpha_G = 3\Delta s/\bar{\lambda}$, $\alpha_I = \bar{\lambda}/\ell_0$	couplings	fold against quantum length and quantum length against inner direction, their product $(v_{flat}/c)^4$
$\tilde{r} = \sqrt{(\bar{\lambda}{\cdot}\ell_0)}$	return depth of the quantum motion	threshold of the mirror mass, $\tilde{r}^2 = \hbar{\cdot}\tau_D/M$
$H_\infty = c{\cdot}\sqrt{\Lambda_s}$	rate of the end state	expansion from three vacuum parts alone, 55.6 km/s/Mpc
R_{max}	largest turnaround	edge beyond which no region binds, around a galaxy v_{flat}/H_∞

Symbol	Name	Meaning
M_N, r_N	limit of the end state	above $M_N = (\sqrt{5/6\pi^2})\cdot M_U$ a fold has no region at rest, $r_N = c/(\sqrt{3}\cdot H_\infty)$
R_E, D_E	Einstein ring	radius of the ring, $D_E = D_l\cdot D_{ls}/D_s$ from the angular distances of lens and source
$M_H = c^3/(2GH)$	Hubble mass	mass of the Hubble sphere at the critical density
δ_c	collapse threshold	linear value at collapse without restoring, $(3/20)\cdot(12\pi)^{2/3}$
d_{eff}	flux dimension	dimension the rule leaves to a form
k	coupling	strength with which two cycles lock in, in kT the Boltzmann constant
φ	phase	of the coupling, k·cos φ its lossy part
$j = a_J/(3\Delta s)$, $a_J = J/(Mc)$	spin number, twist	rotation of a funnel with angular momentum J, j = 1: both horizons coincide
M_{irr}, M_{rot}	partial masses of the rotating fold	legs in the Thales circle over r_0, $M^2 = M_{irr}^2 + M_{rot}^2$, the entropy reads only M_{irr}
counting rule	closure per sense of circulation	every state holds one winding per sense of circulation, it holds white dwarfs and neutron stars
l, κ, Ω, ν_z	orbital winding, frequencies	number of nodes of an orbit, frequencies of the radial, the orbital and the perpendicular oscillation, in $R_{ta}^3 = GM/(\kappa\cdot H^2)$ κ the turnaround factor, in the lens the convergence
$\ell_0 = 2\Lambda$	length of action	measure of the inner direction
M_{min}, M_c, M_U	rungs of the ladder	$\varepsilon_L\cdot m_{Pl}$, $\varepsilon_L^{1/3}\cdot m_{Pl}$, m_{Pl}/ε_L
$M_* = m_{Pl}^3/m_p^2$	mass scale of stars	1.853 M⊙
$M_{Gas} = (81/4)\cdot\sigma^4/(Ga_0)$	smallest gas-bearing mass	what the thermal motion σ of the gas holds
R_{ta}, R_Q	turnaround radius, seam	where a region detaches from the expansion, where the local q reaches the cosmic one
L	cell	spatial cell of the torus, today 27.9 Gpc
$\Lambda_s = \pi^4/(15\Lambda^2)$	constant per closure length	the vacuum part in the action, $\rho_s = \Lambda_s\cdot c^2/(8\pi G)$ its density
$Q_{el} = n\cdot e$	charge	winding n of the phase of a node
Ω_b, Ω_c, Ω_m, Ω_{DE}, ω_b, ω_c	densities	shares of the baryons, the mode, matter ($\Omega_m = \Omega_b + \Omega_c$) and dark energy in the critical density ρ_{crit}, $\omega = \Omega\cdot h^2$ with $h = H_0/(100$ km/s/Mpc)
Y_d	mass-to-light	conversion of light into fold, 0.67
$\mathscr{T}$	tide	of the smoothed parent field
T	step, temperature, period	step by one return in the operator $\mathscr{P}$, in kT the temperature (T_U after Unruh, T_I of the return, T_{CMB} of the background radiation), in $T^2 = 4\pi^2r^3/(3c^2\Delta s)$ the orbital period, T^n the torus of n circulations
B	coefficient, binding	coefficient of the backflow, ln B, in $Z\cdot E_H + N\cdot E_n - B$ the binding of the nucleus
$x = E_C/(2E_S)$	fissility	winding against holding in the nucleus
$\Sigma_0 = a_0/(\pi G)$	threshold of surface density	242 M⊙/pc², the centre of a disc at the threshold
Q_T	Toomre measure	stability of a disc, along the slope with $G\cdot\nu(1 - \chi/2)$
ℓ_J, ℓ_D	Jeans and Debye length	where the fold builds up and the winding screens, $\ell_J/\ell_D = \sqrt{(\alpha/\alpha_G)}$
N(N + 1), ν	ladder of periods	couplings of the mirrored field of the shell, N = n + l, and its height $3.44\cdot Z^{2/3}$
L_{Edd}	Eddington limit	radiative transport in the star, $(9/2)\cdot m_p\cdot c^3\cdot\Delta s/r_e^2$
t_{Edd}	time of the limit	growth of a fold at L_{Edd} by a factor e in $(\varepsilon/(1 - \varepsilon))\cdot t_{Edd}$, $t_{Edd} = \frac{2}{3}\cdot(r_e/r^*(m_p))^2\cdot\tau_D$
kT_{ab}, z_{ab}	temperature of release	ionised gas gives its heat to the light as fast as the whole breathes

Symbol	Name	Meaning
λ_{dB}	thermal length of the gas	mean of quantum length and thermal length of the same heat, $\lambda_{dB}^2 = \bar{\lambda}\cdot\lambda_H$
β	share of the gas	in the pressure of a funnel held by light and heat together
$k_{DE} = \rho_{DE}\cdot Gc^2/a_0^2$	time-fixed number	ratio without a clock, vacuum part $k_s = \pi^3/30$
c_{eff}	light from outside	$c\cdot\sqrt{(1 - 6\Delta s/r)}$, delay of the unrolling
$\mathscr{P}$	closure operator	acts on the generator of motion, not as a source
G, c, ħ, e	constants of nature	gravitational constant, speed of light, quantum of action $\hbar = h/2\pi$, elementary charge
M, m, r, R, t	mass, distance, time	mass of the body (m that of the particle, M☉ the solar mass), distance from the centre, envelope, time
m_p, m_e, m_u	masses	proton, electron, atomic mass unit
Z, N, A	nuclear numbers	protons, neutrons, mass number A = Z + N, N also the number of building blocks of a held funnel and N = n + l the step of the shell
n	number of the shell	n + l the step, in $Q_{el} = n\cdot e$ the winding, in Σq^n the number of returns
H, H_0	expansion rate	today H_0 (km/s/Mpc), $\Delta s_H = c/(6H)$ the fold of the horizon
z	redshift	$1 + z = L(t_{ob})/L(t_{em})$, in ż the state, in $u = z^{(a+3)/2}$ the place in the plane
a(t), $\bar{L}$	stretching	measure of the stretching of the loops, $L(t) = \bar{L}\cdot a(t)$
σ	balance, thermal motion	$\Sigma\Delta s - \Delta L = 0$, σ, σ_v the speed of the thermal motion of the gas, as a measure of measurement the standard deviation
S	entropy	S_{eq} in equilibrium, S_Λ of the vacuum part, in $V/(S\cdot R)$ the surface, in $S = 2\sqrt{2}$ Bell's measure of correlation
E	energy	E_C, E_S Coulomb and surface energy of the nucleus, E_H, E_n rest energies of hydrogen atom and neutron, E_G the Gamow energy, E(Δ) the correlation
u, G(ϖ)	return	share of the state at depth ϖ, G(ϖ) its stationary profile
Φ, Φ_N, A, Ψ	potentials	potential of the metric, Newtonian potential, potential of the vortex, second potential of the metric
F, I	generator, identity	vector field of the motion (F_{par} the parametric part), identity operator
b, p	ray, orbit	distance of the light ray from the centre, parameter of the orbit, p also the number of the period
R_d, M_b	disc	disc scale length, baryonic mass of a galaxy
μ, m̄	moments and masses	magnetic moment (μ_B the Bohr magneton), line mass of a filament, mean mass per particle (μ_e per electron) in m_u, m̄ the mean particle mass
ζ, P, O(k)	functions	Riemann's zeta function, associated Legendre function, states of the held body up to shell k
T_α, Q_α	alpha decay	half-life and energy of the decay
ψ, Δω, W	cycles	phase difference of two cycles, difference of their eigenfrequencies, frequency band
i, ln, lg	signs of calculation	imaginary unit, natural and decimal logarithm
AU, pc, fm, dex, mag	units	distance of the Earth from the Sun, parsec (kpc, Mpc, Gpc), 10^{-15} m, power of ten of the logarithm, magnitude

Appendix B: Data

All numbers in this treatise are computed in the model and checked against the simulation. Every measurement names its data in the text, among them SPARC (rotation curves of 175 galaxies, the computation uses the 141 whose curves have quality 1 or 2 and whose inclination is at least 30°), AME2020 (nuclear masses), PubChem (periodic table), Pantheon+ (supernova distances) and DESI DR2 (distances from the sound wave of the early cosmos). The cosmic densities ω_b, ω_c and the temperature of the background radiation are taken from measurement (Planck 2018).

Appendix C: Sources

Data.

- SPARC: F. Lelli, S. S. McGaugh, J. M. Schombert (2016), rotation curves and photometry of 175 disc galaxies.
- AME2020: Atomic Mass Evaluation 2020, nuclear masses and binding energies.
- PubChem: periodic table of the elements (ionisation energy, atomic radius, melting and boiling points).
- Pantheon+SH0ES: D. Scolnic et al. (2022), D. Brout et al. (2022), distances of type Ia supernovae with covariance.
- DESI DR2: DESI Collaboration (2025), distances from baryon acoustic oscillations.
- Planck 2018: cosmological parameters ω_b, ω_c and the temperature of the background radiation.

Mathematical theorems used in the treatise.

- M. Kac (1947): mean return time in a closed space.
- J. Bertrand (1873): only two force laws close every bound orbit.
- V. Fock (1935): Kepler orbits as great circles on the sphere S^3.
- P. A. M. Dirac (1931): product of electric and magnetic charge.
- B. Riemann (1859): mirror equation of the ζ-function.
- The modular group PSL(2,$\mathbb{Z}$) and Dedekind's η-function: symmetries of the torus shape.
- M. Morse: critical points and Euler characteristic.
- The duality of the force laws of Newton and Hooke, $(a + 3)(A + 3) = 4$.
- L. H. Thomas, E. Fermi (1927): the shell as a held funnel, $R \propto Z^{-1/3}$.
- J. H. Lane, R. Emden: the form $n = 3$ of the held funnel, $\omega_3 = 2.018$.
- L. Onsager, H. Casimir: symmetric and antisymmetric part of transport.
- A. S. Eddington: the radiation limit of a star.
- B. Carter (1974): stars at the border of radiation and convection, $\alpha_G \approx \alpha^{12} \cdot (m_e/m_p)^4$.

The author's source. Tabula Smaragdina (conclusion).